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D I S S E R T A T I O N

ZUR ERLANGUNG DES AKADEMISCHEN GRADES EINES
DOKTORS DER NATURWISSENSCHAFTEN (DR. RER. NAT.)

CLASSIFICATION OF GOOD COMPACTIFIED JACOBIANS

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ABSTRACT

Given a nodal curve X, we introduce the notion of a good compactified Jacobian as a proper algebraic space which is the good moduli space of an open connected substack of the stack $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ of rank 1 torsion-free sheaves on X.
After providing a local description of the stack $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ for every nodal curve X, we focus on necklaces of genus $g = 1$, namely nodal curves whose dual graph is an n-agon.
We first give a geometric and combinatorial classification of all good compactified Jacobians of a fixed necklace X.
Subsequently we consider all the stable necklaces in $\overline{\mathcal{M}}_{1,n}$ and we give a combinatorial classification of all the possible choices of a good compactified Jacobian for every necklace in $\overline{\mathcal{M}}_{1,n}$, such that this choice is compatible with degenerations of curves.
In the last part of this thesis, we classify all the good compactified Jacobians of a vine curve with 3 nodes, namely a curve given by two copies of $\mathbb{P}^1$ meeting transversally at 3 nodes.

ZUSAMMENFASSUNG

Für eine nodale Kurve X führen wir den Begriff einer guten kompaktifizierten Jakobischen als einen eigentlichen algebraischen Raum ein, welcher der gute Modul-Raum eines offenen zusammenhängenden Unterstacks des Stacks $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ der torsion-freien Garben von Rang 1 auf X ist.
Nachdem wir eine lokale Beschreibung des Stacks $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ gegeben haben, konzentrieren wir uns auf Halsketten vom Geschlecht $g = 1$, d.h. nodale Kurven, deren dualer Graph ein n-Eck ist.
Zunächst finden wir eine geometrische und kombinatorische Klassifikation aller guten kompaktifizierten Jakobischen einer festen Halskette X.
Anschließend betrachten wir alle stabilen Halsketten in $\overline{\mathcal{M}}_{1,n}$ und finden eine kombinatorische Klassifikation aller möglichen Wahlen einer guten kompaktifizierten Jacobischen für jede Halskette in $\overline{\mathcal{M}}_{1,n}$, sodass diese Wahl mit Degenerationen von Kurven kompatibel ist.
Im letzten Teil dieser Arbeit klassifizieren wir alle guten kompaktifizierten Jacobischen einer Rebenkurve mit 3 Knoten, also einer Kurve, die durch zwei Kopien von $\mathbb{P}^1$ gegeben ist, die sich transversal in 3 Knoten schneiden.

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INTRODUCTION

In this thesis we use methods from algebraic stacks to classify compactified Jacobians of some classes of nodal curves.
Recall that for a smooth projective curve X the degree d Jacobian $J^d(X)$, namely the moduli space of line bundles of degree d on X, is a smooth projective variety.
For all (g, n), such that $2g + n > 2$, we can consider the moduli stack $\mathcal{M}_{g,n}$ of smooth n-pointed stable curves of genus g. For all integers d, there exists a universal Jacobian $\mathcal{J}^d_{1,n}$ endowed with a smooth fibration $\mathcal{J}^d_{1,n} \to \mathcal{M}_{g,n}$ whose fiber over an n-pointed curve $(X, q_1, \ldots, q_n)$ is $J^d(X)$.
Starting from the Deligne–Mumford compactification $\overline{\mathcal{M}}_{g,n}$ of $\mathcal{M}_{g,n}$, one can try to extend the universal Jacobian $\mathcal{J}^d_{1,n}$ to the universal family over $\overline{\mathcal{M}}_{g,n}$

$$\begin{tikzcd} \mathcal{J}^d_{1,n} \arrow[r] \arrow[d] & ? \arrow[d] \\ \mathcal{M}_{g,n} \arrow[r] & \overline{\mathcal{M}}_{g,n} \end{tikzcd} .$$

This leads to consider nodal curves and to study the moduli spaces of line bundles of degree d on nodal curves.
For reducible nodal curves this moduli space has infinitely many connected components, so it is not of finte type, and moreover these components may fail to be proper.
Starting from the work of Oda and Seshadri several approaches have been developed to compactify Jacobians for nodal curves. The central idea is to enlarge the moduli problem by considering not only line bundles but also degenerations of line bundles, which are taken to be rank 1 torsion-free sheaves.
In particular, in [7], Oda and Seshadri introduce a notion of semistability of rank 1 torsion-free sheaves with respect to some numerical polarizations and find compactifications of Jacobians by considering all rank 1 torsion-free sheaves which are semistable with respect to some fixed numerical polarization. These methods give us some possible compactifications of Jacobians.

In [2] Pagani and Tommasi introduce the notion of *fine* compactified Jacobians of X as an open connected proper subscheme of the moduli space of *simple* rank 1 torsion-free sheaves on X, namely rank 1 torsion-free sheaves whose automorphism group is $\mathbb{G}_m$. They provide a complete classification of fine compactified Jacobians for necklaces, namely nodal curves of genus $g = 1$ whose dual graph is an n-agon. They find that the fine compactified Jacobians of a necklace X are also necklaces. If the necklace X has length $n \geq 3$ the fine compactified Jacobians of X are necklaces of length $m = nk, k \in \mathbb{N}$.

The aim of this thesis is to remove the assumption that the sheaves are simple and to classifly all open connected substack of the stack $\underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X)$ of rank 1 torsion-free sheaves on X which admit a proper good moduli space. We refer to a proper algebraic space which is a good moduli spaces of some open connected substack of $\underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X)$ as *good* compactified Jacobian of X.

After providing, in Chapter 2, a local description of the stack $\underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X)$ for any nodal curve X, we focus on necklaces of genus $g = 1$ and, in Chapter 3, we give a complete classification of all the open connected substacks of $\underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X)$ which admit a proper good moduli space.

Given a necklace X and an open connected substack $\mathcal{U}$ of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ the key point to find good compactified Jacobians is to consider the set $\mathcal{S}$ of the most degenerate sheaves of $\mathcal{U}$, namely the non-locally free sheaves of $\mathcal{U}$ whose isomorphism class defines closed points in $\mathcal{U}$. For every sheaf $N \in \mathcal{S}$ we consider the open $\mathcal{U}(N)$ given by all the sheaves in $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ specializing to the sheaf N. The good moduli space of the open $\mathcal{U}(N)$ turns out to be (see Proposition 3.5):

- A nodal $\mathbb{P}^1$, and hence proper, when N is not locally free at all the nodes,
- A nodal curve given by two copies of $\mathbb{A}^1$ meeting at the origin, and hence separated, when N is not locally free at a proper subset of the set of nodes of X.

We study the properties that a collection of opens of the form $\mathcal{U}(N)$ satisfy so that the union is an open connected substack of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ which admits a proper good moduli space. This leads us to introduce the definition of cyclic opens (see Definition 3.15) and equivalently, associating to every sheaf N_i of $\mathcal{S}$ the set of nodes Σ_i where N_i is not locally free, the cyclicity of a collection of opens $\mathcal{U}(N_i)$'s can be expressed by saying that, for some multidegree $\underline{d}$, the sequence $(\underline{d}, \Sigma_1, \ldots, \Sigma_m)$ has to be r-admissible (see Definition 3.20).

Theorem A (Theorem 3.28)**.** *Let X be a necklace of length n and genus $g = 1$. There is a bijection*

$$f\colon \left\{ \begin{array}{c} \textit{Open and connected substacks} \\ \mathcal{U} \subseteq \underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X) \textit{ that admit} \\ \textit{a proper good moduli space} \end{array} \right\} \longrightarrow \bigcup_{r \in \mathbb{N}_{\geq 1}} \left(\left\{ \begin{array}{c} r\textit{-admissible sequences} \\ (\underline{d}, \Sigma_1, \ldots, \Sigma_m) \end{array} \right\}_{/\sim} \right)$$

The proper good moduli space of an open connected substack $\mathcal{U}$ of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ corresponding to the r-admissible sequence $(\underline{d}, \Sigma_1, \ldots, \Sigma_m)$ is a necklace of length m.

similarly to the case of fine compactified Jacobians we have found that all good compactified Jacobians of a necklace are necklaces themselves. In the simple case the length of the fine compactified Jacobians of a necklace X has to be a multiple of the length of X, while for good compactified Jacobians we find that (see Corollary 3.32):

- If $n = 2$ the length of a good compactified Jacobian is either 1 or 2.
- If $n = 3$ the length of a good compactified Jacobian can be every natural number but $4, 5$ and 7.
- If $n = 4$ the length of a good compactified Jacobian can be every natural number but 5.
- If $n \geq 5$ the length of a good compactified Jacobian can be every natural number.

Starting from Chapter 5 we consider families of good compactified Jacobians. This leads to consider good compactified Jacobians which correspond to 1-admissible sequences.

In Chapter 4, we show that the good compactified Jacobians corresponding to 1-admissible sequences can be obtained taking all the rank 1 torsion-free sheaves which are semi-stable with respect to some numerical polarization (see Proposition 4.5).

In Chapter 5 we consider the universal curve $\overline{\mathcal{C}}_{1,n}/\overline{\mathcal{M}}_{1,n}$ over $\overline{\mathcal{M}}_{1,n}$ and a good compactified Jacobian $\overline{\mathcal{J}}_{1,n}$ of degree d for the curve $\overline{\mathcal{C}}_{1,n}/\overline{\mathcal{M}}_{1,n}$. The fibre $\overline{\mathcal{J}}_{1,n}(X)$ over the point $[X] \in \overline{\mathcal{M}}_{1,n}$ is a good compactified Jacobian of degree d for X and the collection, taken over all the points $[X] \in \overline{\mathcal{M}}_{1,n}$, of the good compactified Jacobians $\overline{\mathcal{J}}_{1,n}(X)$ is compatible with curves degenerations or equivalently with morphisms between the corresponding dual graphs.

We introduce the definition of an *admissible* assignment L_n^d (see Definition 5.1), namely a collection of good compactified Jacobians of degree d, one for every necklace of $\overline{\mathcal{M}}_{1,n}$, such that this collection is compatible with degenerations of curves.
Focusing on all the necklaces of length 2 in $\overline{\mathcal{M}}_{1,n}$, we introduce

- An integer valued function f which carries the information, for every necklace X of length 2, of the smallest integer $-e$ such that the line bundles of multidegree $(e+d, -e)$ are in the good compactified Jacobian of X,
- A set $\mathcal{P}$ that carries the information of which necklaces of length 2 have a good compactified Jacobian which contains a sheaf that is not locally free at both nodes.

These two pieces of information are enough to know the good compactified Jacobian associated to every necklace of length 2 and we explain that starting from a pair $(f, \mathcal{P})$ that is compatible (see Definition 5.22) we can construct an admissible assignment.

Theorem B (Theorem 5.23)**.** *There is a bijection*

$$\left\{ \textit{ Admissible assignments } L_n^d \right\} \longrightarrow \left\{ \begin{array}{c} \textit{compatible pairs} \\ (f, \mathcal{P}) \end{array} \right\}.$$

The main part of the Theorem is to explain how we can reconstruct an admissible assignment L_n^d, namely how we can choose a good compactified Jacobian for every necklace of $\overline{\mathcal{M}}_{1,n}$ starting from the compatible pair $(f, \mathcal{P})$. This is done by studying the smoothable good compactified Jacobians of necklaces of length n in $\overline{\mathcal{M}}_{1,n}$ and by giving a numerical characterization (see Proposition 5.13) parallel to the numerical characterization given in [2, Proposition 3.3] to describe smoothable fine compactified Jacobians.

To classify admissible assignments, a crucial role has been played by necklaces of length 2. In [4] it has been shown that the key to classify universal fine compactified Jacobians over $\overline{\mathcal{M}}_{g,n}$ is to classify fine compactified Jacobians for necklaces and vine curves, i.e. curves with two smooth irreducible components meeting in a finite number of nodes.

In Chapter 6 we make a first step towards the analogous classification for good compactified Jacobians by classifying good compactified Jacobians for vine curves with 3 nodes. For any number of nodes a similiar classification seems within reach, but the concrete geometric description is a bit more complicated than the case of 3 nodes and will be taken up in a subsequent work.

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1. PRELIMINARIES

1.1 General framework

We work over the category of schemes of finite type over an algebraically closed field $k = \overline{k}$. The main objects we are going to consider are *nodal curves*, namely complete, connected, reduced schemes of finte type of pure dimension 1 whose points are either smooth or ordinary double points: A point x of a curve X is an *ordinary double point* if and only if

$$\hat{\mathcal{O}}_{X,x} \simeq k[|x,y|]/(xy).$$

We denote by $\hat{\mathcal{O}}_0 := k[|x,y|]/(xy)$ the *standard node*.
When we work with families of nodal curves, we consider *n-pointed* nodal curves, namely nodal curves together with n distinct smooth points of X.
We denote by Γ_X the the dual graph of a nodal curve X. We recall that the *dual graph* Γ_X of a nodal curve X is a graph having one vertex for each irreducible component of X and one edge for each node of X such that an edge is adjacent to a vertex if and only if the corresponding node belongs to the corresponding irreducible component.
Given a graph Γ an orientation $O = (s,t)$ of the graph Γ is an assignment of a direction to each edge e of Γ: For an edge e connecting the vertices v and w, we define the source of the edge e with respesct to O as $s(e,O) = v$ and the target of e with respect to O as $t(e,O) = w$.
Given a subset $V' \subseteq V(\Gamma)$ of the set of vertices $V(\Gamma)$ of Γ and an orientation O of the vertices in V', we denote by $\mathcal{D}(\Sigma', O)$ the *indegree divisor* of the orientation O, namely the divisor on Γ which assigns to each vertex v the value equal to the number of oriented edges e such that $t(e) = v$.
A consistent part of this document focuses on a class of nodal curves called necklaces.

Definition 1.1.
A *necklace* of length n is a nodal curve X whose dual graph is a n-agon.

We denote by $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ the stack of rank 1 torsion-free sheaves on X and by $\underline{\mathrm{Coh}}_1^{\mathrm{tf},d}(X)$ the stack of rank 1 torsion-free sheaves of degree d.

1.2 Rank 1 torsion-free sheaves on nodal curve

In this section we recall the main definitions and properties of rank 1 torsion-free sheaves, we follow mainly [8].
We assume that X is a nodal curve with m irreducible components, that we denote by X_i and n nodes. We denote by I a rank 1 torsion-free sheaf on X and by Σ' the set of nodes of X where I fails to be locally free. We denote by $\pi_{\Sigma'} \colon X_{\Sigma'} \to X$ the normalization of X at Σ'.

Definition 1.2: Rank 1 torsion-free sheaf.
A quasi-coherent sheaf I over a curve X is called

- *rank 1*: if it has rank 1 at every generic point
- *torsion-free*: if it has no embedded components, namely the dimension of the support of every non-zero subsheaf J of I has dimension equal to the dimension of the support of I and the support of I is X.

Remark 1.3. Line bundles and ideal sheaves of points are rank 1 torsion-free sheaves.

Remark 1.4. [2, Section 3.2] Over the standard node $\mathcal{O}_0$ there are exactly two different rank 1 torsion-free sheaves:

- The free module;
- The ideal $(x, y) \subset \mathcal{O}_0$ considered as an $\mathcal{O}_0$-module.

There are important numerical invariants that can be associated to a rank 1 torsion-free, namely the *degree* and *multidegree*.

Definition 1.5: Degree of a sheaf.
The *degree* d of a rank 1 torsion-free sheaf I on a curve X is defined to be

$$\deg(I) \coloneqq \chi(I) - \chi(\mathcal{O}_X),$$

where χ denotes the Euler characteristic.

Definition 1.6: Multidegree.
If I is a rank 1 torsion-free sheaf on a curve X with irreducible components $X_1, \dots, X_m$, we denote by I_{X_i} the maximal torsion free quotient of $I \otimes \mathcal{O}_{X_i}$, and then define the *multidegree* of I by

$$\deg(I) \coloneqq (\deg(I_{X_1}), \dots, \deg(I_{X_n}))$$

Definition 1.2 might seem a bit technical and not easy to work with. In the following we see that rank 1 torsion-free sheaves are strictly related to line bundles.
Relation between rank 1 torsion-free sheaves on a curve and line bundles on a normalization of the curve.
Following [8, Chapter 10, Section 2] we can describe line bundles on X as follows.

Proposition 1.7. *Given a line bundle L on X, we can consider $\pi^* L$ which is the data $(L_1, \dots, L_m)$ of certain line bundles L_i on the irreducible components $X_1, \dots, X_m$ of X. The line bundle L can be described as the data $(L_1, \dots, L_m)$ together with an identification $\alpha_P \colon L_{s(P)} \otimes k(P) \to L_{t(P)} \otimes k(P)$ for every node P of X.*

We have just seen that any line bundle can be described looking at line bundles on a normalization of the curve. Something similiar also happens for sheaves which fail to be locally free at some nodes.

Proposition 1.8. *[7, Proposition 10.1, Proposition 10.2] Let X be a nodal curve. A rank 1, torsion-free sheaf I which fails to be locally free at the nodes Σ' of X has the form $\pi_* L$, where $\pi \colon X_{\Sigma'} \to X$ is a (total or partial) normalization of X at Σ' and L is a line bundle on $X_{\Sigma'}$. The total degree of L and I are related by the formula*

$$d(L) = d(I) - |\Sigma'|$$

The aim of this thesis is to classify open connected substacks of the stack $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ of rank 1 torsion-free sheaves on X which admit a proper good moduli space.

Definition 1.9: Good compactified Jacobian.
Let X be a nodal curve. A *good compactified Jacobian of X* is a proper algebraic space which is the good moduli space of an open connected substack of the stack $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ of rank 1 torsion-free sheaves on X.

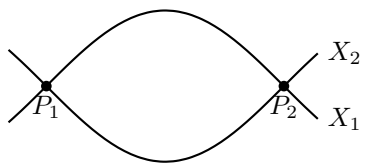
Definition 1.10: Good compactified universal Jacobian.
A good compactified *universal* Jacobian is a good compactified Jacobian in the sense of Definition 1.9 for the universal curve $\overline{\mathcal{C}}_{1,n}/\overline{\mathcal{M}}_{1,n}$ over $\overline{\mathcal{M}}_{1,n}$.

2. LOCAL DESCRIPTION OF $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$

The aim of this Chapter is to give an explicit local description of the stack $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ of rank 1 torsion-free sheaves on X for any nodal curve X around any rank 1 torsion-free sheaf. In the first Section we study the case when X is a necklace of genus $g = 1$ and length 2. In the second Section we use ideas from the explicit example in Section 2.1 to generalize the local description of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ to any nodal curve X.

2.1 Guiding example

We consider a necklace X of genus $g = 1$, consisting of two irreducible components X_1 and X_2 of genus $g = 0$, which intersect at two nodes, P_1 and P_2.



We denote by $\pi_\Sigma \colon X_\Sigma \to X$ the total normalization of X at the set of nodes $\Sigma := \{P_1, P_2\}$. We denote by $O^+ = (s_+, t_+)$ the orientation of the dual graph Γ_X of X such that

- $s_+(P_1) = X_1$
- $s_+(P_2) = X_2$.

We denote the preimages of P_i under the normalization map π_Σ by $\pi_\Sigma^{-1}(P_i) := \{S_i, T_i\}$ and we assume that $S_i \in X_{s_+(P_i)}$ and $T_i \in X_{t_+(P_i)}$, specifically $S_1, T_2 \in X_1$ and $S_2, T_1 \in X_2$. We fix the total degree $d = 0$.

The Example is divided into three parts. We first study degenerations of line bundles of some fixed multidegree $(d, -d)$, then we describe the stack $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ around a sheaf which is not locally free at any of the nodes and finally we use these pieces of information to find all the good compactified Jacobians of X.

We start by studying the stack $\underline{\mathrm{Pic}}^0(X)$ of line bundles on X of total degree 0 which is an open dense substack of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$.

Lemma 2.1. *The stack $\underline{\mathrm{Pic}}(X)$ of line bundles on X can be described as*

$$\underline{\mathrm{Pic}}(X) = \textstyle\bigcup_{d\in\mathbb{Z}} \mathbb{G}_m \times \mathrm{B}\,\mathbb{G}_m \ .$$

Proof. As we have recalled in Proposition 1.7, every line bundle L on X is described by the data (L_1, L_2, x_1, x_2), where

- L_i is a line bundle on X_i of degree d_i, such that $d_1 + d_2 = d = 0$,
- $x_i \colon L_{S_i} \to L_{T_i}$ is an isomorphism.

Moreover $L_i \cong \mathcal{O}_i(d_i \cdot \underline{1})$ and the line bundle $(\mathcal{O}_1, \mathcal{O}_2, 1, 1)$ is isomorphic to $(\mathcal{O}_1, \mathcal{O}_2, \alpha, \alpha)$, for every $\alpha \in k^\times$. Therefore we can describe every line bundle $L = (\mathcal{O}_1(d_1 \cdot \underline{1}), \mathcal{O}_2(d_2 \cdot \underline{1}), x_1, x_2)$ of multidegree (d_1, d_2) such that $d_1 + d_2 = 0$ as

$$(\mathcal{O}_1(d_1 \cdot \underline{1}), \mathcal{O}_2(-d_1 \cdot \underline{1}), x, 1)$$

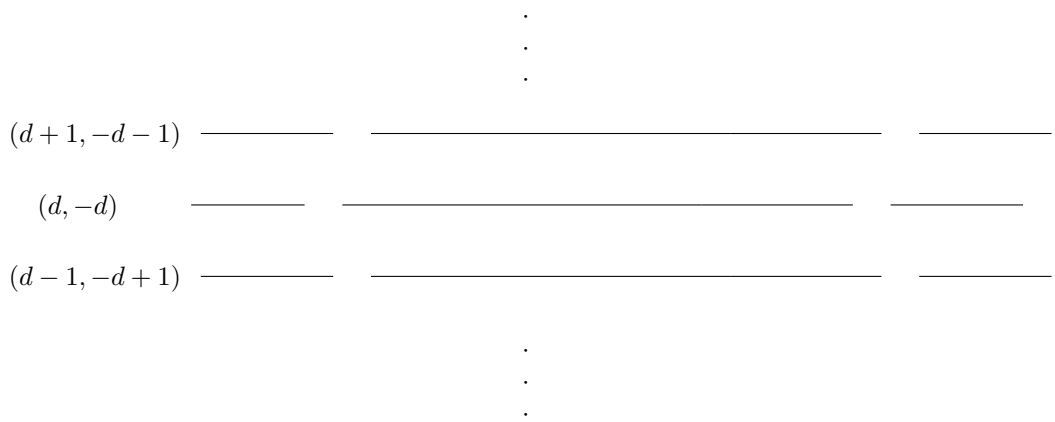
where $x := \frac{x_1}{x_2} \in k^\times$.
This means that any line bundle is identified by an integer and an element of $k^\times$, namely

$$\underline{\mathrm{Pic}}(X) = \bigcup_{d \in \mathbb{Z}} \mathbb{G}_m \times \mathrm{B}\mathbb{G}_m \ .$$

□

Notation 2.2.
Pictorially we represent $\mathbb{G}_m$ as:

similarly, we represent $\underline{\mathrm{Pic}}^0(X)$ as



Notation 2.3.
We denote by L_{x_1,x_2} the line bundle $(\mathcal{O}_1, \mathcal{O}_2, x_1, x_2)$ of multidegree $(0,0)$.

Remark 2.4. $\underline{\mathrm{Pic}}(X)$ is an explicit example of a moduli space which is not of finite type and whose irreducible components are not proper.

From Proposition 1.8 we know that all the rank 1 torsion-free sheaves on X are given by the push-forward of a line bundle on the total or partial normalization of X at some subset of the set of nodes Σ. Therefore the singular locus of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ is given by:

- Sheaves that fail to be locally free at one node P_j which are the push-forward of a line bundle of multidegree (e_1, e_2) on the partial normalization X_j of X at P_j such that $e_1 + e_2 = -1$. We denote this sheaf by $N_j^{(e_1,e_2)}$.
- Sheaves that fail to be locally free at both nodes P_1 and P_2 which are the push-forward of a line bundle of multidegree (e_1, e_2) on the normalization X_Σ of X at Σ such that $e_1 + e_2 = -2$. We denote this sheaf by $N_{12}^{(e_1,e_2)}$.

In the first part of this Example we study the closure in $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ of the stack $\mathcal{J}^d(X)$ of line bundles on X of multidegree $(d, -d)$.

Definition 2.5.
We refer to sheaves in the closure of $\mathcal{J}^d(X)$ in $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ as sheaves which can be obtained as *flat limits* of line bundles of multidegree $(d, -d)$ or as *degenerations* of line bundles of multidegree $(d, -d)$.

For simplicity, we fix the multidegree $(d, -d) = (0, 0)$.

Lemma 2.6. *The closure in $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ of the stack $\mathcal{J}^0(X)$ of line bundles of multidegree $(0,0)$ on X contains the non-locally free sheaves $N_1^{(0,-1)}, N_2^{(0,-1)}, N_1^{(-1,0)}, N_2^{(-1,0)}, N_{12}^{(-1,-1)}, N_{12}^{(-2,0)}$ and $N_{12}^{(0,-2)}$.*

Proof. Every line bundle of multidegree $(0,0)$ on X can be expressed as $(\mathcal{O}_1, \mathcal{O}_2, x_1, x_2)$. For every $x_1, x_2 \in k^\times = \mathrm{Isom}(L_{S_i}, L_{T_i})$, we consider the map $\psi^0_{x_1,x_2}$ defined as

$$\psi^0_{x_1,x_2}{}^{1} : : \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) \to k_{P_1} \oplus k_{P_2}$$

that sends a section (f_1, f_2) to $(x_1 f_1(S_1) - f_2(T_1), x_2 f_2(S_2) - f_1(T_2))$.
Taking the kernel of this map gives back the line bundle

$$L_{x_1,x_2} = (\mathcal{O}_1, \mathcal{O}_2, x_1, x_2)$$

of multidegree $(0,0)$.
The key point is to study what happens when we allow x_1 and x_2 to be 0:

- If $x_1 = 0$ and $x_2 \neq 0$, taking the kernel of the map ψ^0_{0,x_2} gives the equation $f_2(T_1) = 0$, which means that the sheaf that we find is the push-forward of the line bundle $\mathcal{O}_1 \oplus \mathcal{O}_2(-T_1)$, hence we find the sheaf $N_1^{(0,-1)}$,
- If $x_1 \neq 0$ and $x_2 = 0$, taking the kernel of the map $\psi^0_{x_1,0}$ gives the equation $f_1(T_2) = 0$, hence we find the sheaf that is the push-forward of the line bundle $\mathcal{O}_1(-T_2) \oplus \mathcal{O}_2$, namely the sheaf $N_2^{(-1,0)}$,
- If $x_1 = x_2 = 0$ we have $f_2(T_1) = f_2(T_2) = 0$, hence taking the kernel we obtain the sheaf which is the push-forward of the line bundle $\mathcal{O}_1(-T_2) \oplus \mathcal{O}_2(-T_1)$, hence the sheaf $N_{12}^{(-1,-1)}$.

Taking $y_1, y_2 \in k$ we can define a similiar map:

$$\phi^0_{y_1,y_2} : : \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) \to k_{P_1} \oplus k_{P_2}$$

that sends a section (f_1, f_2) to $(f_1(S_1) - y_1 f_2(T_1), f_2(S_2) - y_2 f_1(T_2))$.
Taking the kernel of this map when $y_1, y_2 \in k^\times$ gives the line bundle

$$L_{y_1{}^{-1},y_2{}^{-1}} = (\mathcal{O}_1, \mathcal{O}_2, y_1{}^{-1}, y_2{}^{-1})$$

of multidegree $(0,0)$, in particular, for every $\alpha, \beta \in k^\times$ we have

$$\mathrm{Ker}(\psi^0_{\alpha,\beta}) = \mathrm{Ker}(\phi^0_{\alpha^{-1},\beta^{-1}}).$$

Analyzing as above what happens when we consider either $y_1 = 0$ or $y_2 = 0$, we have the following situations:

- If $y_1 = 0$ and $y_2 \neq 0$, taking the kernel of the map ϕ^0_{0,y_2} gives the equation $f_1(S_1) = 0$, which means that the sheaf that we find is the push-forward of the line bundle $\mathcal{O}_1(-S_1) \oplus \mathcal{O}_2$, namely we find the sheaf $N_1^{(-1,0)}$,
- If $y_1 \neq 0$ and $y_2 = 0$, taking the kernel of the map $\phi^0_{y_1,0}$ gives the equation $f_2(S_2) = 0$, which gives us the sheaf that is the push-forward of the line bundle $\mathcal{O}_1 \oplus \mathcal{O}_2(-S_2)$, namely the sheaf $N_2^{(0,-1)}$,
- If $y_1 = y_2 = 0$, taking the kernel we have $f_1(S_1) = f_2(S_2) = 0$, namely we obtain the sheaf which is the push-forward of the line bundle $\mathcal{O}_1(-S_1) \oplus \mathcal{O}_2(-S_2)$, namely the sheaf $N_{12}^{(-1,-1)}$.

similarly we can define the map

$$\eta^0_{x_1,y_2} : \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) \to k_{P_1} \oplus k_{P_2}$$

that sends a section (f_1, f_2) to $(x_1 f_1(S_1) - f_2(T_1), f_2(S_2) - y_2 f_1(T_2))$.
Taking the kernel of this map when $x_1, y_2 \in k^\times$ gives the line bundle

$$L_{x_1,y_2{}^{-1}} = (\mathcal{O}_1, \mathcal{O}_2, x_1, y_2{}^{-1})$$

[1] For every integer d we can define the map $\psi^d_{x_1,x_2} : \pi_{\Sigma,*}(\mathcal{O}_1(d \cdot \underline{1}) \oplus \mathcal{O}_2(-d \cdot \underline{1})) \to k_{P_1} \oplus k_{P_2}$ that sends (f_1, f_2) to $(x_1 f_1(S_1) - f_2(T_1), x_2 f_1(S_2) - f_2(T_2))$

of multidegree $(0,0)$.
When we consider either $x_1 = 0$ or $y_2 = 0$, we have the following situations:

- If $x_1 = 0$ and $y_2 \neq 0$, taking the kernel of the map η^0_{0,y_2} gives the equation $f_2(T_1) = 0$, which means that the sheaf that we find is the push-forward of the line bundle $\mathcal{O}_1 \oplus \mathcal{O}_2(-T_1)$, namely we find the sheaf $N_1^{(0,-1)}$,
- If $x_1 \neq 0$ and $y_2 = 0$, taking the kernel of the map $\eta^0_{x_1,0}$ gives the equation $f_2(S_2) = 0$, which gives us the sheaf that is the push-forward of the line bundle $\mathcal{O}_1 \oplus \mathcal{O}_2(-S_2)$, namely the sheaf $N_2^{(0,-1)}$,
- If $x_1 = y_2 = 0$ taking the kernel we have $f_2(T_1) = f_2(S_2) = 0$, namely we obtain the sheaf which is the push-forward of the line bundle $\mathcal{O}_1 \oplus \mathcal{O}_2(-T_1 - S_2)$, namely the sheaf $N_{12}^{(0,-2)}$.

We can also define the map

$$\gamma^0_{y_1,x_2} \colon : \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) \to k_{P_1} \oplus k_{P_2}$$

that sends a section (f_1, f_2) to $(f_1(S_1) - y_1 f_2(T_1), x_2 f_2(S_2) - f_1(T_2))$. Taking the kernel of this map when $y_1, x_2 \in k^\times$ gives the line bundle

$$L_{{y_1}^{-1},x_2} = (\mathcal{O}_1, \mathcal{O}_2, {y_1}^{-1}, x_2)$$

of multidegree $(0,0)$.
When we consider either $y_1 = 0$ or $x_2 = 0$, we have the following situations:

- If $y_1 = 0$ and $x_2 \neq 0$, taking the kernel of the map γ^0_{0,y_2} gives the equation $f_1(S_1) = 0$, which means that the sheaf that we find is the push-forward of the line bundle $\mathcal{O}_1(-S_1) \oplus \mathcal{O}_2$, namely we find the sheaf $N_1^{(-1,0)}$,
- If $y_1 \neq 0$ and $x_2 = 0$, taking the kernel of the map $\gamma^0_{y_1,0}$ gives the equation $f_1(T_2) = 0$, which gives us the sheaf that is the push-forward of the line bundle $\mathcal{O}_1(-T_2) \oplus \mathcal{O}_2$, namely the sheaf $N_2^{(-1,0)}$,
- If $y_1 = x_2 = 0$ taking the kernel we have $f_1(S_1) = f_1(T_2) = 0$, namely we obtain the sheaf which is the push-forward of the line bundle $\mathcal{O}_1(-S_1 - T_2) \oplus \mathcal{O}_2$, namely the sheaf $N_{12}^{(-2,0)}$.

Putting all these pieces of information together we find all the sheaves which are listed in the Lemma. □

Remark 2.7. We fixed the multidegree $(0,0)$, and the case of multidegree $(d,-d)$ for any integer d can be obtained from the $(0,0)$ case by tensoring with the line bundle $(\mathcal{O}_1(d), \mathcal{O}_2(-d), 1, 1)$.

Remark 2.8. The key point to keep in mind from the proof of Lemma 2.6 is that, starting from a fixed multidegree $(d,-d)$ and $\alpha_1, \alpha_2 \in \mathrm{Isom}(L_{S_i}, L_{T_i}) = k^\times$, we can define a suitable map and describe the line bundle L_{α_1,α_2} as the kernel of this map. By allowing α_1 and α_2 to degenerate to elements in $\mathrm{Hom}(L_{S_i}, L_{T_i})$, namely by allowing $\alpha_1 = 0$ and $\alpha_2 = 0$ and taking the kernel again, we find degenerations of line bundles of multidegree $(d,-d)$.

Remark 2.9. This construction will be generalized to every nodal curve in Proposition 2.23.

In the second part of this example we fix a sheaf N that is not locally free at any of the nodes and describe the stack $\mathrm{Coh}_1^{\mathrm{tf}}(X)$ locally around this sheaf.

Proposition 2.10. *Let $N \coloneqq N_{\Sigma}^{\underline{e}}$ be a sheaf that fails to be locally free at all the nodes of X. There is an étale morphism of stacks*

$$\theta_N \colon [\mathrm{Spec}(R)/\mathbb{G}_m^2] \to \underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X),$$

where

- $R = k[x_1, y_1, x_2, y_2]/(x_1y_1, x_2y_2)$,
- $(\lambda_1, \lambda_2) \cdot (x_1, y_1, x_2, y_2) := (\lambda_1 x_1 \lambda_2^{-1}, \lambda_2 y_1 \lambda_1^{-1}, \lambda_2 x_2 \lambda_1^{-1}, \lambda_1 y_2 \lambda_2^{-1})$,

such that the image of θ_N is the open substack in $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ consisting of all the sheaves specializing to N.

Proof. Without loss of generality we prove the Proposition for the sheaf $N := N_{1,2}^{(-1,-1)}$. From the proof of Lemma 2.6 we get that the sheaf N can be described as the kernel of suitable maps. The first map that we can consider is

$$\psi^0_{x_1,x_2} \colon \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) \to k_{P_1} \oplus k_{P_2},\ (f_1, f_2) \mapsto\ (x_1 f_1(S_1) - f_2(T_1), x_2 f_2(S_2) - f_1(T_2)).$$

whose kernel defines the sheaf I_{x_1,x_2} that fits into the short exact sequence

$$0 \to I_{x_1,x_2} \to \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) \xrightarrow{\psi^0_{x_1,x_2}} k_{P_1} \oplus k_{P_2} \to 0. \tag{2.1}$$

If both x_1 and x_2 are different from 0, we have that I_{x_1,x_2} is a line bundle of multidegree $(0,0)$. If $x_1 = 0, x_2 \neq 0$, we have $I_{0,x_2} = N_1^{(0,-1)}$, since by definition of I_{0,x_2} we must have $f_2(P_1) = 0$. If $x_1 \neq 0, x_2 = 0$, we find the sheaf $N_2^{(-1,0)}$ and if $x_1 = x_2 = 0$, we find N. Observe that the sheaf I_{x_1,x_2} is isomorphic to the sheaf $I_{\lambda_1 x_1 \lambda_2{}^{-1}, \lambda_2 x_2 \lambda_1{}^{-1}}$ for $\lambda_1, \lambda_2 \in k^\times$. Indeed we have the following:

$$\begin{tikzcd}
0 \arrow[r] & I_{x_1,x_2} \arrow[r] & \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) \arrow[r] \arrow[d, "g"] & k_{P_1} \oplus k_{P_2} \arrow[r] \arrow[d, "h"] & 0 \\
0 \arrow[r] & I_{\lambda_1 x_1 \lambda_2{}^{-1}, \lambda_2 x_2 \lambda_1{}^{-1}} \arrow[r] & \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) \arrow[r] & k_{P_1} \oplus k_{P_2} \arrow[r] & 0,
\end{tikzcd}$$

where

- $g \colon \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) \to \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2),\ (f_1, f_2) \mapsto (\lambda_2 f_1, \lambda_1 f_2)$,
- $h \colon k_{P_1} \oplus k_{P_2} \to k_{P_1} \oplus k_{P_2},\ (\beta_1, \beta_2) \mapsto (\lambda_1 \beta_1, \lambda_2 \beta_2)$.

The equality $h \circ \psi^0_{x_1,x_2} = \psi^0_{\lambda x_1, \lambda^{-1} x_2} \circ g$ implies that $I_{x_1,x_2} \simeq I_{\lambda x_1, \lambda^{-1} x_2}$.
This allows us to define a morphism of stacks

$$[\mathrm{Spec}(k[x_1, x_2])/\mathbb{G}_m^2] \to \underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$$

that sends a point (x_1, x_2) to the sheaf I_{x_1,x_2} that fits into the short exact sequence 2.1
Now we consider the map $\phi^0_{y_1,y_2}$ and the sheaf I_{y_1,y_2} defined by the short exact sequence

$$\begin{tikzcd}
0 \arrow[r] & I_{y_1,y_2} \arrow[r] & \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) \arrow[r, "\phi^0_{y_1,y_2}"] & k_{P_1} \oplus k_{P_2} \arrow[r] & 0 \\
 & & (f_1, f_2) \arrow[r, mapsto] & (f_1(S_1) - y_1 f_2(T_1), f_2(S_2) - y_2 f_1(T_2)). &
\end{tikzcd} \tag{2.2}$$

We can see that $I_{y_1,y_2} \simeq I_{\lambda_2 y_1 \lambda_1{}^{-1}, \lambda_1 y_2 \lambda_2{}^{-1}}$, for every $\lambda_1, \lambda_2 \in k^\times$:

$$\begin{tikzcd}
0 \arrow[r] & I_{y_1,y_2} \arrow[r] & \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) \arrow[r] \arrow[d, "g"] & k_{P_1} \oplus k_{P_2} \arrow[r] \arrow[d, "h"] & 0 \\
0 \arrow[r] & I_{\lambda_2 y_1 \lambda_1{}^{-1}, \lambda_1 y_2 \lambda_2{}^{-1}} \arrow[r] & \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) \arrow[r] & k_{P_1} \oplus k_{P_2} \arrow[r] & 0,
\end{tikzcd}$$

where

- $g \colon \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) \mapsto \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2),\ (f_1, f_2) \mapsto (\lambda_1 f_1, \lambda_2 f_2)$,

- $h\colon k_{P_1} \oplus k_{P_2} \to k_{P_1} \oplus k_{P_2},\ (\beta_1, \beta_2) \mapsto (\lambda_2\beta_1, \lambda_1\beta_2)$.

The equality $h \circ \psi^0_{y_1,y_2} = \psi^0_{\lambda x_1, \lambda^{-1}x_2} \circ g$ implies that $I_{x_1,x_2} \simeq I_{\lambda x_1, \lambda^{-1}x_2}$. Therefore we define a morphism

$$[\mathrm{Spec}(k[y_1, y_2])/\mathbb{G}_m^2] \to \underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$$

that sends a point (y_1, y_2) to the sheaf I_{y_1,y_2} that fits into the short exact sequence 2.2. Furthermore we can consider the map $\eta^{-1}_{x_1,y_2}$ and the sheaf I_{x_1,y_2} defined by

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I_{x_1,y_2} & \longrightarrow & \pi_{\Sigma,*}(\mathcal{O}_1(-1) \oplus \mathcal{O}_2(+1)) & \xrightarrow{\eta^{-1}_{x_1,y_2}} & k_{P_1} \oplus k_{P_2} & \longrightarrow & 0 \\ & & & & (f_1, f_2) & \longmapsto & (x_1 f_1(S_1) - f_2(T_1), f_2(S_2) - y_2 f_1(T_2)), & & \end{array} \tag{2.3}$$

We see that $I_{x_1,y_2} \simeq I_{\lambda_1 x_1 \lambda_2{}^{-1}, \lambda_1 y_2 \lambda_2{}^{-1}}$: As above we can see that $I_{x_1,y_2} \simeq I_{\lambda_2 y_1 \lambda_1{}^{-1}, \lambda_1 y_2 \lambda_2{}^{-1}}$, for every $\lambda_1, \lambda_2 \in k^\times$:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I_{x_1,y_2} & \longrightarrow & \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) & \longrightarrow & k_{P_1} \oplus k_{P_2} & \longrightarrow & 0 \\ & & & & \downarrow g & & \downarrow h & & \\ 0 & \longrightarrow & I_{\lambda_1 x_1 \lambda_2{}^{-1}, \lambda_1 y_2 \lambda_2{}^{-1}} & \longrightarrow & \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) & \longrightarrow & k_{P_1} \oplus k_{P_2} & \longrightarrow & 0, \end{array}$$

where

- $g\colon \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) \mapsto: \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2),\ (f_1, f_2) \mapsto (\lambda_2 f_1, \lambda_2 f_2)$,
- $h\colon k_{P_1} \oplus k_{P_2} \to k_{P_1} \oplus k_{P_2},\ (\beta_1, \beta_2) \mapsto (\lambda_1\beta_1, \lambda_1\beta_2)$.

and we define a morphism

$$[\mathrm{Spec}(k[x_1, y_2])/\mathbb{G}_m^2] \to \underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$$

that sends a point (x_1, y_2) to the sheaf I_{x_1,y_2} that fits into the short exact sequence 2.3. Finally we consider the map $\gamma^1_{y_1,y_2}$ and the sheaf I_{y_1,x_2} defined by

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I_{y_1,x_2} & \longrightarrow & \pi_{\Sigma,*}(\mathcal{O}_1(+1) \oplus \mathcal{O}_2(-1)) & \xrightarrow{\gamma^1_{y_1,x_2}} & k_{P_1} \oplus k_{P_2} & \longrightarrow & 0 \\ & & & & (f_1, f_2) & \longmapsto & (f_1(S_1) - y_1 f_2(T_1), x_2 f_2(S_2) - f_1(T_2)). & & \end{array} \tag{2.4}$$

As above we can see that $I_{y_1,x_2} \simeq I_{\lambda_2 y_1 \lambda_1{}^{-1}, \lambda_2 x_2 \lambda_1{}^{-1}}$, for every $\lambda_1, \lambda_2 \in k^\times$:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I_{y_1,x_2} & \longrightarrow & \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) & \longrightarrow & k_{P_1} \oplus k_{P_2} & \longrightarrow & 0 \\ & & & & \downarrow g & & \downarrow h & & \\ 0 & \longrightarrow & I_{\lambda_2 y_1 \lambda_1{}^{-1}, \lambda_2 x_2 \lambda_1{}^{-1}} & \longrightarrow & \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) & \longrightarrow & k_{P_1} \oplus k_{P_2} & \longrightarrow & 0, \end{array}$$

where

- $g\colon \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) \mapsto: \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2),\ (f_1, f_2) \mapsto (\lambda_1 f_1, \lambda_1 f_2)$,
- $h\colon k_{P_1} \oplus k_{P_2} \to k_{P_1} \oplus k_{P_2},\ (\beta_1, \beta_2) \mapsto (\lambda_2\beta_1, \lambda_2\beta_2)$.

We have $I_{y_1,x_2} \simeq I_{\lambda_2 y_1 \lambda_1{}^{-1}, \lambda_2 x_2 \lambda_1{}^{-1}}$ and we define a morphism

$$[\mathrm{Spec}(k[y_1, x_2])/\mathbb{G}_m^2] \to \underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$$

that sends a point (y_1, x_2) to the sheaf I_{y_1,x_2} that fits into the short exact sequence 2.4.
We claim that the glueing of these 4 morphisms gives the morphism of the statement. The scheme $\mathrm{Spec}(k[x_1, y_1, x_2, y_2]/(x_1y_1, x_2y_2))$ is the union of 4 copies of $\mathbb{A}^2$, defined by 2 coordinates (t_1, t_2), where $t_i \in \{x_i, y_i\}$. Consider a point (x_1, y_1, x_2, y_2). Let $I \subseteq \{P_1, P_2\}$ be the set of nodes such that $x_i = y_i = 0$.
We have

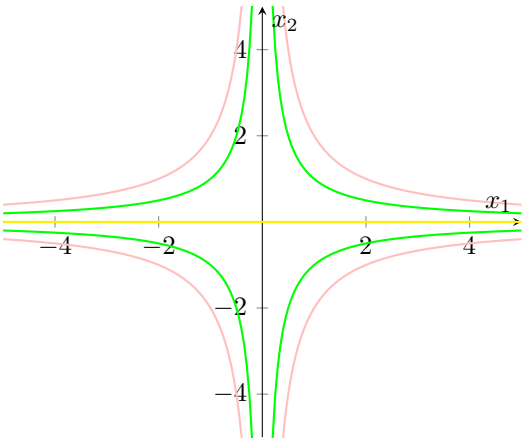
- $\mathrm{Ker}(\psi^0_{0,0}) = \mathrm{Ker}(\phi^0_{0,0}) = \mathrm{Ker}(\eta^{-1}_{0,0}) = \mathrm{Ker}(\gamma^1_{0,0}) = N$,
- $\mathrm{Ker}(\psi^0_{0,x_2}) = \mathrm{Ker}(\eta^{-1}_{0,y_2}) = N_1$,
- $\mathrm{Ker}(\psi^0_{x_1,0}) = \mathrm{Ker}(\gamma^1_{y_1,0}) = N_2$,
- $\mathrm{Ker}(\phi^0_{0,y_2}) = \mathrm{Ker}(\gamma^1_{0,x_2}) = N_1$,
- $\mathrm{Ker}(\phi^0_{y_1,0}) = \mathrm{Ker}(\eta^{-1}_{x_1,0}) = N_2$.

We claim that this morphism is étale and we prove this claim more generally in Section 2 (see Proposition 2.28 for more details). The image of θ_N in $\underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X)$ is therefore open and it is given by all the sheaves that specialize to N by construction. □

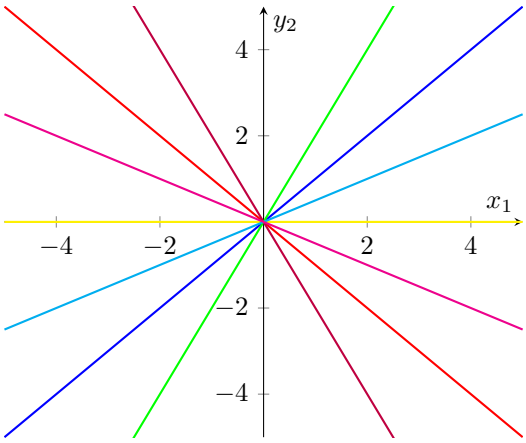
Remark 2.11. Given any sheaf $N^{(d-1,-d-1)}_{12}$ we can generalize the result of Proposition 2.10 by tensoring with the line bundle $(\mathcal{O}_1(d), \mathcal{O}_2(-d), 1, 1)$.
Therefore, under the morphism θ_N, we have that

- the origin is the sheaf $N = N^{(d-1,-d-1)}_{12}$,
- the points of $k[x_1]$ outside the origin represent $N^{(d,-d-1)}_2$,
- the points of $k[y_1]$ outside the origin represent $N^{(d-1,-d)}_2$,
- the points of $k[x_2]$ outside the origin represent $N^{(d-1,-d)}_1$,
- the points of $k[y_2]$ outside the origin represent $N^{(d,-d-1)}_1$,
- on $k[x_1, x_2]$ we have an anti-diagonal action and the points outside the axes parametrize line bundles of multidegree $(d, -d)$,
- on $k[y_1, y_2]$ we have an anti-diagonal action and the points outside the axes parametrize line bundles of multidegree $(d, -d)$,
- on $k[x_1, y_2]$ we have a diagonal action and the points outside the axes parametrize line bundles of multidegree $(d + 1, -d - 1)$,
- on $k[x_2, y_1]$ we have a diagonal action and the points outside the axes parametrize line bundles of multidegree $(d - 1, -d + 1)$.

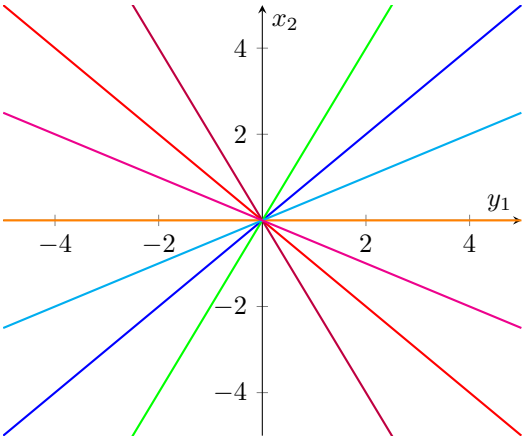
In the last part of this example we use what we have figured out so far in order to find all the open connected substacks of $\underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X)$ which admit a proper good moduli space.



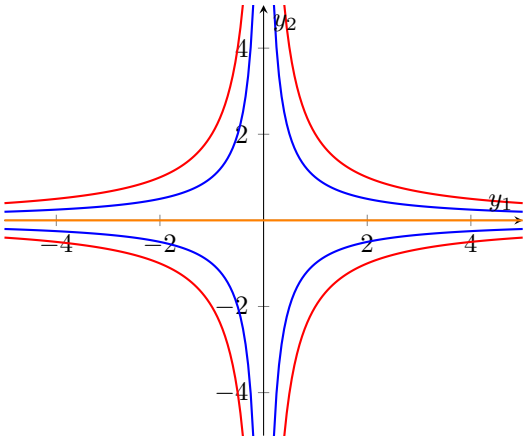
(a) Orbits of the action of $\mathbb{G}_m^2$ on $k[x_1, x_2]$



(b) Orbits of the action of $\mathbb{G}_m^2$ on $k[x_1, y_2]$



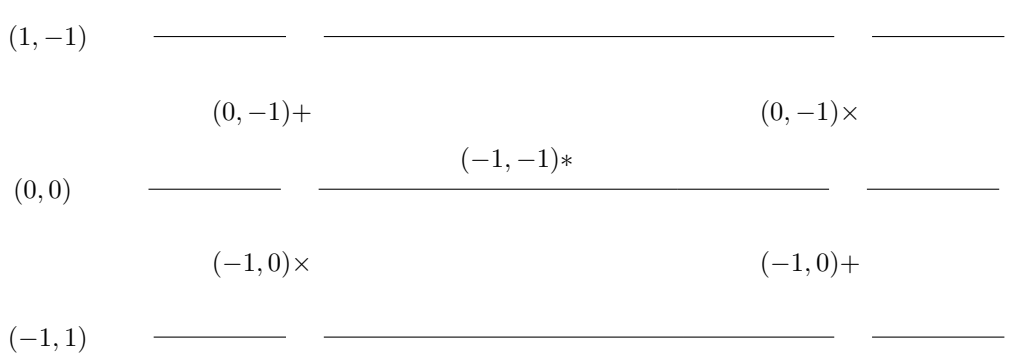
(c) Orbits of the action of $\mathbb{G}_m^2$ on $k[y_1, x_2]$



(d) Orbits of the action of $\mathbb{G}_m^2$ on $k[y_1, y_2]$

Definition 2.12.
Given a sheaf $N_{12}^{\underline{e}}$ we call $\mathcal{U}(N_{12}^{\underline{e}})$ the open in $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ which is the image of the morphism $\theta_{N_{12}^{\underline{e}}}$ of Proposition 2.10.

Remark 2.13. If we consider the sheaf $N_{12}^{(-1,-1)}$, we represent the étale neighborhood of $N_{12}^{(-1,-1)}$ as:



where we have denoted by $+$ the sheaves that are not locally free at the node P_1, by $\times$ the sheaves that are not locally free at the node P_2 and by $*$ the sheaf $N_{12}^{(-1,-1)}$ that is not locally free at both nodes. For every sheaf I which is not a line bundle we have written the multidegree of the line bundle L such that I is the push-forward of L.
The two copies of $\mathbb{G}_m$ representing line bundles of multidegree respectively $(1,-1)$ and $(-1,1)$ come from the copies of $\mathbb{A}^2$ which are $\mathrm{Spec}(k[x_1,y_2])$ and $\mathrm{Spec}(k[y_1,x_2])$ where the action of $\mathbb{G}_m^2$ is the diagonal action.
The copy of $\mathbb{G}_m$ representing line bundles of multidegree $(0,0)$ comes from the two copies of $\mathbb{A}^2$, which are $\mathrm{Spec}(k[x_1,x_2])$ and $\mathrm{Spec}(k[y_1,y_2])$ where the action of $\mathbb{G}_m^2$ is anti-diagonal.
By definition, a k-point (x_1,x_2), with $x_i \neq 0$ for $i=1,2$ of $\mathrm{Spec}(k[x_1,x_2])$, represents the line bundle L_{x_1,x_2}.
Analogously a k-point (y_1,y_2) of $\mathrm{Spec}(k[y_1,y_2])$ with $y_i \neq 0$ for $i=1,2$ represents the line bundle $L_{y_1^{-1},y_2^{-1}}$.

Proposition 2.14. *Consider the sheaf $N_{12}^{\underline{e}}$. The open $\mathcal{U}(N_{12}^{\underline{e}})$ admits a proper good moduli space which is a nodal $\mathbb{P}^1$ with 1 node.*

Proof. First of all we observe that the GIT quotient $\mathrm{Spec}(k[x_1x_2,y_1y_2]/(x_1y_1,x_2y_2))//\mathbb{G}_m^2$ is $\mathrm{Spec}(k[t_1,t_2])/(t_1t_2)$, where $t_1=x_1y_2$ and $t_2=y_1x_2$. In particular, since $\mathrm{Spec}(R)$ is given by 4 copies of $\mathbb{A}^2$, we have:

- $\mathrm{Spec}(k[x_1,x_2])//\mathbb{G}_m^2$ is $\mathbb{A}^1=\mathrm{Spec}(k[x_1x_2])$,
- $\mathrm{Spec}(k[y_1,y_2])//\mathbb{G}_m^2$ is $\mathbb{A}^1=\mathrm{Spec}(k[y_1y_2])$,
- $\mathrm{Spec}(k[x_1,y_2])//\mathbb{G}_m^2$ is a point,
- $\mathrm{Spec}(k[y_1,x_2])//\mathbb{G}_m^2$ is a point.

In our moduli space, when $x_1,x_2,y_1,y_2 \in k^\times$, the identification $x_i \mapsto y_i^{-1}$ implies that the two copies of $\mathbb{A}^1$ obtained as GIT quotients are identified outside the origin by sending t to t^{-1}. Moreover, the origins of the two copies of $\mathbb{A}^1$ and the two points from the other two quotients are also identified. We thus obtain a nodal $\mathbb{P}^1$ as moduli space, where the copy of $\mathbb{G}_m$ obtained by removing the node represents line bundles of multidegree $(0,0)$, and all the other rank 1 torsion-free sheaves that specialize to $N_{12}^{\underline{e}}$ get identified in the node. $\square$

We have seen that if we want a sheaf $N_{\Sigma}^{\underline{e}}$ which is not locally free at all the nodes to be in the open connected substack of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ which admits a proper good moduli space, then this open is $U(N_{\Sigma}^{\underline{e}})$.

If we do not allow sheaves which are not locally free at all the nodes to be in the open connected substack of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ that admits a proper good moduli space, we end up considering only simple sheaves. This case has been already analyzed in [2], where the authors give a complete classification of all the open proper connected substack of the stack $\underline{\mathrm{Simp}}(X)$ of rank 1 torsion-free simple sheaves on X for a necklace curve X of genus $g = 1$.

Lemma 2.15. *Let $\overline{J}_X$ be a fine compactified Jacobian for a necklace X of genus $g = 1$ and length 2. There exists a sheaf N which is not locally free at all the nodes such that*

$$\overline{J}_X = \theta_N(U),$$

where θ_N is the morphism of Proposition 2.10 and U is the open obtained by removing the plane $\mathrm{Spec}(k[x_1, x_2])$ *from* $\mathrm{Spec}(k[x_1, y_1, x_2, y_2]/(x_1y_1, x_2y_2))$.

Proof. From [2, Lemma 3.7], we know that a fine compactified Jacobian $\overline{J}_X$ for a necklace of genus $g = 1$ and length 2 is given by the union

$$\overline{J}_X := \mathcal{J}^d(X) \cup N_1^{(d,-d-1)} \cup \mathcal{J}^{d+1}(X) \cup N_2^{(d,-d-1)},$$

for some integer d.
We consider the sheaf $N := N_{12}^{(d,-d-2)}$ and the morphism θ_N of Proposition 2.10. The image under θ_N of the open defined by removing the plane $\mathrm{Spec}(k[x_1, x_2])$ is precisely $\overline{J}_X$. □

Remark 2.16. We obtain a result similar to Lemma 2.15 if we consider the open obtained by removing the plane $\mathrm{Spec}(k[y_1, y_2])$ from $\mathrm{Spec}(k[x_1, y_1, x_2, y_2]/(x_1y_1, x_2y_2))$.
In both cases, we obtain an open whose proper good moduli space is a necklace of length 2.

Remark 2.17. In this example, by studying $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ locally around a sheaf that is not locally free at any of the nodes of the necklace X, we have found all open connected substacks of the stack $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ which admit a proper good moduli space. This proper good moduli space turns out to be a necklace whose length can be either 1 or 2.

Remark 2.18. The main points to keep in mind from the second part of this example are that it was crucial to study the stack $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ around a sheaf that is not locally free at any of the nodes, and that the open connected substacks of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ admitting a proper good moduli space have as their good moduli space a necklace.

2.2 Local description

Let X be a nodal curve. Starting from the open immersion

$$\iota\colon \underline{\mathrm{Pic}}(X) \to \underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$$

from the stack $\underline{\mathrm{Pic}}(X)$ of line bundles on X to the stack $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ of rank 1 torsion-free sheaves on X, we give an étale local description of the stack $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$. The main result is Proposition 2.28.

Notation 2.19.
Given a nodal curve X, we fix an orientation $O_0 = (s_0, t_0)$ of the dual graph Γ_X of X and, for every node $P \in \Sigma$, we call $s_0(P)$ and $t_0(P)$ the preimages of the node P under the normalization map $\pi_\Sigma\colon X_\Sigma \to X$.
Given any other orientation $O = (s, t)$ of Γ_X we refer to the preimages of the node P under the normalization map π_Σ as $s(P)$ and $t(P)$, where

- $s(P) = s_0(P)$ if the orientations O_0 and O coincide for the node P,
- $s(P) = t_0(P)$ if the orientations O_0 and O do not coincide for the node P.

As we have recalled in Section 1.2, any line bundle L on X can be described by giving

- A line bundle $L' = \pi_\Sigma^* L$ on the total normalization X_Σ of X at Σ,

- An identification $\alpha_P : L'_{s(P)} \to L'_{t(P)}$ for every node $P \in \Sigma$.

Notation 2.20.
We denote by $\mathcal{L}$ the universal line bundle on $X_\Sigma \times \underline{\mathrm{Pic}}(X_\Sigma)$, namely the line bundle such that for every morphism $f_T \colon T \to \underline{\mathrm{Pic}}(X_\Sigma)$, which corresponds to a line bundle L_{f_T} on $X_\Sigma \times T$, we have

$$(\mathrm{id}_{X_\Sigma} \times f_T)^* \mathcal{L} = L_{f_T}.$$

For every point $Q \in X_\Sigma$, we denote by $\mathcal{L}_Q$ the restriction of the universal line bundle $\mathcal{L}$ to $\{Q\} \times \underline{\mathrm{Pic}}(X_\Sigma)$, namely if we denote by $i_Q \colon \mathrm{Spec}(k) \to X_\Sigma$ the inclusion morphism of Q in X_Σ, we define

$$\mathcal{L}_Q \coloneqq (i_Q \times \mathrm{id}_{\underline{\mathrm{Pic}}(X_\Sigma)})^* \mathcal{L}.$$

Recall that given two line bundles $\mathcal{L}_1$ and $\mathcal{L}_2$ over an algebraic stack $\mathcal{Y}$, one can define the stack $\underline{\mathrm{Isom}}_{\mathcal{Y}}(\mathcal{L}_1, \mathcal{L}_2)$ as

$$\underline{\mathrm{Isom}}_{\mathcal{Y}}(\mathcal{L}_1, \mathcal{L}_2)(T) \coloneqq \langle f \colon T \to \mathcal{Y}, f^*\mathcal{L}_1 \simeq f^*\mathcal{L}_2 \rangle.$$

In this chapter we assume $\mathcal{Y} = \underline{\mathrm{Pic}}(X_\Sigma)$, $\mathcal{L}_1 = \mathcal{L}_{s(P)}$ and $\mathcal{L}_2 = \mathcal{L}_{t(P)}$ and we consider

$$\textstyle\prod_{P \in \Sigma} \underline{\mathrm{Isom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)}),$$

namely the stack whose T-points are pairs $(L, \{\alpha_P\}_{P \in \Sigma})$ where

- L is a line bundle on $X_\Sigma \times T$,
- For every $P \in \Sigma$, α_P is an identification of L restricted to $s(P) \times T$ with L restricted to $t(P) \times T$.

The normalization map $\pi_\Sigma \colon X_\Sigma \to X$ induces a morphism

$$\tilde{\pi}_\Sigma^* \colon \underline{\mathrm{Pic}}(X) \to \underline{\mathrm{Pic}}(X_\Sigma)$$

defined on T-points by sending a T-point of $\underline{\mathrm{Pic}}(X)$, namely a line bundle L on $X \times T$, to $\pi_\Sigma^* L$ which is the line bundle over $X_\Sigma \times T$ defined by

$$\begin{tikzcd} \pi_\Sigma^* L \arrow[r] \arrow[d] & L \arrow[d] \\ X_\Sigma \times T \arrow[r, "\pi_\Sigma \times id_T"] & X \times T \end{tikzcd}.$$

Since we have

$$\begin{tikzcd} \mathrm{Spec}(k) \arrow[r, "i_{s(P)}"] \arrow[dr, "i_P"'] & X_\Sigma \arrow[d] & \mathrm{Spec}(k) \arrow[l, "i_{t(P)}"'] \arrow[dl, "i_P"] \\ & X & \end{tikzcd},$$

we can lift the above morphism $\tilde{\pi}_\Sigma^*$ to a morphism $\overline{\pi}_\Sigma^*$,

$$\begin{tikzcd} & \prod_{P \in \Sigma} \underline{\mathrm{Isom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)}) \arrow[d] \\ \underline{\mathrm{Pic}}(X) \arrow[ur, "\overline{\pi}_\Sigma^*"] \arrow[r, "\tilde{\pi}_\Sigma^*"] & \underline{\mathrm{Pic}}(X_\Sigma). \end{tikzcd}$$

defined on T-points by

$$L \mapsto (\pi_\Sigma^* L, i_{s(P)}^* \pi_\Sigma^* L \simeq i_{t(P)}^* \pi_\Sigma^* L).$$

Lemma 2.21. *Let X be a nodal curve. For every orientation $O = (s, t)$ of the dual graph Γ_X of X, the morphism*

$$\tilde{\theta}_O\colon \prod_{P\in\Sigma} \underline{\mathrm{Isom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)}) \to \underline{\mathrm{Pic}}(X),$$

defined by sending a T-point $(L, \{\alpha_P\}_{P\in\Sigma})$ to the the line bundle given by the kernel of the map

$$\psi\colon \pi_{\Sigma,*}L \mapsto \oplus_{P\in\Sigma}(\pi_{\Sigma,*}L)_P,\ \ f \mapsto f_{t(P)} - \alpha_P(f_{s(P)}), \tag{2.5}$$

is an isomorphism of stacks.

Proof. We observe that since we have

$$(\pi_{\Sigma,*}L)_P = L_{s(P)} \oplus L_{t(P)},$$

the kernel of ψ is a line bundle on $X \times T$ so the morphism is well-defined.
We show that the morphism $\tilde{\theta}_O$ is the inverse morphism of $\overline{\pi}^*_\Sigma$. The morphism $\tilde{\theta}_O$ sends a T-point $(L, \{\alpha_P\}_{P\in\Sigma})$ to the line bundle $L' := \ker(\psi)$ on $X \times T$. The morphism $\pi^*_\Sigma\pi_{\Sigma,*}L \to L$ modulo torsion is an isomorphism. Therefore we have a canonical morphism

$$\pi^*_\Sigma L' \hookrightarrow L$$

of line bundles of the same multidegree and therefore they are equal. Moreover the pullback of L' is equipped with an identification at the node for every node P in Σ and by construction these identifications are the isomorphisms α_P's.
Conversely, if we start with a line bundle L on $X \times T$, via the pullback morphism we get a line bundle $\pi^*_\Sigma L$ on $X_\Sigma \times T$ equipped with an isomorphism α_P between the restriction of $\pi^*_\Sigma L$ to $s(P) \times T$ with the restriction of $\pi^*_\Sigma L$ to $t(P) \times T$, for every node $P \in \Sigma$. Applying the morphism $\tilde{\theta}_O$ to $(\pi^*_\Sigma L, \{\alpha_P\}_{P\in\Sigma})$ we have to the compute the kernel of $\psi\colon \pi_{\Sigma,*}(\pi^*_\Sigma L) \mapsto \oplus_{P\in\Sigma}(\pi^*_\Sigma L)_P$. We have that $L \subseteq \ker(\psi)$ since by assumption α_P is the identification between the pullback of L restricted to to $s(P) \times T$ and the restriction of $\pi^*_\Sigma L$ to $t(P) \times T$. Moreover modulo torsion we have that $\pi_{\Sigma,*}(\pi^*_\Sigma L)$ is isomorphisc to L hence we conclude that the kernel of ψ is L.
Thus θ is an isomorphism. □

Remark 2.22. In the example of Section 2.1, we described line bundles on a necklace of length 2 by looking at the kernel of maps of the form of Equation (2.5). Subsequently, we observed that one could consider α_P to be a homomorphism instead of an isomorphism. Following this idea, for every orientation $O = (s, t)$ of Γ_X, we consider the stack

$$\prod_{P\in\Sigma} \underline{\mathrm{Hom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)})$$

and extend the isomorphism $\tilde{\theta}_O$ to a morphism

$$\theta_O\colon \prod_{P\in\Sigma} \underline{\mathrm{Hom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)}) \to \underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X).$$

Proposition 2.23. *Let X be a nodal curve and $O = (s, t)$ an orientation of the dual graph Γ_X of X. The morphism*

$$\theta_O\colon \prod_{P\in\Sigma} \underline{\mathrm{Hom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)}) \to \underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X) \tag{2.6}$$

defined by sending a T-point $(L, \{\alpha_P\}_{P\in\Sigma})$ to the kernel of the morphism

$$\psi\colon \pi_{\Sigma,*}L \mapsto \oplus_{P\in\Sigma}(i^*_P\pi_{\Sigma,*}L),\ \ f \mapsto f_{t(P)} - \alpha_P(f_{s(P)})$$

is an unramified monomorphism.

Proof. We first show that the morphism θ_O is unramified, by showing that it is injective on tangent spaces. We recall that an element of the tangent space of $\prod_{P\in\Sigma} \underline{\mathrm{Hom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)})$ is a $k[\epsilon]/(\epsilon^2)$-valued point, namely a pair $(L', \{\alpha'_P\}_{P\in\Sigma})$ where

- L' is a line bundle on $X_\Sigma \times \mathrm{Spec}(k[\epsilon]/(\epsilon^2))$,
- α'_P is a homomorphism of free $k[\epsilon]/(\epsilon^2)$-modules of rank 1, for every node $P \in \Sigma$.

Given a node $P \in \Sigma$, by choosing local coordinates for X around P, we can fix an isomorphism

$$\hat{O}_{X,P} \simeq k[|x,y|]/(xy),$$

namely we can identify $\hat{O}_{X,P}$ with the standard node $\hat{O}_0$ whose closed point we denote by P_0.
We recall that there is a unique rank 1 torsion-free sheaf on the standard node which is not locally free at the node, whose representation as a $k[|x,y|]/(xy)$-module is

$$I \coloneqq \langle x \cdot e_2 = y \cdot e_1 = 0 \rangle.$$

Moreover, from [1, Definition 3.12], we know that the local presentation of the mini-versal deformation of I as a $k[|u,v|]/(uv) \otimes k[|x,y|]/(xy)$-module is

$$\mathcal{I} \coloneqq \langle e_1, e_2 \colon y \cdot e_1 = -u \cdot e_2, x \cdot e_2 = -v \cdot e_1 \rangle.$$

Now we consider a $k[\epsilon]/(\epsilon^2)$-point, namely a pair (L', α') and we denote by (L, α) the reduction mod ϵ of (L', α').
If $\alpha \neq 0$ then (L, α) is a k-point of $\underline{\mathrm{Isom}}(\mathcal{L}_{s(P_0)}, \mathcal{L}_{t(P_0)})$ and from Lemma 2.21, we know that θ_O restricts to an embedding on $\underline{\mathrm{Isom}}(\mathcal{L}_{s(P_0)}, \mathcal{L}_{t(P_0)})$. Therefore we can focus on the case when $\alpha = 0$. Under our choice of coordiantes, we can describe ψ locally as

$$(k[|x|] \oplus k[|y|]) \otimes k[\epsilon]/(\epsilon^2) \to k[\epsilon]/(\epsilon^2),\ (f(x,\epsilon), g(y,\epsilon)) \to f(0,\epsilon) - \alpha' g(0,\epsilon). \tag{2.7}$$

As a $k[\epsilon]/(\epsilon^2)[|x,y|]/(xy)$-module, the kernel of ψ is generated by $e_1 = (x, 0)$ and $e_2 = (-\alpha', -1)$, where $y \cdot e_1 = 0$ and $x \cdot e_2 = -\alpha' \cdot e_1$, namely:

$$\langle e_1, e_2 \colon y \cdot e_1 = 0, x \cdot e_2 = -\alpha' \cdot e_1 \rangle.$$

In particular if $\alpha' \neq 0$ we get a non-trivial local deformation because it is the restriction of the versal deformation to a non-zero tangent vector.
If $\alpha' = 0$ the kernel of ψ is a sheaf which is canonically isomorphic to the push-forward of the line bundle $L(-t(P_0))$. This implies that we can reconstruct the line bundle L first by applying the pullback morphism, then by reducing modulo torsion, and finally by twisting by $\mathcal{O}(t(P_0))$. Therefore if we assume that L is not deformed trivially also $\pi_* L(-t(P_0))$ is not deformed trivially.
Thus θ_O is unramified.

We want to prove that θ_O is a monomorphism, namely we want to show that given a scheme T that maps to $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ and to $\prod_{P \in \Sigma} \underline{\mathrm{Hom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)})$ such that

$$\begin{tikzcd} T \arrow[d] \arrow[dr] & \\ \prod_{P \in \Sigma} \underline{\mathrm{Hom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)}) \arrow[r, "\theta_O"'] & \underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X) \end{tikzcd},$$

we can reconstruct the T-point $(\mathcal{L}_T, \{\alpha_P\}_{P \in \Sigma})$ of $\prod_{P \in \Sigma} \underline{\mathrm{Hom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)})$ from the T-point $\mathcal{F}_T$ of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$.
It is sufficient to consider a covering of $\prod_{P \in \Sigma} \underline{\mathrm{Hom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)})$. Therefore we assume $T = \mathrm{Spec}(R)$ and that

$$\mathrm{Spec}(R) \to \prod_{P \in \Sigma} \underline{\mathrm{Hom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)})$$

is smooth.
We want to show that we can assume that R is a complete local ring.
We first observe that the generic point U of $\mathrm{Spec}(R)$ maps to $\prod_{P \in \Sigma} \underline{\mathrm{Isom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)})$ and hence the image of the generic point is open and it has to intersect $\prod_{P \in \Sigma} \underline{\mathrm{Isom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)})$.
We have

- A line bundle on $X \times U$, given by the morphism $U \to \prod_{P \in \Sigma} \underline{\mathrm{Isom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)})$

- A line bundle on $(X_\Sigma \setminus \pi_\Sigma^{-1}(\Sigma)) \times \mathrm{Spec}(R)$, from the map $\mathrm{Spec}(R) \to \underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$.

These two line bundles coincide on $(X_\Sigma \setminus \pi_\Sigma^{-1}(\Sigma)) \times U$. By smoothness and the fact that the dimension of the complement is bigger than one, we conclude that the morphism

$$U \to \prod_{P\in\Sigma} \underline{\mathrm{Isom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)}) \to \underline{\mathrm{Pic}}(X_\Sigma) \tag{2.8}$$

extends canonically to a morphsim

$$\mathrm{Spec}(R) \to \underline{\mathrm{Pic}}(X_\Sigma).$$

Now we consider

$$\begin{array}{ccc} U & \longrightarrow & \prod_{P\in\Sigma} \underline{\mathrm{Hom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)}) \\ \downarrow & & \downarrow \\ \mathrm{Spec}(R) & \longrightarrow & \underline{\mathrm{Pic}}(X_\Sigma) \end{array}.$$

Since the morphism $\prod_{P\in\Sigma} \underline{\mathrm{Hom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)}) \to \underline{\mathrm{Pic}}(X_\Sigma)$ is affine, to check whether the morphism from U to $\prod_{P\in\Sigma} \underline{\mathrm{Hom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)})$ extends to the closed points we can check this locally at the completion of these points.
Therefore we conclude that it is enough to consider complete local rings.

As we did in the first part of this proof, for some fixed node P in Σ, by choosing local coordinates, we fix an isomorphism

$$\hat{O}_{X,P} \simeq k[|x, y|]/(xy).$$

We consider a morphism $f\colon \mathrm{Spec}(R) \to \underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$, corresponding to a family $\mathcal{F}_R$ over $X \times \mathrm{Spec}(R)$ such that locally around P the family $\mathcal{F}_R$ is isomorphic to $\mathcal{I}_R := \mathcal{I} \otimes_{k[|u,v|]/(uv)} R$. We have a morphism from $\mathrm{Spf}(R)$ to $\mathrm{Spf}(k[|u, v|]/(uv))$ which is the local deformation space of the unique rank 1 torsion-free sheaf I which is not locally free at the node. We have seen, while proving that the morphism is unramified, that a $\mathrm{Spec}(R)$-point of $\prod_{P\in\Sigma} \underline{\mathrm{Hom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)})$ induces a morphism from $\mathrm{Spec}(R)$ to the restriction to $u = 0$ of the representation of the miniversal deformation of I as a $k[|u, v, x, y|]/(uv, xy)$-module.
Moreover we observe that

$$\mathcal{I}|_{k[|y,v|]} \simeq \langle e_1, e_2 |\ ve_1 = 0 = ye_1\rangle \simeq k[|y, v|]e_2 \oplus k \cdot e_1,$$

which has y-torsion. If we consider $\mathcal{I}|_{k[|y,v|]}$, modulo the y-torsion part, we get a free module on $k[|y, v|]$.
So if we consider the restriction to $u = 0$ and the morphism of $k[|x, y, v|]/(xy)$-modules

$$\phi\colon \mathcal{I} \to (\mathcal{I}/x\mathcal{I})/(y\text{-tor}) \simeq k[|y, v|]$$

whose kernel is $xk[|x, v|]$, we have the exact sequence of $k[|x, y, v|]/(xy)$-modules:

$$0 \to xk[|x, v|] \to \mathcal{I} \to k[|y, v|] \to 0.$$

Considering also the exact sequence $0 \to \mathcal{I} \to k[|v, y|] \oplus k[|v, x|] \to k[|v|] \to 0$, we can write

the diagram

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & x \cdot k[|v,x|] \arrow[r, hook] \arrow[d] & k[|v,x|] \arrow[r] \arrow[d] & k[|v|] \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathcal{I} \arrow[r] \arrow[d] & k[|v,y|] \oplus k[|v,x|] \arrow[r] \arrow[d] & k[|v|] \arrow[r] & 0 \\
 & k[|v,y|] \arrow[r, "="] \arrow[d] & k[|v,y|] \arrow[d] & & \\
 & 0 & 0 & &
\end{tikzcd}$$

From this diagram we get that $E := k[|v,y|] \oplus k[|v,x|]$ is the pushout of

$$\begin{tikzcd}
xk[|v,x|] \arrow[r] \arrow[d] & k[|v,x|] \\
\mathcal{I} &
\end{tikzcd}.$$

Denoting by $\mathcal{L}'$ the kernel of ϕ, by $\mathcal{L}'(P) := k[|v,x|]$ and by $\mathcal{L}'' := k[|v,y|]$, starting from

$$\begin{tikzcd}
\mathcal{L}' \arrow[r] \arrow[d] & \mathcal{L}'(P) \arrow[d] \\
\mathcal{I} \arrow[r] & E
\end{tikzcd} \tag{2.9}$$

and taking the cokernels in the first two rows of diagram (2.9) above we get that, from properties of the pushouts, these cokernels are isomorphic and therefore, denoting by $\mathcal{L}'' := k[|v,y|]$, we have

$$\begin{tikzcd}
\mathcal{L}' \arrow[r] \arrow[d] & \mathcal{L}'(P) \arrow[r] \arrow[d] & \mathcal{L}'(P)_P \arrow[d] \\
\mathcal{I} \arrow[r] \arrow[d] & E \arrow[r] \arrow[d] & \mathcal{L}'(P)_P \\
\mathcal{L}'' \arrow[r] & \mathcal{L}'' &
\end{tikzcd} \tag{2.10}$$

where the push-out E splits and moreover, since there are no morphism from $\mathcal{L}''$ to $\mathcal{L}'$, the splitting is canonical.
So far we have discussed what happens for the universal family. If we consider the family $\mathcal{F}_R$ that locally around P is isomorphic to $\mathcal{I}_R$, we get the diagram

$$\begin{tikzcd}
\mathcal{L}'_R \arrow[r] \arrow[d] & \mathcal{L}'_R(P) \arrow[r] \arrow[d] & \mathcal{L}'_R(P)_P \arrow[d] \\
\mathcal{I}_R \arrow[r] \arrow[d] & \mathcal{L}'_R(P) \oplus \mathcal{L}''_R \arrow[r] \arrow[d] & \mathcal{L}'_R(P)_P \\
\mathcal{L}''_R \arrow[r] & \mathcal{L}''_R &
\end{tikzcd} \tag{2.11}$$

which gives the exact sequence

$$0 \to \mathcal{I}_R \to \mathcal{L}'_R(P) \oplus \mathcal{L}''_R \to \mathcal{L}'_R(P)_P \to 0,$$

where $\mathcal{L}'_R(P)$ is locally free as a $k[|x|]$-module and $\mathcal{L}''_R$ is locally free as a $k[|y|]$-module. From this we can therefore find the R-point of $\prod_{P\in\Sigma} \underline{\mathrm{Hom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)})$ considering the morphism α_P given by considering the composition

$$\mathcal{L}''_R \to \mathcal{L}'_R(P) \oplus \mathcal{L}''_R \to \mathcal{L}'_R(P)_P$$

restricted to P. □

Remark 2.24. In the above proof if for the node P we consider the orientation O' such that

- $s'(P) = t(P)$, and
- $t'(P) = s(P)$,

when we consider the kernel of the morphism (2.7), we get the restriction to $v = 0$ of the local presentation $\mathcal{I}$ of the mini-versal deformation of the ideal I.

Now we fix a multidegree $\underline{d}$ and we consider the restriction of the morphism θ_O of Proposition 2.23 to $\underline{\mathrm{Hom}}^{\underline{d}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)})$, namely the stack whose T-points are pairs $(L, \{\alpha_P\}_{P\in\Sigma})$ where

- L is a line bundle on $X_\Sigma \times T$ of multidegree $\underline{d}$,
- For every $P \in \Sigma$, α_P is a homomorphism from L restricted to $s(P) \times T$ to L restricted to $t(P) \times T$.

Notation 2.25.
We denote by $\theta_O^{\underline{d}}$ the restriction of the morphism of Proposition 2.23 to $\underline{\mathrm{Hom}}^{\underline{d}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)})$.

Proposition 2.26. *Given a nodal curve X and an orientation $O = (s, t)$ of the dual graph Γ_X, the image of*

$$\prod_{P\in\Sigma\setminus\Sigma'} \underline{\mathrm{Isom}}^{\underline{d}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)}) \times \prod_{P\in\Sigma'} \{0\}$$

in $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ via the morphism $\theta_O^{\underline{d}}$ is isomorphic to $\overline{\pi}_{\Sigma',*}(\underline{\mathrm{Pic}}^{\underline{e}}(X_{\Sigma'}))$, namely it is given by the push-forward of line bundles of multidegree $\underline{e}$ on the partial normalization $X_{\Sigma'}$, where $\underline{e} = \underline{d} - \mathcal{D}(\Sigma', O)$.

Proof. Let T be a scheme and $(L, \{\alpha_P\}_{P\in\Sigma})$ be a T-point of

$$\prod_{P\in\Sigma\setminus\Sigma'} \underline{\mathrm{Isom}}^{\underline{d}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)}) \times \prod_{P\in\Sigma'} \{0\},$$

where

- L a line bundle on $X_\Sigma \times T$ of multidegree $\underline{d}$,
- $\alpha_P = 0$, for every $P \in \Sigma'$.

The image of $(L, \{\alpha_P\}_{P\in\Sigma})$ is $\pi_{\Sigma,*}(L(-t(\Sigma')))$. Namely it is the push-forward of the line bundle $L(-t(\Sigma'))$ whose multidegree is

$$\underline{e} = \underline{d} - \mathcal{D}(\Sigma', O).$$

□

We want to define a stack that identifies partial zero sections of stacks of the form $\prod_{P\in\Sigma} \underline{\mathrm{Hom}}^{\underline{d}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)})$ for some orientation $O = (s, t)$ and multidegree $\underline{d}$ when the images in $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ via θ_O coincide.

Definition 2.27.
Consider the set

$$\mathrm{ParOr} := \{(\Sigma', O') |\ \Sigma' \subseteq \Sigma, O' \text{ orientation of } \Sigma'\}$$

which is partially ordered as follows: $(\Sigma_1, O_1) \leq (\Sigma_2, O_2)$ if and only if

- $\Sigma_1 \subseteq \Sigma_2$,
- $O_1(P) = O_2(P)$, for every node $P \in \Sigma_1$

If $(\Sigma_1, O_1) \leq (\Sigma_2, O_2)$ we identify

$$\prod_{P \in \Sigma_1} \underline{\mathrm{Hom}}^{\underline{e}+\mathcal{D}(\Sigma_1, O)}(\mathcal{L}_{s_1(P)}, \mathcal{L}_{t_1(P)})$$

with

$$\prod_{P \in \Sigma_2 \setminus \Sigma_1} \{0\} \times \prod_{P \in \Sigma_1} \underline{\mathrm{Hom}}^{\underline{e}+\mathcal{D}(\Sigma_1, O)}(\mathcal{L}_{s_2(P)}, \mathcal{L}_{t_2(P)}).$$

Given a fixed multidegree $\underline{e}$ we define $\mathcal{X}_{\underline{e}}$ to be the stack

$$\mathcal{X}_{\underline{e}} := \varinjlim_{\mathrm{ParOr}} \prod_{P \in \Sigma'} \underline{\mathrm{Hom}}^{\underline{e}+\mathcal{D}(\Sigma', O)}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)}).$$

Proposition 2.28. *The morphisms $\theta_O^{\underline{e}+\mathcal{D}(\Sigma, O)}$'s induce an étale morphism*

$$\theta_{\underline{e}} \colon \mathcal{X}_{\underline{e}} \to \underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X).$$

The image of the above morphism is the smallest $\mathrm{Pic}^{\underline{0}}(X)$-stable open neighborhood of $\overline{\pi}_{\Sigma',*}(\underline{\mathrm{Pic}}^{\underline{e}}(X_{\Sigma'}))$.

Proof. For every node $P \in \Sigma$ we can fix an isomorphism

$$\hat{O}_{X,P} \simeq k[|x, y|]/(xy).$$

In Proposition 2.23 we have seen that the morphism θ_O identifies the tangent space of $\prod_{P \in \Sigma} \underline{\mathrm{Hom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)})$ with the subspace of the deformation space which is the preimage of the subspace of the local deformation space restricted to one branch of the node $k[|u, v|]/(uv)$. In Remark 2.24 we have observed that taking the opposite orientation of O we find the subspace which is the preimage of the restriction to the other branch of the node $k[|u, v|]/(uv)$. Thus we conclude that $\theta_O^{\underline{d}}$ is étale.
The image of the morphism is therefore open and $\mathrm{Pic}^{\underline{0}}(X)$-stable by construction and any open neighborhood of $\overline{\pi}_{\Sigma',*}(\underline{\mathrm{Pic}}^{\underline{e}}(X_{\Sigma'}))$ it is the smallest because every open has to intersect some $\underline{\mathrm{Pic}}^{\underline{d}}(X)$'s in the image. □

In the last part of this section we show that the stack $\prod_{P \in \Sigma} \underline{\mathrm{Hom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)})$ can be described as a global quotient stack.

Notation 2.29.
We denote by $\mathrm{Pic}(X_\Sigma)$ the scheme whose T-points are pairs (L, α_i) where

- L is a line bundle on $X_\Sigma \times T$,
- $\alpha \colon L_{z_i \times T} \to \mathcal{O}_T$ is an isomorphism, where z_i is a fixed smooth point of the irreducible component X_i of X.

Lemma 2.30. *The morphism*

$$\mathrm{Pic}(X) \to \underline{\mathrm{Pic}}(X)$$

defined by sending a T-point (L, α) of $\mathrm{Pic}(X)$ to the line bundle L is a $\mathbb{G}_m^{\pi_0(X)}$-torsor.

Proof. Observe that, if X has connected components $X_1, \ldots, X_m$, we have

$$\underline{\mathrm{Pic}}(X) = \prod_{i=1,\ldots,m} \underline{\mathrm{Pic}}(X_i).$$

Therefore it is enough to prove that, for every i, $\mathrm{Pic}(X_i) \to \underline{\mathrm{Pic}}(X_i)$ is a $\mathbb{G}_m$-torsor and this is true because $\mathrm{Pic}(X_i) \simeq \mathrm{Isom}(\mathcal{L}|_{z_i}, \mathcal{O}_{\underline{\mathrm{Pic}}(X_i)})$, which is a $\mathbb{G}_m$-torsor. □

Notation 2.31.
We denote by L_Q the restriction of $\mathcal{L}_Q$ to $\mathrm{Pic}(X_\Sigma)$, via the morphism of Lemma 2.30.

Proposition 2.32. *There is an isomorphism of stacks*

$$\underline{\mathrm{Hom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)}) \to \left[\mathrm{Hom}(L_{s(P)}, L_{t(P)})/\mathbb{G}_m^{\pi_0(X_\Sigma)}\right].$$

Proof. We consider the diagram

$$\begin{array}{ccc} \prod_{P\in\Sigma} \mathrm{Hom}(L_{s(P)}, L_{t(P)}) & \longrightarrow & \prod_{P\in\Sigma} \underline{\mathrm{Hom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)}) \\ \downarrow & & \downarrow \\ \mathrm{Pic}_{X_\Sigma} & \longrightarrow & \underline{\mathrm{Pic}}_{X_\Sigma}. \end{array}$$

Since the morphism $\mathrm{Pic}_{X_\Sigma} \to \underline{\mathrm{Pic}}_{X_\Sigma}$ is a $\mathbb{G}_m^{\pi_0(X_\Sigma)}$-torsor we have an isomorphism of stacks

$$\underline{\mathrm{Hom}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)}) \to \left[\mathrm{Hom}(L_{s(P)}, L_{t(P)})/\mathbb{G}_m^{\pi_0(X_\Sigma)}\right]$$

where for a line bundle $L \in \mathrm{Pic}(X_\Sigma)$ we have that $\underline{\lambda} \in \mathbb{G}_m^{\pi_0(X_\Sigma)} = \mathrm{Aut}(L)$ acts on $(L_1, \ldots, L_m, \alpha_P)$ as

$$\underline{\lambda} \cdot (L_1, \ldots, L_m, \alpha_P) := (\lambda_1 \mathcal{L}_1, \ldots, \lambda_m \mathcal{L}_m, \lambda_{s(P)} \alpha_P (\lambda_{t(P)})^{-1}).$$

□

2.3 The case of nodal curves with rational components

In this section we give an explicit description of the stack

$$\mathcal{X}_e = \varinjlim_{\mathrm{ParOr}} \prod_{P\in\Sigma'} \underline{\mathrm{Hom}}^{\underline{e}+\mathcal{D}(\Sigma',O)}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)})$$

for the case of nodal curves whose irreducible components, that we denote by $X_1, \ldots, X_{\pi_0(X_\Sigma)}$, are rational. We fix an orientation $O_0 = (s_0, t_0)$ of the dual graph Γ_X of X. We denote by $\Sigma := \{P_1, \ldots, P_n\}$ the set of nodes. For every node $P_i \in \Sigma$ we denote by S_i and T_i the preimages of P_i under the normalization map π_Σ, such that $S_i \in s_0(P_i)$ and $T_i \in t_0(P_i)$.

Notation 2.33.
Let $m := \pi_0(X_\Sigma)$. Given a multidegree $\underline{e} = (e_1, \ldots, e_m)$, a subset $\Sigma' \neq \emptyset$ of the set of nodes Σ and an orientation $O = (s, t)$ of the dual graph Γ_X of X, we denote by

- $N_{\Sigma'}^{\underline{e}}$ the unique rank 1 torsion-free sheaf on X that is the push-forward of a line bundle of multidegree $\underline{e}$ on the partial normalization $X_{\Sigma'}$ of X at Σ', which is a sheaf that fails to be locally free at the set of nodes $\Sigma' \subseteq \Sigma$,
- $R_{\Sigma'}$ the ring $(k[x_i, y_i]_{\{i|P_i\in\Sigma'\}})/(x_i y_i)$,
- $T_{\Sigma'}$ the subgroup of $\mathbb{G}_m^m$ consisting of sequences $\underline{\lambda} = (\lambda_1, \ldots, \lambda_m)$ with the property that $\lambda_i = \lambda_j$ if $X_i \cap X_j$ contains some node not in Σ',
- $\mathbb{A}_O^{|\Sigma'|} = \mathrm{Spec}(k[t_i]_{\{i|P_i\in\Sigma'\}})$ where
  - $t_i = x_i$, if $s(P_i) = s_0(P_i)$,
  - $t_i = y_i$, if $s(P_i) = t_0(P_i)$.
- For every irreducible component X_i of X we fix a smooth point $z_i \in X_i$ and we consider $\mathcal{O}_i(d_i \cdot z_i)$.

Lemma 2.34. *Let $\underline{d} = (d_1, \ldots, d_m)$ be a multidegree and O an orientation of the dual graph Γ_X of a nodal curve X with rational irreducible components. The unramified morphism $\theta_O^{\underline{d}}$ of Proposition 2.23 is*

$$\theta_O^{\underline{d}}\colon [\mathbb{A}_O^n/\mathbb{G}_m^m] \to \underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X),$$

which sends a k-point (t_i) to the sheaf $I_{(t_i)}$ defined by the short exact sequence

$$0 \to I_{(t_i)} \to \pi_{\Sigma,*}(\mathcal{O}_1(d_1 \cdot z_1) \oplus \cdots \oplus \mathcal{O}_m(d_m \cdot z_m)) \xrightarrow{\psi_{(t_i)}} k_{P_1} \oplus \cdots \oplus k_{P_n} \to 0, \tag{2.12}$$

where the map $\pi_{P_i} \circ \psi_{(t_i)}\colon \pi_{\Sigma,*}(\mathcal{O}_1(d_1 \cdot z_1) \oplus \cdots \oplus \mathcal{O}_m(d_m \cdot z_m)) \to k_{P_i}$ sends $(f_1, \ldots, f_m)$ to

$$\begin{cases} f_{s(P_i)}(S_i) - t_i f_{t(P_i)}(T_i), & \text{if } t_i = x_i, \\ t_i f_i(S_i) - f_{i+1}(T_i), & \text{if } t_i = y_i. \end{cases}$$

Proof. We have that $\underline{\mathrm{Pic}}^{\underline{d}}(X_\Sigma) \simeq \prod \underline{\mathrm{Pic}}^{d_i}(\mathbb{P}^1) \simeq \prod B\mathbb{G}_m$, thus we have

$$\underline{\mathrm{Hom}}^{\underline{d}}(\mathcal{L}_{s(P)}, \mathcal{L}_{t(P)}) \simeq \left[\mathrm{Hom}^{\underline{d}}(L_{s(P)}, L_{t(P)})/\mathbb{G}_m^m\right]$$

and moreover we can give explicit coordinates, namely

- $\mathrm{Hom}^{\underline{d}}(L_{S_i}, L_{T_i}) \cong \mathbb{A}^1 := \mathrm{Spec}(k[x_i])$,
- $\mathrm{Hom}^{\underline{d}}(L_{T_i}, L_{S_i}) \cong \mathbb{A}^1 := \mathrm{Spec}(k[y_i])$

and $\underline{\lambda} \in \mathbb{G}_m^m$ acts on $\mathrm{Hom}^{\underline{d}}(L_{S_i}, L_{T_i})$ and on $\mathrm{Hom}^{\underline{d}}(L_{T_i}, L_{S_i})$ by

- $\underline{\lambda} \cdot \sigma_i \in \mathrm{Hom}^{\underline{d}}(L_{S_i}, L_{T_i}) := \lambda_{s_0(P_i)} \sigma_i (\lambda_{t_0(P_i)})^{-1}$,
- $\underline{\lambda} \cdot \sigma_i \in \mathrm{Hom}^{\underline{d}}(L_{T_i}, L_{S_i}) := \lambda_{t_0(P_i)} \sigma_i (\lambda_{s_0(P_i)})^{-1}$.

Therefore, given a fixed orientation $O = (s, t)$ of Γ_X, we have

$$\textstyle\prod_{P \in \Sigma} \underline{\mathrm{Hom}}^{\underline{d}}(L_{s(P)}, L_{t(P)}) \cong [\mathbb{A}_O^n/\mathbb{G}_m^m] = [\mathrm{Spec}(k[t_1, \ldots, t_n])/\mathbb{G}_m^m],$$

where $t_i = x_i$ if $s(P_i) = s_0(P_i)$ and $t_i = y_i$ if $s(P_i) = t_0(P_i)$. □

Proposition 2.35. *Let $\underline{e}$ be a multidegree. The étale morphism of Proposition 2.28 is*

$$\theta_{\underline{e}}\colon [\mathrm{Spec}(R_\Sigma)/\mathbb{G}_m^m] \to \mathrm{Coh}_1^{\mathrm{tf}}(X), \tag{2.13}$$

where the action of $\mathbb{G}_m^m$ on $\mathrm{Spec}(R_\Sigma)$ is given on k-points by

- $\underline{\lambda} \cdot x_i = \lambda_{s_0(P_i)} x_i (\lambda_{t_0(P_i)})^{-1}$,
- $\underline{\lambda} \cdot y_i = (\lambda_{s_0(P_i)})^{-1} y_i \lambda_{t_0(P_i)}$.

Proof. We want to prove that, given the ordered set

$$\mathrm{ParOr} := \{(\Sigma', O') |\ \Sigma' \subseteq \Sigma, O' \text{ orientation of } \Sigma'\}$$

we have

$$\varinjlim_{\mathrm{ParOr}} [\mathbb{A}_O^n/\mathbb{G}_m^m] = [\mathrm{Spec}(k[x_1, y_1, \ldots, x_n, y_n]/(x_1 y_1, \ldots, x_n y_n))/\mathbb{G}_m^m],$$

where the action of $\mathbb{G}_m^m$ on $X_n := \mathrm{Spec}(k[x_1, y_1, \ldots, x_n, y_n]/(x_1 y_1, \ldots, x_n y_n))$ is defined on k-points by

$$(\lambda_1, \ldots, \lambda_m) \cdot (x_1, y_1, \ldots, x_n, y_n) = (\lambda_{s_0(P_i)} x_i (\lambda_{t_0(P_i)})^{-1}, (\lambda_{s_0(P_i)})^{-1} y_i \lambda_{t_0(P_i)}).$$

Using that

$$\varinjlim_{\mathrm{ParOr}} [\mathbb{A}_O^n/\mathbb{G}_m^m] = \left[(\varinjlim_{\mathrm{ParOr}} \mathbb{A}_O^n)/\mathbb{G}_m^m\right],$$

it is enough to prove that

$$\varinjlim_I \mathbb{A}_O^n = \mathrm{Spec}(k[x_1, y_1, \ldots, x_n, y_n]/(x_1 y_1, \ldots, x_n y_n))$$

To do this we denote by $X_n := \mathrm{Spec}(k[x_1, y_1, \dots, x_n, y_n]/(x_1y_1, \dots, x_ny_n))$ and for every orientation O of the dual graph Γ_X of X, we define the morphism

$$\gamma_O \colon \mathbb{A}^n_O = \mathrm{Spec}(k[t_1, \dots, t_n]) \to X_n$$

defined by sending $(t_1, \dots, t_n)$ to $(x_1, y_1, \dots, x_n, y_n)$ such that

- $x_i = 0$ if and only if $t_i = y_i$,
- $y_i = 0$ if and only if $t_i = x_i$.

We want to show that two points $(t'_{\Sigma'}, 0_{\Sigma\setminus\Sigma'})$ and $(t'_{\Sigma''}, 0_{\Sigma\setminus\Sigma''})$ respectively of $\mathbb{A}^n_O$ and of $\mathbb{A}^n_{O'}$ have the same image in $\underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X)$ if and only if they have the same image in X_n. Indeed we have that $(t'_{\Sigma}, 0_{\Sigma\setminus\Sigma'})$ and $(t'_{\Sigma''}, 0_{\Sigma\setminus\Sigma''})$ have the same image in $\underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X)$ if and only if

- $\Sigma' = \Sigma''$,
- $t_i = t'_i$, for every $i \in \Sigma \setminus \Sigma'$.

Namely we have $O|_{\Sigma\setminus\Sigma'} = O'|_{\Sigma\setminus\Sigma'}$ which is equivalent to say that the images of the points $(t'_{\Sigma}, 0_{\Sigma\setminus\Sigma'})$ and $(t'_{\Sigma''}, 0_{\Sigma\setminus\Sigma''})$ via respectively the morphisms γ_O and $\gamma_{O'}$ coincide in X_n. $\square$

Corollary 2.36. *Let Σ' be a proper subset of Σ and $\underline{e}$ be a multidegree. There is an étale morphism*

$$\theta_{\Sigma',\underline{e}} \colon [\mathrm{Spec}(R_{\Sigma'})/T_{\Sigma'}] \to \mathrm{Coh}^{\mathrm{tf}}_1(X), \tag{2.14}$$

where the action of $T_{\Sigma'} \subseteq \mathbb{G}^m_m$ on $\mathrm{Spec}(R_{\Sigma'})$ is given on k-points by

- $\underline{\lambda} \cdot x_i = \lambda_{s_0(P_i)} x_i (\lambda_{t_0(P_i)})^{-1}$,
- $\underline{\lambda} \cdot y_i = (\lambda_{s_0(P_i)})^{-1} y_i \lambda_{t_0(P_i)}$.

The image of the above morphism is the smallest $\mathrm{Pic}^{\underline{0}}(X)$-stable open neighborhood of $N^{\underline{e}}_{\Sigma'}$.

Proof. Define $\underline{e}' := \underline{e} - \mathcal{D}(\Sigma \setminus \Sigma', O^+)$ and consider the sheaf $N^{\underline{e}'}_{\Sigma}$ that fails to be locally free at all the nodes Σ of X.
For every orientation O of $\Gamma_{X_{\Sigma'}}$ we have the unramified morphism

$$\theta^{\underline{d}}_{O'} \colon [\mathbb{A}^n_{O'}/\mathbb{G}^m_m] \to \underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X),$$

where O' is the orientation of Γ_{X_Σ} such that $O'|_{\Sigma'} = O$ and $O'|_{\Sigma\setminus\Sigma'} = O_0$.
We consider the open in $\mathbb{A}^n_{O'}$ defined by the equations $t_i \neq 0$ for every i such that $P_i \notin \Sigma'$. The image of this open is given by sheaves that specialize to $N^{\underline{e}'}_{\Sigma'}$.
We can restrict every morphism $\theta^{\underline{d}}_{O'}$ to $\mathbb{G}^{m-|\Sigma'|}_m \times \mathbb{A}^{|\Sigma'|}_{(s,t)}$. We get $2^{|\Sigma'|}$ different morphisms

$$\left[(\mathbb{G}^{m-|\Sigma'|}_m \times \mathbb{A}^{|\Sigma'|}_{(s,t)})/\mathbb{G}^m_m\right] \to \underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X).$$

Define $\mathcal{X} = \mathbb{G}^{m-|\Sigma'|}_m \times \mathbb{A}^{|\Sigma'|}, G = \mathbb{G}^m_m$ and $H = \mathbb{G}^{m-|\Sigma'|}_m$. From [9, Remark 2.4] we have that given a stack $\mathcal{X}$ with an action of the group G and a normal subgroup H of G, there is an isomorphism of stacks

$$[\mathcal{X}/G] \simeq [(\mathcal{X}/H)/(G/H)].$$

Therefore when $\mathcal{X} = \mathbb{G}^{m-|\Sigma'|}_m \times \mathbb{A}^{|\Sigma'|}, G = \mathbb{G}^m_m$ and $H = \mathbb{G}^{m-|\Sigma'|}_m$, we get a morphism

$$\overline{\theta}^{\underline{d}}_{O'} \colon \left[\mathbb{A}^{|\Sigma'|}_{(s,t)}/T_{N^{\underline{e}}_{\Sigma'}}\right] \to \underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X)$$

because $\mathbb{G}^m_m/\mathbb{G}^{m-|\Sigma'|}_m \simeq T_{N^{\underline{e}}_{\Sigma'}}$. We prove that

$$\varinjlim_I \mathbb{A}^{|\Sigma'|}_O = \mathrm{Spec}(R_{\Sigma'}).$$

For every orientation O of the dual graph $\Gamma_{X_{\Sigma'}}$ of $X_{\Sigma'}$, we define the morphism

$$\gamma_O \colon \mathbb{A}_O^{|\Sigma'|} \to \mathrm{Spec}(R_{\Sigma'})$$

defined by sending $(t_i)_{i\in\Sigma'}$ to $(x_i, y_i)_{i\in\Sigma'}$ such that

- $x_i = 0$ if and only if $t_i = y_i$,
- $y_i = 0$ if and only if $t_i = x_i$.

We have that two points $(t_{\Sigma_1}, 0_{\Sigma'\setminus\Sigma_1})$ and $(t'_{\Sigma_2}, 0_{\Sigma'\setminus\Sigma_2})$ respectively of $\mathbb{A}_O^{|\Sigma'|}$ and of $\mathbb{A}_{O'}^{|\Sigma'|}$ have the same image in $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ if and only if

- $\Sigma_1 = \Sigma_2$,
- $t_i = t'_i$, for every $i \in \Sigma' \setminus \Sigma_1$,

namely, the images of the points $(t_{\Sigma_1}, 0_{\Sigma'\setminus\Sigma_1})$ and $(t'_{\Sigma_2}, 0_{\Sigma'\setminus\Sigma_2})$ via respectively the morphisms γ_O and $\gamma_{O'}$ coincide in $\mathrm{Spec}(R_{\Sigma'})$. Therefore we have

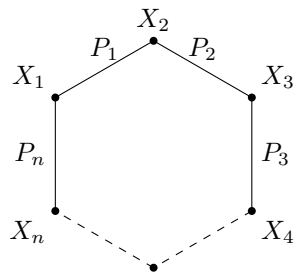
$$\varinjlim_I \mathbb{A}_O^{|\Sigma'|} = \mathrm{Spec}(R_{\Sigma'}).$$

$\square$

3. THE CASE OF NECKLACES OF GENUS 1

Given a necklace X of length n and genus $g = 1$, we classify all open connected substacks of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ that admit a proper good moduli space. We first give a geometric description in Proposition 3.17 and then a combinatorial classification in Theorem 3.28.

Throughout this Chapter, we denote by $\Sigma = \{P_1, \ldots, P_n\}$ the set of nodes of X and by $X_1, \ldots, X_n$ the irreducible components of X. Our convention is that the node $P_i \in X_i \cap X_{i+1}$. We denote by $\{S_i, T_i\}$ the preimages of P_i under the normalization map π_Σ, such that $S_i \in X_i$ and $T_i \in X_{i+1}$.
The dual graph Γ_X of X is



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We define the *clockwise orientation* O^+ as the orientation of the dual graph Γ_X of X determined by $s(P_i, O^+) = s_+(P_i) = X_i$ and $t(P_i, O^+) = t_+(P_i) = X_{i+1}$, for every $P_i \in \Sigma$.
We define the *anticlockwise orientation* O^- as the orientation of the dual graph Γ_X of X determined by $s(P_i, O^-) = s_-(P_i) = X_{i+1}$ and $t(P_i, O^-) = t_-(P_i) = X_i$, for every $P_i \in \Sigma$.

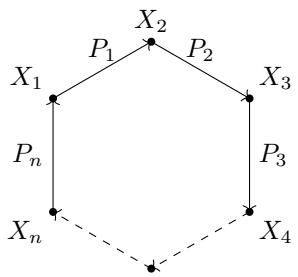


Fig. 3.1: (a) Dual graph of X with clockwise orientation.

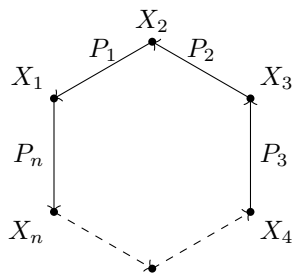


Fig. 3.2: (b) Dual graph of X with anticlockwise orientation.

3.1 Geometric classification

In this Section, given a necklace X of genus $g = 1$, we give a geometric classification of all open connected substacks of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ which admit a proper good moduli space. The main result is Proposition 3.17 in which we find that all good compactified Jacobians of a necklace are necklaces.

Following the Notation of Section 2.3, for every rank 1 torsion-free sheaf N in $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$, we can give an explicit description of the étale morphism θ_N of Proposition 2.28.

Lemma 3.1. *Let $\underline{d} = (d_1, \dots, d_n)$ be a multidegree and $O = (s, t)$ an orientation of the dual graph Γ_X of a necklace X of length n. The unramified morphism $\theta_O^{\underline{d}}$ of Proposition 2.23 is*

$$\theta_O^{\underline{d}} \colon [\mathbb{A}_O^n / \mathbb{G}_m^n] \to \underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X),$$

which sends a k-point (t_i) to the sheaf $I_{(t_i)}$ defined by the short exact sequence

$$0 \to I_{(t_i)} \to \mathcal{O}_1(d_1 \cdot z_1) \oplus \cdots \oplus \mathcal{O}_n(d_n \cdot z_n) \xrightarrow{\psi_{(t_i)}} k_{P_1} \oplus \cdots \oplus k_{P_n} \to 0, \tag{3.1}$$

where the map $\pi_{P_i} \circ \psi_{(t_i)} \colon \mathcal{O}_1(d_1 \cdot z_1) \oplus \cdots \oplus \mathcal{O}_n(d_n \cdot z_n) \to k_{P_i}$ sends $(f_1, \dots, f_n)$ to

$$\begin{cases} f_i(S_i) - t_i f_{i+1}(T_i), & \text{if } t_i = x_i, \\ t_i f_i(S_i) - f_{i+1}(T_i), & \text{if } t_i = y_i. \end{cases}$$

Proof. This is Lemma 2.34 in the case of a necklace X. In particular, we have that $m = n$ and $\mathbb{A}_O^n = \mathrm{Spec}(k[t_i]_{\{i \mid P_i \in \Sigma\}})$ where

- $t_i = x_i$, if $s(P_i) = s_+(P_i) = X_i$,
- $t_i = y_i$, if $s(P_i) = t_+(P_i) = X_{i+1}$.

□

Corollary 3.2. *The image of a k-point (t_i) of $[\mathbb{A}_O^n / \mathbb{G}_m^n]$ via the morphism $\theta_O^{\underline{d}}$ is the sheaf $N_{\Sigma'}^{\underline{e}}$ on X, where*

- $\Sigma' = \{P_i \in \Sigma \mid t_i = 0\}$,
- $\underline{e} = \underline{d} - \mathcal{D}(\Sigma', O)$.

In particular the line bundles of multidegree $\underline{d}$ are in the image of the morphism $\theta_O^{\underline{d}}$.

Proof. We proved this in general in Proposition 2.26. □

Proposition 3.3. *Let X be a necklace of genus $g = 1$, Σ' a subset of the set of nodes Σ of X and $\underline{e}$ a multidegree. The étale morphisms of Proposition 2.35 and Corollary 2.36 for the sheaf $N_{\Sigma'}^{\underline{e}}$ is*

$$\theta_{N_{\Sigma'}^{\underline{e}}} \colon [\mathrm{Spec}(R_{\Sigma'}) / T_{\Sigma'}] \to \mathrm{Coh}_1^{\mathrm{tf}}(X), \tag{3.2}$$

where the action of $T_{\Sigma'} \subseteq \mathbb{G}_m^n$ on $\mathrm{Spec}(R_{\Sigma'})$ is given on k-points by

- $\underline{\lambda} \cdot x_i = \lambda_{s_+(P_i)} x_i (\lambda_{t_+(P_i)})^{-1} = \lambda_i x_i (\lambda_{i+1})^{-1}$,
- $\underline{\lambda} \cdot y_i = (\lambda_{s_+(P_i)})^{-1} y_i \lambda_{t_+(P_i)} = (\lambda_i)^{-1} y_i \lambda_{i+1}$.

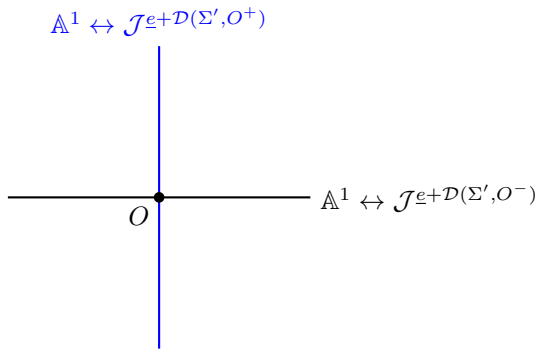
Proof. This is Proposition 2.35 in the case of a necklace X where as fixed orientation O_0 of Γ_X we take the clockwise orientation O^+. Therefore we have $s_0(P_i) = s_+(P_i) = X_i$ and $t_0(P_i) = t_+(P_i) = X_{i+1}$ and $T_{\Sigma'}$ is the subgroup of $\mathbb{G}_m^n$ consisting of sequences $\underline{\lambda} = (\lambda_1, \dots, \lambda_n)$ with the property that $\lambda_i = \lambda_{i+1}$ if $P_i \notin \Sigma'$. □

Notation 3.4.
We denote by

- $N^{\underline{e}}_{\Sigma'}$ the unique rank 1 torsion-free sheaf on X which is the push-forward of a line bundle of multidegree $\underline{e}$ on the partial normalization $X_{\Sigma'}$ of X at Σ',
- $\mathcal{U}(N^{\underline{e}}_{\Sigma'})$ the open $\mathrm{Im}(\theta_{N^{\underline{e}}_{\Sigma'}}) \subseteq \underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X)$, where $\theta_{N^{\underline{e}}_{\Sigma'}}$ is the étale morphism of Proposition 3.3,
- $U(N^{\underline{e}}_{\Sigma'})$ the moduli space of the open $\mathcal{U}(N^{\underline{e}}_{\Sigma'})$.

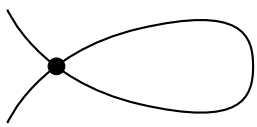
Proposition 3.5. *Consider the sheaf $N^{\underline{e}}_{\Sigma'}$ and the open $\mathcal{U}(N^{\underline{e}}_{\Sigma'})$.*

- *If $\Sigma' \subsetneq \Sigma$ the open $\mathcal{U}(N^{\underline{e}}_{\Sigma'})$ admits a separated good moduli space $U(N^{\underline{e}}_{\Sigma'})$ which is a nodal curve given by two copies of $\mathbb{A}^1$ meeting at the origin:*



  In $U(N^{\underline{e}}_{\Sigma'})$ the points outside the origin of one of the two copies of $\mathbb{A}^1$ represent line bundles of multidegree $\underline{d}_1 := \underline{e} + \mathcal{D}(\Sigma', O^+)$ and the points outside the origin of the other copy of $\mathbb{A}^1$ represent line bundles of multidegree $\underline{d}_2 := \underline{e} + \mathcal{D}(\Sigma', O^-)$. All the other sheaves in $\mathcal{U}(N^{\underline{e}}_{\Sigma'})$ get contracted to the node O in the moduli space.

- *If $\Sigma' = \Sigma$, the open $\mathcal{U}(N^{\underline{e}}_{\Sigma})$ admits a proper good moduli space $U(N^{\underline{e}}_{\Sigma})$ which is a nodal $\mathbb{P}^1$. In $U(N^{\underline{e}}_{\Sigma})$ the points outside the node represent line bundles of multidegree $\underline{d} := \underline{e} + \mathcal{D}(\Sigma, O^+)$. All the other sheaves of $\mathcal{U}(N^{\underline{e}}_{\Sigma'})$ get contracted to the node in the moduli space.*



Proof. We want to understand the moduli space $U(N^{\underline{e}}_{\Sigma'})$ of the open $\mathcal{U}(N^{\underline{e}}_{\Sigma'})$ in $\underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X)$, which is by definition the image in $\underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X)$ of $[\mathrm{Spec}(R_{\Sigma'})/T_{\Sigma'}]$ via the étale morphism $\theta_{N^{\underline{e}}_{\Sigma'}}$.
We start by observing that the scheme $\mathrm{Spec}(R_{\Sigma'})$ is given by the union of $2^{|\Sigma'|}$ copies of $\mathbb{A}^{|\Sigma'|}$, defined by $|\Sigma'|$ coordinates (t_i), where $t_i \in \{x_i, y_i\}$. Namely, these $2^{|\Sigma'|}$ copies are

$$\mathbb{A}^{|\Sigma'|}_O = \mathrm{Spec}(k[t_i]_{\{i|P_i\in\Sigma'\}})$$

where $O = (s,t)$ is the orientation of the nodes of Σ' such that $s(P_i) = X_i$ if and only if $t_i = x_i$.
In particular, for the clockwise orientation O^+, we have

$$\mathbb{A}^{|\Sigma'|}_{O^+} = \mathrm{Spec}(k[x_i]_{\{i|P_i\in\Sigma'\}}),$$

and, for the anticlockwise orientation O^-, we have

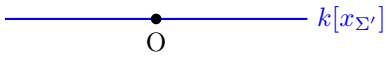
$$\mathbb{A}^{|\Sigma'|}_{O^-} = \mathrm{Spec}(k[y_i]_{\{i|P_i\in\Sigma'\}}).$$

For every copy $\mathbb{A}^{|\Sigma'|}_O$, we study the GIT quotient $\mathbb{A}^{|\Sigma'|}_O//T_{\Sigma'}$:

- If we consider the clockwise orientation O^+, we have $\mathbb{A}^{|\Sigma'|}_{O^+} = \mathrm{Spec}(k[x_i]_{\{i|P_i\in\Sigma'\}})$ and the action of $T_{\Sigma'}$ on k-points is given by

$$\underline{\lambda}\cdot(x_i) = (\lambda_i x_i(\lambda_{i+1})^{-1}).$$

  Therefore the ring of invariants is $\mathbb{A}^1 = \mathrm{Spec}(k[x_{\Sigma'}])$, where $x_{\Sigma'} := \prod_i x_i$.
  All the points with at least one coordinate equal to zero have the point $\underline{0}$ in the closure of their $T_{\Sigma'}$-orbit and therefore they get identified in the GIT quotient.
  The points which are not contracted in the quotient are the points with all coordinates different from zero. If we consider the open $U_+ = \mathbb{G}_m^{|\Sigma'|}$ in $\mathbb{A}^{|\Sigma'|}_{O^+}$ consisting of these points with all the coordinates different from zero, we have that the quotient $U_+//T_{\Sigma'}$ is $\mathbb{G}_m$ and this represents line bundles of multidegree $\underline{e} + \mathcal{D}(\Sigma', O^+)$.



(3.3)

  In the picture (3.3) above, the points outside the origin represent line bundles of multidegree $\underline{e} + \mathcal{D}(\Sigma', O^+)$ and all the sheaves $N^{\underline{e}'}_{\Sigma''}$, with

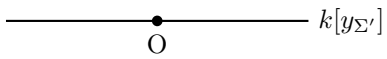
  - $\Sigma'' \subseteq \Sigma'$,
  - $\underline{e}' = \underline{e} + \mathcal{D}(\Sigma' \setminus \Sigma'', O^+)$,

  are identified in the node O.

- If we consider the anticlockwise orientation O^-, we have $\mathbb{A}^{|\Sigma'|}_{O^-} = \mathrm{Spec}(k[y_i]_{\{i|P_i\in\Sigma'\}})$ and the action of $T_{\Sigma'}$ on k-points is given by

$$\underline{\lambda}\cdot(y_i) = (\lambda_{i+1} y_i(\lambda_i)^{-1}).$$

  Therefore the ring of invariants is $\mathbb{A}^1 = \mathrm{Spec}(k[y_{\Sigma'}])$, where $y_{\Sigma'} := \prod_i y_i$.
  All points with at least one coordinate equal to zero have the point $\underline{0}$ in the closure of their $T_{\Sigma'}$-orbit and therefore they get identified in the GIT quotient. The points which are not contracted in the quotient are the points with all coordinates different from zero.
  If we consider the open $U_- = \mathbb{G}_m^{|\Sigma'|}$ in $\mathbb{A}^{|\Sigma'|}_{O^-}$ consisting of these points with all coordinates different from zero, we have that the quotient $U_-//T_{\Sigma'}$ is $\mathbb{G}_m$ and this represents line bundles of multidegree $\underline{e} + \mathcal{D}(\Sigma', O^-)$.



(3.4)

  In the picture (3.4) above, the points outside the origin represent line bundles of multidegree $\underline{e} + \mathcal{D}(\Sigma', O^+)$ and all the sheaves $N^{\underline{e}'}_{\Sigma''}$, with

  - $\Sigma'' \subseteq \Sigma'$,
  - $\underline{e}' = \underline{e} + \mathcal{D}(\Sigma' \setminus \Sigma'', O^-)$,

  are identified in the node O.

- If we consider an orientation $O \notin \{O^+, O^-\}$ the ring of invariant is k.
  This means that in the moduli space all the points of $\mathbb{A}^{|\Sigma'|}_{O}$ get identified in one point, namely:

  - The line bundles of multidegree $\underline{d} = \underline{e} + \mathcal{D}(\Sigma', O)$,
  - All sheaves $N^{\underline{e}'}_{\Sigma''}$ which are not locally free at some set of nodes $\Sigma'' \subseteq \Sigma'$ and are the push-forward of a line bundle of multidegree $\underline{e} + \mathcal{D}(\Sigma' \setminus \Sigma'', O)$.

Observe that each point of the ring $R_{N^{\underline{e}}_{\Sigma'}}$ is defined by $2|\Sigma'|$ coordinates $(x_i, y_i)_{\{i|\ P_i \in \Sigma'\}}$ and for every i at least one coordinate between x_i and y_i is equal to 0. All points $(x_i, y_i)_{\{i|\ P_i \in \Sigma'\}}$ with at least one of the coordinates x_i's equal to zero and one of the coordinates y_j's equal to zero belong to some $\mathbb{A}^{|\Sigma'|}_O$ for some orientation $O \notin \{O^+, O^-\}$ and we have just seen that the GIT quotient $\mathbb{A}^{|\Sigma'|}_O // T_{N^{\underline{e}}_{\Sigma'}}$ is just a point. This means that

- All sheaves of $\mathcal{U}(N^{\underline{e}}_{\Sigma'})$ which are not line bundles,
- All line bundles of multidegree $\underline{e} + \mathcal{D}(\Sigma', O)$ for any orientation $O \notin \{O^+, O^-\}$,

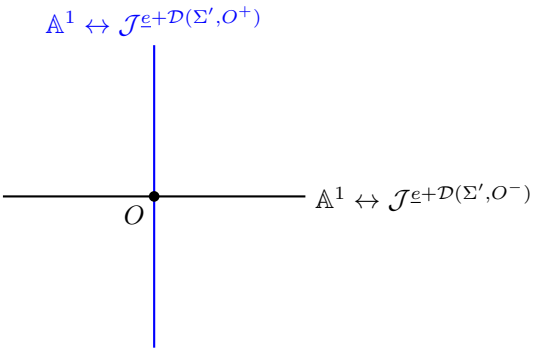
are contracted to the same point in $U(N^{\underline{e}}_{\Sigma'})$.
Computing the GIT quotient $\mathrm{Spec}(R_{\Sigma'})//T_{\Sigma'}$ and denoting by $R^{T_{\Sigma'}}_{\Sigma'}$ the ring of invariants, we have

$$\mathrm{Spec}(R_{\Sigma'})//T_{\Sigma'} = \mathrm{Spec}(R^{T_{\Sigma'}}_{\Sigma'}) = \mathrm{Spec}(k[x_{\Sigma'}, y_{\Sigma'}]/x_{\Sigma'} y_{\Sigma'})$$

where $x_{\Sigma'} := \prod_{P_i \in \Sigma'} x_i$ and $y_{\Sigma'} := \prod_{P_i \in \Sigma'} y_i$.
The two copies of $\mathbb{A}^1$ namely $\mathrm{Spec}\, k[x_{\Sigma'}]$ and $\mathrm{Spec}\, k[y_{\Sigma'}]$ come from the quotients $\mathbb{A}^{|\Sigma'|}_{O^+} // T_{N^{\underline{e}}_{\Sigma'}}$ and $\mathbb{A}^{|\Sigma'|}_{O^-} // T_{N^{\underline{e}}_{\Sigma'}}$.
Therefore the points of $\mathrm{Spec}\, k[x_{\Sigma'}]$ and $\mathrm{Spec}\, k[y_{\Sigma'}]$ outside the origin correspond respectively to line bundles of multidegree

- $\underline{d}_1 = \underline{e} + \mathcal{D}(\Sigma', O^+)$,
- $\underline{d}_2 = \underline{e} + \mathcal{D}(\Sigma', O^-)$,

and, as claimed, if $\Sigma' \subsetneq \Sigma$, the moduli space is



If $\Sigma' = \Sigma$, we have $\mathcal{D}(\Sigma, O^+) = \mathcal{D}(\Sigma, O^-)$ which implies

$$\underline{d}_1 = \underline{e} + \mathcal{D}(\Sigma', O^+) = \underline{e} + \mathcal{D}(\Sigma', O^-) = \underline{d}_2$$

In particular the k-point $(x_1, \dots, x_n)$ of $\left[\mathbb{A}^n_{O^+}/\mathbb{G}^n_m\right]$ with $x_i \neq 0$ for every $i = 1, \dots, n$ is mapped to the line bundle

$$L = (\mathcal{O}_1(d_1 \cdot \underline{1}), \dots, \mathcal{O}_n(d_n \cdot \underline{1}), x_1, \dots, x_n)$$

where x_i is an isomorphism $L_{S_i} \to L_{T_i}$ for every $i = 1, \dots, n$.
If for every $i = 1, \dots, n$ we define $y_i := x_i^{-1}$, namely $y_i \colon L_{T_i} \to L_{S_i}$, we can write

$$L = (\mathcal{O}_1(d_1 \cdot \underline{1}), \dots, \mathcal{O}_n(d_n \cdot \underline{1}), y_1^{-1}, \dots, y_n^{-1}).$$

This means that the k-point $(y_1^{-1}, \dots, y_n^{-1})$ of $\left[\mathbb{A}^n_{O^-}/\mathbb{G}^n_m\right]$ is mapped via the morphism $\theta^{\underline{d}}_{O^-}$ to the line bundle L. Therefore we have the identification of $\mathrm{Spec}(k[x_{\Sigma'}])$ and $\mathrm{Spec}(k[y_{\Sigma'}])$ outside the origin defined by sending $x_{\Sigma'} \mapsto (y_{\Sigma'})^{-1}$. Hence we get a nodal $\mathbb{P}^1$ as moduli space. $\square$

Corollary 3.6. *Given the sheaf $N^{\underline{e}}_{\Sigma'}$, the line bundles in $\mathcal{U}(N^{\underline{e}}_{\Sigma'})$ have multidegree*

$$\underline{e} + \mathcal{D}(\Sigma', O),$$

where O is an orientation of Σ'.
The line bundles which are not contracted to the node of $U(N^{\underline{e}}_{\Sigma'})$ have multidegree $\underline{e} + \mathcal{D}(\Sigma', O^+)$ and $\underline{e} + \mathcal{D}(\Sigma', O^-)$, where O^+ is the clockwise orientation and O^- the anticlockwise orientation of the dual graph Γ_X of X.

Definition 3.7.
Let $\underline{d}$ be a fixed multidegree.

- A sheaf, which is not a line bundle, is of type 1 for $\underline{d}$ if it is in the image of $\theta^{\underline{d}}_{O^+}$.
- A sheaf, which is not a line bundle, is of type 2 for $\underline{d}$ if it is in the image of $\theta^{\underline{d}}_{O^-}$.
- A sheaf, which is not a line bundle, is of type 3 if it is not in the image of $\theta^{\underline{d}}_{O^+}$ and $\theta^{\underline{d}}_{O^-}$.

Remark 3.8. If a sheaf N is of type 1 or 2 for the multidegree $\underline{d}$, the line bundles of multidegree $\underline{d}$ are not contracted to the node of $U(N)$.
If a sheaf N is of type 3 for the multidegree $\underline{d}$, the line bundles of this multidegree are contracted to the node of $U(N)$.

Remark 3.9. A sheaf $N^{\underline{e}}_{\Sigma'}$, with $\Sigma' \subsetneq \Sigma$, is

- of type 1 for exactly one multidegree, which is $\underline{e} + \mathcal{D}(\Sigma', O^+)$,
- of type 2 for exactly one multidegree, which is $\underline{e} + \mathcal{D}(\Sigma', O^-)$,
- of type 3 for $2^{|\Sigma'|} - 2$ distinct multidegrees, which are of the form $\underline{e} + \mathcal{D}(\Sigma', O)$, where O is any orientation different from the clockwise and anticlockwise orientations.

If $\Sigma' = \Sigma$, we have

$$\underline{e} + \mathcal{D}(\Sigma, O^+) = \underline{e} + \mathcal{D}(\Sigma, O^-),$$

and the sheaf $N^{\underline{e}}_{\Sigma}$ is both of type 1 and 2 for the multidegree $\underline{e} + \mathcal{D}(\Sigma, O^+)$.

The aim of this Chapter is to classify all open connected substacks of $\underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X)$ which admit a proper good moduli space for a necklace X. In Proposition 3.5 we have just found some examples, namely the opens $\mathcal{U}(N^{\underline{e}}_{\Sigma})$ for a sheaf $N^{\underline{e}}_{\Sigma}$ which fails to be locally free at all the nodes of X.
We observe that if $\mathcal{U}$ is an open connected substack of $\underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X)$ which admits a proper good moduli space and if $N^{\underline{e}}_{\Sigma'}$ is a sheaf in $\mathcal{U}$, then $\mathcal{U}(N^{\underline{e}}_{\Sigma'}) \subseteq \mathcal{U}$ because any open neighborhood of $N^{\underline{e}}_{\Sigma'}$ in $\mathcal{U}(N^{\underline{e}}_{\Sigma'})$ is dense, hence all points in $\mathcal{U}(N^{\underline{e}}_{\Sigma'})$ are specializations of points of $\mathcal{U}(N^{\underline{e}}_{\Sigma'}) \cap \mathcal{U}$, so they have to be contained in $\mathcal{U}$ if U admits a proper good moduli space. This implies that if we consider the set $\mathcal{S}$ of all the sheaves which are not line bundles in $\mathcal{U}$, we have

$$\mathcal{U} = \bigcup_{N \in \mathcal{S}} \mathcal{U}(N).$$

We have to figure out which unions of opens of the form $\mathcal{U}(N^{\underline{e}}_{\Sigma'})$ are connected and admit a proper good moduli space.
As a first step we introduce the notion of *consecutive* opens and show that the union of two consecutive opens admits a *separated* moduli space.

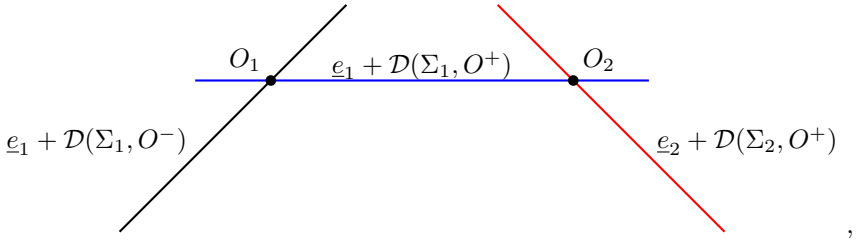
Definition 3.10.
Two opens $\mathcal{U}(N^{\underline{e}_1}_{\Sigma_1})$ and $\mathcal{U}(N^{\underline{e}_2}_{\Sigma_2})$ are *consecutive* if

$$\underline{e}_1 + \mathcal{D}(\Sigma_1, O^+) = \underline{e}_2 + \mathcal{D}(\Sigma_2, O^-).$$

In other words, there exists a multidegree $\underline{d} := \underline{e}_1 + \mathcal{D}(S_1, O^+)$ such that $N^{\underline{e}_1}_{\Sigma_1}$ and $N^{\underline{e}_2}_{\Sigma_2}$ are respectively of type 1 and of type 2 for $\underline{d}$.

Remark 3.11. If two opens $\mathcal{U}(N_1)$ and $\mathcal{U}(N_2)$ are consecutive, the union $\mathcal{U}(N_1) \cup \mathcal{U}(N_2)$ is an open connected substack of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$.

Lemma 3.12. *Let $\mathcal{U}(N_{\Sigma_1}^{\underline{e}_1})$ and $\mathcal{U}(N_{\Sigma_2}^{\underline{e}_2})$ be two consecutive opens. If $\Sigma_1 \cup \Sigma_2$ is not a partition of Σ, the open $\mathcal{U}(N_{\Sigma_1}^{\underline{e}_1}) \cup \mathcal{U}(N_{\Sigma_2}^{\underline{e}_2})$ admits a separated good moduli space which is a union of a $\mathbb{P}^1$ and two affine lines attached to the points 0 and ∞ of $\mathbb{P}^1$*

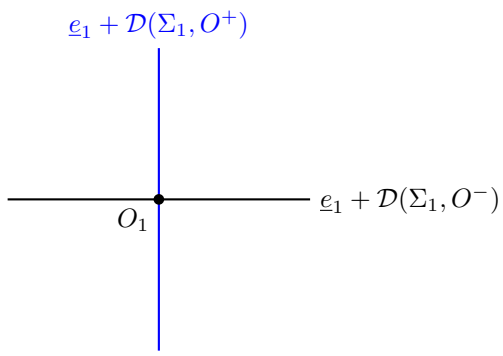


, (3.5)

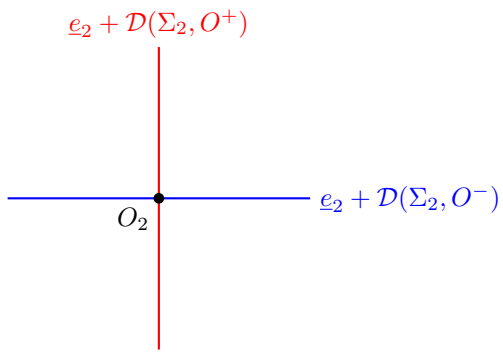
where:

- *The points that are black, different from O_1, represent line bundles of multidegree $\underline{e}_1 + \mathcal{D}(\Sigma_1, O^-)$,*
- *The points that are blue, different from O_1 and O_2, represent line bundles of multidegree $\underline{e}_1 + \mathcal{D}(\Sigma_1, O^+)$,*
- *The points that are red, different from O_2, represent line bundles of multidegree $\underline{e}_2 + \mathcal{D}(\Sigma_2, O^+)$,*
- *The point O_1 represents all the sheaves in $\mathcal{U}(N_{\Sigma_1}^{\underline{e}_1})$ different from the line bundles of multidegree $\underline{e}_1 + \mathcal{D}(\Sigma_1, O^-)$ and $\underline{e}_1 + \mathcal{D}(\Sigma_1, O^+)$,*
- *The point O_2 represents all the sheaves in $\mathcal{U}(N_{\Sigma_1}^{\underline{e}_1})$ different from the line bundles of multidegree $\underline{e}_2 + \mathcal{D}(\Sigma_2, O^-)$ and $\underline{e}_2 + \mathcal{D}(\Sigma_2, O^+)$.*

Proof. From Proposition 3.5, we already know that the moduli space $U(N_{\Sigma_1}^{\underline{e}_1})$ is



where all the sheaves of $\mathcal{U}(N_{\Sigma_1}^{\underline{e}_1})$ different from the line bundles of multidegree $\underline{e}_1 + \mathcal{D}(\Sigma_1, O^+)$ and $\underline{e}_1 + \mathcal{D}(\Sigma_1, O^-)$ get identified and are represented by the point O_1.
Analogously, the moduli space $U(N_{\Sigma_2}^{\underline{e}_2})$ is



where all the sheaves of $\mathcal{U}(N_{\Sigma_2}^{\underline{e}_2})$ different from the line bundles of multidegree $\underline{e}_2 + \mathcal{D}(\Sigma_2, O^+)$ and $\underline{e}_2 + \mathcal{D}(\Sigma_2, O^-)$ get identified and are represented by the point O_2.
From the hypothesis that the opens $\mathcal{U}(N_{\Sigma_1}^{\underline{e}_1})$ and $\mathcal{U}(N_{\Sigma_2}^{\underline{e}_2})$ are consecutive we get that

$$\underline{e}_1 + \mathcal{D}(\Sigma_1, O^+) = \underline{e}_2 + \mathcal{D}(\Sigma_2, O^-).$$

Define

$$\underline{d} \coloneqq \underline{e}_1 + \mathcal{D}(\Sigma_1, O^+).$$

By definition N_1 is of type 1 for $\underline{d}$ and N_2 is of type 2 for $\underline{d}$. This means that N_1 is in the image of the morphism $\theta_{O^+}^{\underline{d}}$ and N_2 is in the image of the morphism $\theta_{O^-}^{\underline{d}}$. Namely N_1 is the origin of $k[x_\Sigma]$ and N_2 is the origin of $k[y_\Sigma]$ and, as explained in the proof if Proposition 3.5, we have the identification $x_\Sigma \to (y_\Sigma)^{-1}$ which tells us how the two moduli spaces $U(N_{\Sigma_1}^{\underline{e}_1})$ and $U(N_{\Sigma_2}^{\underline{e}_2})$ are glued together and we get the moduli space (3.5) of the statement. □

Remark 3.13. If in Lemma 3.12 we assume that $\Sigma_1 \cup \Sigma_2$ is a partition of Σ, we have

$$\mathcal{D}(\Sigma, O^+) = \mathcal{D}(\Sigma, O^-) = (1, \dots, 1),$$

hence we can write

$$\mathcal{D}(\Sigma_1, O^+) + \mathcal{D}(\Sigma_2, O^+) = \mathcal{D}(\Sigma_1, O^-) + \mathcal{D}(\Sigma_2, O^-),$$

which implies

$$\mathcal{D}(\Sigma_2, O^+) - \mathcal{D}(\Sigma_2, O^-) = \mathcal{D}(\Sigma_1, O^-) - \mathcal{D}(\Sigma_1, O^+).$$

Since by assumption we have

$$\underline{e}_1 + \mathcal{D}(\Sigma_1, O^+) = \underline{e}_2 + \mathcal{D}(\Sigma_2, O^-),$$

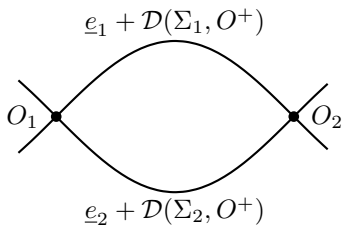
we get

$$\underline{e}_1 + \mathcal{D}(\Sigma_1, O^-) = \underline{e}_2 + \mathcal{D}(\Sigma_2, O^+).$$

This means that if we define

$$\underline{d} \coloneqq \underline{e}_1 + \mathcal{D}(\Sigma_1, O^-) = \underline{e}_2 + \mathcal{D}(\Sigma_2, O^+),$$

the sheaf N_1 is of type 2 for $\underline{d}$ and N_2 is of type 1 for $\underline{d}$. Namely N_1 is in the image of the morphism $\theta_{O^-}^{\underline{d}}$ and N_2 is in the image of the morphism $\theta_{O^+}^{\underline{d}}$. Therefore N_2 is the origin of $k[x_\Sigma]$ and N_1 is the origin of $k[y_\Sigma]$ and, as explained in the proof if Proposition 3.5, we have the identification $x_\Sigma \to (y_\Sigma)^{-1}$.
This implies that the good moduli space of the open $\mathcal{U}(N_{\Sigma_1}^{\underline{e}_1}) \cup \mathcal{U}(N_{\Sigma_2}^{\underline{e}_2})$ is a necklace of length 2 and in particular it is proper.



Remark 3.14. We want to introduce now the definition of cyclic opens with the idea in mind that when we consider m cyclic opens $\mathcal{U}_1, \dots, \mathcal{U}_m$ we will find a *proper* good moduli space.
We first introduce the formal definition and then we will prove in Proposition 3.17 that this is the right definition to determine when a union of opens $\mathcal{U}_i$'s admits a proper good moduli space.

Definition 3.15.
The opens $\mathcal{U}_1 \coloneqq \mathcal{U}(N_{\Sigma_1}^{\underline{e}_1}), \dots, \mathcal{U}_m \coloneqq (N_{\Sigma_m}^{\underline{e}_m})$ are *cyclic* if

- $\mathcal{U}_i$ and $\mathcal{U}_{i+1}$ are consecutive for every $i \in \mathbb{Z}/m\mathbb{Z}$,
- $\mathcal{U}_i \cap \mathcal{U}_j = \emptyset$ for every i, j that are not consecutive in $\mathbb{Z}/m\mathbb{Z}$.

Remark 3.16. Two consecutive opens as in Remark 3.13 are cyclic.

Proposition 3.17. *Let X be a necklace of genus $g = 1$.*
An open connected substack $\mathcal{U}$ of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ admits a proper good moduli space if and only if there exist m cyclic opens $\mathcal{U}(N_{\Sigma_1}^{\underline{e}_1}), \dots, \mathcal{U}(N_{\Sigma_m}^{\underline{e}_m})$ such that

$$\mathcal{U} = \mathcal{U}(N_{\Sigma_1}^{\underline{e}_1}) \cup \dots \cup \mathcal{U}(N_{\Sigma_m}^{\underline{e}_m}).$$

The proper good moduli space is a necklace of length m.

Proof. Let us first show that if

$$\mathcal{U}(N_1) \coloneqq \mathcal{U}(N_{\Sigma_1}^{\underline{e}_1}), \dots, \mathcal{U}(N_m) \coloneqq \mathcal{U}(N_{\Sigma_m}^{\underline{e}_m})$$

are m cyclic opens, then the union

$$\mathcal{U} \coloneqq \mathcal{U}(N_1) \cup \dots \cup \mathcal{U}(N_m)$$

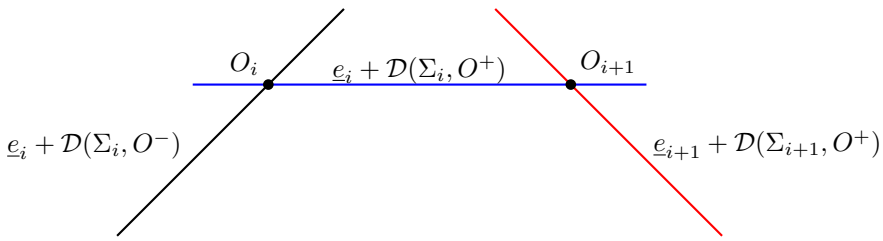
is an open connected substack of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ which admits a proper good moduli space.
We observe that $\mathcal{U}$ is open because it is given by the union of the opens $\mathcal{U}(N_i)$'s and it is connected because the opens $\mathcal{U}(N_i)$'s are consecutive by assumption.
If $m = 1$, the assumption that the $\mathcal{U}(N_i)$'s are cyclic implies that

$$\underline{e}_1 + \mathcal{D}(\Sigma_1, O^+) = \underline{e}_1 + \mathcal{D}(\Sigma_1, O^-)$$

which means that $\mathcal{D}(\Sigma_1, O^+) = \mathcal{D}(\Sigma_1, O^-)$ and then $\Sigma_1 = \Sigma$. Namely the sheaf N_1 is not locally free at all the nodes. In this case

$$\mathcal{U} = \mathcal{U}(N_{\Sigma}^{\underline{e}_1})$$

and, as we have seen in Proposition 3.5, $\mathcal{U}(N_{\Sigma}^{\underline{e}_1})$ admits a proper good moduli space which is a nodal $\mathbb{P}^1$.
If $m > 1$ all the sheaves N_i's fail to be locally free at some proper subset of Σ and, for every $i = 1, \dots, m$ the good moduli space $U(N_i)$ is a nodal curve consisting of two copies of $\mathbb{A}^1$ meeting at the origin, as we have seen in Proposition 3.5. Moreover, from Lemma 3.12, we know that the moduli space of the union $\mathcal{U}(N_i) \cup \mathcal{U}(N_{i+1})$ of two consecutive open is



Since by assumption the opens $U(N_1), \dots, U(N_m)$ are cyclic, also $U(N_1)$ and $U(N_m)$ are consecutive. This implies that, iterating the construction of Lemma 3.12, the moduli space of the open $\mathcal{U}(N_1) \cup \dots \cup \mathcal{U}(N_m)$ is a closed chain of m copies of $\mathbb{P}^1$, namely a necklace of length m. We conclude that $\mathcal{U}$ is an open connected substack of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ which has a proper good moduli space which is a necklace of length m.
Assume now that $\mathcal{U}$ is an open connected substack of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ that has a proper good moduli space. If N is a sheaf in $\mathcal{U}$ which is not a line bundle, then $\mathcal{U}(N) \subseteq \mathcal{U}$ because $\mathcal{U}$ is open and all the points of $\mathcal{U}(N)$ specialize to N. Consider the set $\mathcal{S}$ of the most degenerate sheaves in $\mathcal{U}$, namely

$$\mathcal{S} \coloneqq \{N \in \mathcal{U} \mid \nexists N' \in \mathcal{U} \text{ such that } \mathcal{U}(N) \subseteq \mathcal{U}(N')\}.$$

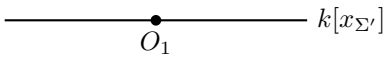
If $N \in \mathcal{U}$ and it is not a line bundle, there exists a sheaf $N' \in \mathcal{S}$ such that $\mathcal{U}(N) \subseteq \mathcal{U}(N')$, because of the definition of $\mathcal{S}$. This implies that $\mathcal{U}$ is given by the union taken over all the sheaves N of $\mathcal{S}$ of the opens $\mathcal{U}(N)$. We claim that the set

$$\mathcal{U}(\mathcal{S}) := \{\mathcal{U}(N) |\ N \in \mathcal{S}\}$$

can be ordered in a cyclic way.
We want to understand when two sheaves can be both in $\mathcal{S}$.
Let $N_1 := N^{\underline{e}_1}_{\Sigma_1}$ and $N_2 := N^{\underline{e}_2}_{\Sigma_2}$ be two sheaves in $\mathcal{S}$. Let us assume that $\mathcal{U}(N_1) \cap \mathcal{U}(N_2) \neq \emptyset$, equivalently both these sheaves can be obtained as degenerations of line bundles of some multidegree $\underline{d}$. We claim that N_1 and N_2 have to be one of type 1 and the other of type 2 for the multidegree $\underline{d}$.
To prove the claim we start by showing that neither N_1 nor N_2 can be of type 3 for the multidegree $\underline{d}$: If N_1 is of type 3 for the multidegree $\underline{d}$, we would have that N_1 and the line bundles of multidegree $\underline{d}$ get identified in the quotient in a unique point. Since the line bundles of multidegree $\underline{d}$ get contracted in the moduli space, we must have that also N_2 is of type 3 for the multidegree $\underline{d}$. This implies that the sheaf N_2 and the line bundles of multidegree $\underline{d}$ get identified in a unique point in the moduli space. Therefore the sheaves N_1 and N_2 would be identified in the quotient, but by assumption the sheaves N_1 and N_2 are in $\mathcal{S}$ and so, in order to have a good quotient, they cannot be identified since they do not have a common specialization. We conclude that the sheaves N_1 and N_2 have to be either of type 1 or of type 2 for $\underline{d}$.
We show now that the sheaves N_1 and N_2 cannot be of the same type for $\underline{d}$: Assume by contradiction that N_1 and N_2 are of the same type for $\underline{d}$. Without loss of generality we can assume that these sheaves are both of type 1 for $\underline{d}$ and hence we have

$$\underline{d} = \underline{e}_1 + \mathcal{D}(\Sigma_1, O^+) = \underline{e}_2 + \mathcal{D}(\Sigma_2, O^+).$$

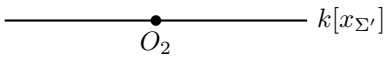
The sheaves N_1 and N_2 are both in the image of the morphism $\theta^{\underline{d}}_{O^+}$, therefore in $U(N_1)$ there is a copy of $\mathbb{A}^1$



where

- The points outside the origin represent line bundles of multidegree $\underline{d}$,
- All the sheaves of $\mathcal{U}(N_1)$ different from line bundles of multidegree $\underline{d}$ and $\underline{d} - \mathcal{D}(\Sigma_1, O^+) + \mathcal{D}(\Sigma_1, O^-)$ get identified to the node O_1.

Similarly, in $U(N_2)$ there is a copy of $\mathbb{A}^1$



where

- The points outside the origin represent line bundles of multidegree $\underline{d}$,
- All the sheaves of $\mathcal{U}(N_2)$ different from line bundles of multidegree $\underline{d}$ and $\underline{d} - \mathcal{D}(\Sigma_2, O^+) + \mathcal{D}(\Sigma_2, O^-)$ get identified to the node O_2.

There are two possibilities:

1. There are some sheaves in $\mathcal{U}(N_1) \cap \mathcal{U}(N_2)$ that get identified to both O_1 and O_2. This implies that we cannot have a good quotient because we are assuming that N_1 and N_2 are both in $\mathcal{S}$ and so they do not have a common specialization by assumption.
2. The are no sheaves in $\mathcal{U}(N_1) \cap \mathcal{U}(N_2)$ that get identified to both O_1 and O_2. This implies that we get a copy of $\mathbb{A}^1$ with double origin in the moduli space, which is not separated.

We conclude that N_1 and N_2 have to be of different type for $\underline{d}$, namely if $N_1, N_2 \in \mathcal{S}$ and $\mathcal{U}(N_1) \cap \mathcal{U}(N_2) \neq \emptyset$, there exists a multidegree $\underline{d}$ such that N_1 and N_2 are one of type 1 and one of type 2 for $\underline{d}$.
Assume now that a sheaf $N_{\Sigma_1}^{\underline{e}_1}$, which is of type 1 for a multidegree $\underline{d}$, is in $\mathcal{S}$. If $\Sigma_1 = \Sigma$, from Proposition 3.5, we know that we get a nodal $\mathbb{P}^1$ as good moduli space which is proper and in particular we must have that $|\mathcal{S}| = 1$.
If $\Sigma_1 \subsetneq \Sigma$, the good moduli space of $\mathcal{U}(N)$ is, as we have seen in Proposition 3.5, a nodal curve where one of the two copies of $\mathbb{G}_m$ that we get removing the node of $U(N)$ represents line bundles of multidegree $\underline{d}$. In order to get a proper quotient also a sheaf N_2 of type 2 for $\underline{d}$ has to appear in $\mathcal{S}$: In this case in the moduli space we get

$$O_1 \cup \mathbb{G}_m \cup O_2 = \mathbb{P}^1,$$

where O_i is the node of $U(N_i)$.
Now we explain how we can order the set $\mathcal{U}(\mathcal{S})$ in a cyclic way: Pick a sheaf in $\mathcal{S}$ and call this N_1, this is of type 1 for a multidegree $\underline{d}_1$. We know that in $\mathcal{S}$ there is a sheaf of type 2 for $\underline{d}_1$, we call this sheaf N_2. The sheaf N_2 is of type 1 for some multidegree $\underline{d}_2$, then there is a sheaf in $\mathcal{S}$ of type 2 for $\underline{d}_2$, we call this sheaf N_3 and so on. This process has to be finite since we are assuming that the quotient is proper. The last sheaf N_m of $\mathcal{S}$ that we pick is of type 1 for some multidegree $\underline{d}_m$. Because of the discussion above, there must be a sheaf of type 2 for $\underline{d}_m$. The only possibility is that this sheaf is N_1 because we know that every sheaf N_i is of type 2 for $\underline{d}_{i-1}$, for $i \geq 2$ and every sheaf is of type 2 for a unique multidegree. We conclude that $\mathcal{U}(N_1)$ and $\mathcal{U}(N_m)$ are consecutive and hence we can order $\mathcal{U}(\mathcal{S})$ in a cyclic way. □

3.2 Combinatorial description for necklaces of genus 1

In this Section we provide a combinatorial description of good compactified Jacobians of a necklace X of genus $g = 1$, which extends the one given by Pagani and Tommasi in [2, Lemma 3.7] for fine compactified Jacobians.

Let $\mathcal{U}$ be an open connected substack of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ which admits a proper good moduli space. From Proposition 3.17, we have that

$$\mathcal{U} = \mathcal{U}(N_{\Sigma_1}^{\underline{e}_1}) \cup \cdots \cup \mathcal{U}(N_{\Sigma_m}^{\underline{e}_m}),$$

where

- $\{N_{\Sigma_1}^{\underline{e}_1}, \ldots, N_{\Sigma_m}^{\underline{e}_m}\}$ is the set of the most degenerate sheaves of $\mathcal{U}$,
- $\mathcal{U}(N_{\Sigma_i}^{\underline{e}_i})$ and $\mathcal{U}(N_{\Sigma_{i+1}}^{\underline{e}_{i+1}})$ are consecutive for every $i \in \mathbb{Z}/m\mathbb{Z}$.

Starting from $\mathcal{U} = \mathcal{U}(N_{\Sigma_1}^{\underline{e}_1}) \cup \cdots \cup \mathcal{U}(N_{\Sigma_m}^{\underline{e}_m})$ we can define a pair

$$(\underline{d}, \underline{\Sigma}) \coloneqq (\underline{d}, \Sigma_1, \ldots, \Sigma_m),$$

where $\underline{d} \coloneqq \underline{e}_1 + \mathcal{D}(\Sigma_1, O^-)$.

Definition 3.18.
Let $(\underline{d}, \underline{\Sigma}) = (\underline{d}, \Sigma_1, \ldots, \Sigma_m)$ be a vector, where $\underline{d} \in \mathbb{Z}^n$ and $\Sigma_i \subseteq \Sigma$, for every $i = 1, \ldots, m$. We define the open $\mathcal{U}(\underline{d}, \underline{\Sigma})$ in $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ as

$$\mathcal{U}(\underline{d}, \underline{\Sigma}) \coloneqq \mathcal{U}(N_{\Sigma_1}^{\underline{e}_1}) \cup \cdots \cup \mathcal{U}(N_{\Sigma_m}^{\underline{e}_m}),$$

where

- $\underline{e}_1 \coloneqq \underline{d} - \mathcal{D}(\Sigma_1, O^-)$,
- $\underline{e}_{i+1} \coloneqq \underline{e}_i + \mathcal{D}(\Sigma_i, O^+) - \mathcal{D}(\Sigma_{i+1}, O^-)$ for every $i = 1, \ldots, m-1$,

Remark 3.19. Using the notation

$$\mathcal{D}'(\Sigma_i, O^+) = \mathcal{D}(\Sigma_i, O^+) - \mathcal{D}(\Sigma_i, O^-),$$

for every $i = 1, \dots, m$, we can write

$$\underline{e}_i = \underline{e}_1 + \mathcal{D}'(\Sigma_1, O^+) + \mathcal{D}'(\Sigma_2, O^+) \cdots + \mathcal{D}'(\Sigma_{i-1}, O^+) - \mathcal{D}(\Sigma_i, O^-). \tag{3.6}$$

The opens $\mathcal{U}(N^{\underline{e}_i}_{\Sigma_i})$ and $\mathcal{U}(N^{\underline{e}_{i+1}}_{\Sigma_{i+1}})$ for every $i = 1, \dots, m-1$ are consecutive by construction but in general the opens $\mathcal{U}(N^{\underline{e}_1}_{\Sigma_1}), \dots, \mathcal{U}(N^{\underline{e}_m}_{\Sigma_m})$ are not cyclic and therefore the open $\mathcal{U}(\underline{d}, \underline{\Sigma})$ does not have a proper good moduli space. There are two obstructions that may occour:

- Two opens $\mathcal{U}(N^{\underline{e}_i}_{\Sigma_i})$ and $\mathcal{U}(N^{\underline{e}_j}_{\Sigma_j})$ with i and j non consecutive in $\mathbb{Z}/m\mathbb{Z}$ are not disjoint,
- $\mathcal{U}(N^{\underline{e}_1}_{\Sigma_1})$ and $\mathcal{U}(N^{\underline{e}_m}_{\Sigma_m})$ are not consecutive.

We now introduce the definition of *r-admissible sequences*, which are sequences

$$\underline{\Sigma} = (\Sigma_1, \dots, \Sigma_m),$$

where $\Sigma_i \subseteq \Sigma$, for every $i = 1, \dots, m$.
These sequences must satisfy two conditions, and we will prove in Theorem 3.28, that these conditions ensure that when we consider the open $\mathcal{U}(\underline{d}, \underline{\Sigma})$ in $\underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X)$ for some multidegree $\underline{d}$, the two aforementioned obstructions do not occour.

Definition 3.20.
A sequence $\underline{\Sigma} := (\Sigma_i : i \in \mathbb{Z}/m\mathbb{Z})$ of m subsets of $\{1, \dots, n\}$ is *r-admissible* if the two following conditions are satisfied:

1. $\#\{l \colon t \in \Sigma_l\} = r$, for every $t = 1, \dots, n$;
2. For any sequence $\underline{\sigma} := (\sigma_j : j \in \mathbb{Z}/rn\mathbb{Z})$ obtained from $\underline{\Sigma}$ by choosing an ordering of the elements in each Σ_i, the numbers $\{1, \dots, n\}$ do not appear all and with the same multiplicity $r' \in \{1, \dots, r-1\}$ in $r'n$ consecutive positions of $\underline{\sigma}$.

Moreover, we say that an r-admissible sequence is *simple* if $|\Sigma_i| = 1$ for every $i \in \mathbb{Z}/m\mathbb{Z}$.
We refer to r as the *multiplicity* of the r-admissible sequence and to m as the *length* of the r-admissible sequence.

Remark 3.21. If $(\Sigma_1, \dots, \Sigma_m)$ is a 1-admissible sequence, the second condition of Definition 3.20 is trivially satisfied because there is not a natural number r' that satisfies $1 \leq r' < 1$. Moreover the set $\{\Sigma_1, \dots, \Sigma_m\}$ is a partition of the set of nodes Σ.

Definition 3.22.
Let $\underline{d}$ be a vector in $\mathbb{Z}^n$. A pair $(\underline{d}, \underline{\Sigma})$ is (simple) *r-admissible* if $\underline{\Sigma}$ is a (simple) r-admissible sequence.

The aim of the Chapter is to construct open connected substacks of $\underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X)$ which admit proper good moduli space starting from a pair $(\underline{d}, \underline{\Sigma})$. We introduce now an equivalence relation $\sim$ on the set of r-admissible pairs and we will show in Theorem 3.28 that, given two admissible pairs $(\underline{d}, \underline{\Sigma})$ and $(\underline{d}', \underline{\Sigma}')$, we have the equality

$$\mathcal{U}(\underline{d}, \underline{\Sigma}) = \mathcal{U}(\underline{d}', \underline{\Sigma}')$$

if and only if $(\underline{d}, \underline{\Sigma}) \sim (\underline{d}', \underline{\Sigma}')$.

Definition 3.23.
On the set of r-admissible pairs, we define an equivalence relation $\sim$, by saying that:

$$(\underline{d}, \underline{\Sigma}) \sim (\underline{d}', \underline{\Sigma}')$$

if and only if there exists a natural number k such that

- $\underline{d}' = \underline{d} + [\mathcal{D}(\Sigma_1, O^+) - \mathcal{D}(\Sigma_1, O^-)] + \cdots + [\mathcal{D}(\Sigma_k, O^+) - \mathcal{D}(\Sigma_k, O^-)]$,

- $\underline{\Sigma}' = \tau^k \underline{\Sigma}$,

where τ^k is the cyclic permutation that sends i to $i+k$ for every $i = 1, \dots, n$.

Example 3.24. (Examples and non-examples of r-admissible sequences.)

1. The sequence

$$(\{1, 2, \dots, n\})$$

is 1-admissibile because every number from 1 to n appears once and, since it is 1-admissible, as we explained in Remark 3.21, the second condition of Definition 3.20 is satisfied.

2. The sequence

$$(\{1\}, \dots, \{1\}, \dots, \{n\}, \dots, \{n\})$$

where every natural number from 1 to n appears r times in a row is r-admissible if $n \geq 3$. If $n = 2$ and $r > 1$ the above sequence is not r-admissible because in this case we have

$$(\{1\}, \dots, \{1\}, \{2\}, \dots \{2\})$$

and therefore the numbers 1 and 2 appears in two consecutive position of the sequence and this is not compatible with the second condition of Definition 3.20.

3. The sequence

$$(\{1\}, \{1, 2\}, \{2\}, \{2, 3\}, \{3\}, \{1, 3\})$$

is not 3-admissible: The second condition of Definition 3.20 is not satisfied because, by choosing the ordering $\{2\}, \{1\}$ for the second entry of the sequence, the ordering $\{3\}, \{2\}$ for the fourth entry and the ordering $\{1\}, \{3\}$ for the last entry of the sequence, we get

$$(\{1\}, \{2\}, \{1\}, \{2\}, \{3\}, \{2\}, \{3\}, \{1\}, \{3\})$$

where the numbers $1, 2$ and 3 appears in 3 consecutive entries of the sequence.

Lemma 3.25. *If $n = 2$, there are r-admissible sequences only for $r = 1$.*
If $n \geq 3$, there are r-admissible sequences for every natural number r.

Proof. If $n = 2$, we cannot have a sequence where the numbers 1 and 2 both appear $r > 1$ times without appearing in 2 consecutive positions. Therefore r has to be 1 and the only 1-admissible sequences are $(\{1, 2\})$, $(\{1\}, \{2\})$ and $(\{2\}, \{1\})$.
If $n \geq 3$, for every natural number r, we can consider the r-admissible sequence

$$(\{1\}, \dots, \{1\}, \dots, \{n\}, \dots, \{n\})$$

where every natural number from 1 to n appears r times in a row. □

Lemma 3.26. *Let $(\Sigma_1, \dots, \Sigma_m)$ be an r-admissible sequence of length m, where $r \geq 2$ and $\Sigma_i \subseteq \{1, \dots, n\}$, for every $i = 1, \dots, m$. Then*

$$\frac{n \cdot r}{n-1} \leq m \leq n \cdot r.$$

Proof. By assumption the sequence $(\Sigma_1, \dots, \Sigma_m)$ is r-admissible, namely all the numbers from 1 to n appear in precisely r entries of $(\Sigma_1, \dots, \Sigma_m)$. This implies that

$$|\Sigma_1| + \dots + |\Sigma_m| = n \cdot r.$$

Since $|\Sigma_i| \geq 1$ for every $i = 1, \dots, m$, we get the inequality

$$m \leq |\Sigma_1| + \cdots + |\Sigma_m| = n \cdot r.$$

From the second condition of Definition 3.20, we have that, if $r \geq 2$, every Σ_i must have cardinality smaller than or equal to $n-1$, because all the numbers from 1 to n cannot appear in the same entry of $(\Sigma_1, \dots, \Sigma_m)$. This implies that also the inequality

$$\frac{n \cdot r}{n-1} \leq m \text{ holds.}$$

□

Lemma 3.27. *If $n = 2$, the r-admissible sequences have length $m \in \{1, 2\}$.*
If $n = 3$, there are r-admissible sequences of length m for every positive natural number $m \notin \{4, 5, 7\}$.
If $n = 4$, there are r-admissible sequences of length m for every positive natural number $m \neq 5$.
If $n \geq 5$, there are r-admissible sequences of length m for every positive natural number m.

Proof. If there is an r-admissible sequence $(\Sigma_1, \dots, \Sigma_m)$ of length m then any sequence obtained by substituting any Σ_i by a partition of Σ_i is still an r-admissible sequence. This implies that, if there is an r-admissible sequence of length m, then there are r-admissible sequences of any length between m and $|\Sigma_1| + \cdots + |\Sigma_m|$. In particular, since for every positive natural number n the sequence $(\{1, \dots, n\})$ is 1-admissible, for every natural number n we have 1-admissible sequences of every length between 1 and n.
Now we analyze in more details what happens for every natural number $n \geq 2$.

- If $n = 2$, as we have seen in Lemma 3.25, there are only 1-admissible sequences and so the length m has to be either 1 or 2.
- When $n \geq 3$ for every $r \geq 2$ we consider the sequence

$$(\{1\}, \dots, \{1\}, \{1, 2\}, \{2\}, \dots, \{2\}, \dots, \{n-1, n\}, \{n\}, \dots \{n\}, \{n, 1\}), \tag{3.7}$$

  where the first $r-2$ positions are the singleton $\{1\}$, the position $r-1$ is the set $\{1, 2\}$, the next $r-2$ positions are the singleton $\{2\}$, then we have the set $\{2, 3\}$ and so on, so this sequence has length $n \cdot (r-1)$.
- If $n = 3$, the sequences of the form (3.7) are r-admissible for every $r \geq 4$:
  - If $r = 2$, we have the sequence

$$(\{1, 2\}, \{2, 3\}, \{1, 3\})$$

    which is not 2-admissible because from this sequence, by choosing an ordering of every entry, we can get the sequence

$$(\{2\}, \{1\}, \{2\}, \{3\}, \{1\}, \{3\})$$

    where the numbers $1, 2$ and 3 appear in 3 consecutive entries, therefore condition 2 of Definition 3.20 is not satisfied.
  - If $r = 3$, we have the sequence

$$(\{1\}, \{1, 2\}, \{2\}, \{2, 3\}, \{3\}, \{1, 3\})$$

    which is not 3-admissible as explained in Example 3.24.
  - If $r \geq 4$, we have the sequence

$$(\{1\}, \dots, \{1\}, \{1, 2\}, \{2\}, \dots, \{2\}, \{2, 3\}, \{3\}, \dots, \{3\}, \{1, 3\})$$

    which has length $m = 3(r-1)$ and is r-admissible:
    1. The first condition of Definition 3.20 is satisfied because all numbers $1, 2, 3$ appear r times.

2. To prove the second condition of Definition 3.20, we consider any sequence $\underline{\sigma}$ obtained by choosing an ordering of the entries $\{1,2\},\{2,3\}$ and $\{1,3\}$ and we prove that we cannot find $3r'$ consecuitve entries where all numbers $1,2,3$ appear with the same multiplicity $r' < r$. Since $r \geq 4$ we cannot find the numbers $1,2$ and 3 in 3 consecutive positions of some sequence $\underline{\sigma}$: If we take two consecutive entries where 1 and 2 appear there is at least one entry different from the number 3 both in the entry immediately before and in the entry immediately after the entries with the numbers 1 and 2 that we have chosen. Therefore to find $r'n$ consecutive entries where the numbers $1,2$ and 3 appear with the same multiplicity r', we must have $r' > 1$. The best attempt that we can do to find $3r'$ consecutive entries of the sequence where all numbers $1,2,3$ appear with multiplicity r' consists in taking the $(r-2)$-th, $(r-1)$-th, r-th and $(r+1)$-th entry, where both the number 1 and the number 2 appear two times. But now we have to choose other consecutive entries in order to have also entries where the number 3. This implies in particular that we have to take all entries from the first to the $(2r-2)$-th entry as well and yet all the entries are different from 3 and the numbers 1 and 2 appear with multiplicity $r-1$. This implies that now we are looking for a sequence of multiplicity $r-1$ and in particular that we want to have also $r-1$ entries equal to 3. Therefore we also have to consider the $(3r)$-th and $(2r-1)$-th entry and the number 3 appears at most in two entries (for the sequence $(\underline{\sigma})$ where the order of the set $\{2,3\}$ is $\{2\},\{3\}$ and the order of the set $\{1,3\}$ is $\{1\},\{3\}$). At this moment we are considering the entries from position 1 to position $2r-1$ and the entry $3r$. But we are assuming $r \geq 4$ which implies that we need at least another entry equal to 3. So we need to keep on taking consecuitve entries and either we choose the $(3r-1)$-th entry which is 2 or the $(2r-1)$-th entry which is 1, but then in the first case there are r entries equal to 2 and in the second case there are r entries equal to 1 which implies that we cannot find $3r'$ consecutive entries of the sequence where $1,2,3$ appear all with multiplicity $r' < r$.

These sequences have length $m = 3(r-1)$. Therefore there are r-admissible sequences of any length between $3(r-1)$ and $3r$, for every $r \geq 4$. This implies that there are r-admissible sequences of any length $m \geq (r-1)\cdot 3 = (4-1)\cdot 3 = 9$ and we know that there are r-admissible sequences of length $1 \leq m \leq 3$. We have to understand whether there are r-admissible sequences of length $4 \leq m \leq 8$. We claim that the length m can be 6 and 8 but cannot be $4,5$ and 7.
The length $m = 6$ is possible because we have the simple 2-admissible sequence

$$(\{1\},\{1\},\{2\},\{2\},\{3\},\{3\})$$

and the length $m = 8$ is possible as well because we have the 3-admissible sequence

$$(\{1\},\{1\},\{1,2\},\{2\},\{2\},\{3\},\{3\},\{3\}).$$

If we want to find r-admissible sequences of length $m \leq 7$, the multiplicity r has to be smaller than 5: If $r \geq 5$ then, from Lemma 3.26, $m \geq (n\cdot r)/2 \geq (3\cdot 5)/2 > 7$.
We show now that we cannot find r-admissible sequences of length $4,5$ and 7 with $r = 4$.
If there is a 4-admissible sequence of length m, then $|\Sigma_1|+\cdots+|\Sigma_m| = r\cdot n = 4\cdot 3 = 12$, and for every i we have $|\Sigma_i| \leq 2$, namley we must be able to write 12 as the sum of m numbers smaller than or equal to 2. Therefore to get 4-admissible sequences of either length 4 or 5 we should be able to write 12 respectively as a sum of 4 and 5 natural numbers smaller than or equal to 2 and this is not possible.
If there is a 4-admissible sequence of length 7, we should be able to write 12 as the sum of 7 numbers smaller than or equal to 2. The only possibility is to write

$$12 = 2 + 2 + 2 + 2 + 2 + 1 + 1.$$

This means that there must be at least 3 consecutive entries of $\underline{\Sigma}$ which are subsets of $\{1, 2, 3\}$ and have cardinality 2. Without loss of generality we can assume that these three consecutive entries are $\Sigma_1, \Sigma_2, \Sigma_3$ and we assume that $\Sigma_1 = \{1, 2\}$. This implies that $\Sigma_2 = \Sigma_3 = \{1, 2\}$ because, for condition 2 of Definition 3.20, the numbers $1, 2$ and 3 cannot appear all in two consecutive entries of the sequence. The number 3, which has to appear in four entries of the vector $\underline{\Sigma}$, is not an element of Σ_1, Σ_2 and Σ_3 and cannot be an element of Σ_4 and Σ_7 because otherwise we could find all the numbers from 1 to 3 in two consecutive entries of $\underline{\Sigma}$ and this is not allowed by the second condition of Definition 3.20. This implies that 3 could be an element only of at most two entries of $\underline{\Sigma}$ and hence we cannot have 4-admissible sequences of length 7. Considering $r = 3$, in order to find 3-admissible sequences of length 7 we have to see how we can write 9 as a sum of 7 numbers smaller than or equal to 2. The only possiblity is

$$9 = 2 + 2 + 1 + 1 + 1 + 1 + 1,$$

which means that two entries of the sequence we are looking for must be a set with two elements and all the other entries are singletons. We distinguish between two cases:

- The two subsets with two elements are equal, namely without loss of generality we assume they are the set $\{1, 2\}$,
- The two subsets with two elements are different, namely without loss of generality we assume they are the set $\{1, 2\}$ and the set $\{2, 3\}$.

In the first case if we assume that the entries with the subset $\{1, 2\}$ are consecutive then the singletons 3 has to appear in the only three entries which are not consecutive to the entries with $\{1, 2\}$. Namely, without loss of generality, we have

$$(\{1, 2\}, \{1, 2\}, ?, \{3\}, \{3\}, \{3\}, ?).$$

But at this point neither the singleton $\{1\}$ nor the singleton $\{2\}$ can be the third entry of the sequence because in both cases we would find all numbers $1, 2, 3$ from the second to the fourth entry and this does not respect condition 2 of Definition 3.20.
If the entries with the subset $\{1, 2\}$ are not consecutive than the singleton $\{3\}$ cannot be neither the entry immediately before nor the entry immediately after the entries with $\{1, 2\}$. This implies that there are at most two entries of a sequence with 7 entries equal to $\{3\}$. Therefore we cannot find a 3-admissible sequence of length 7 where two entries are subsets with 2 elements and are equal.
We consider now the case with two different subsets of two elements. We assume these subsets are $\{1, 2\}$, $\{2, 3\}$. We observe that this two subsets cannot be two consecutive entries of the sequence because otherwise all numbers $1, 2, 3$ would appear in two consecutive entries of the sequence. We assume that the first entry is $\{1, 2\}$ and that there is precisely one entry which is a singleton between the first entry and the entry with $\{2, 3\}$, namely we assume

$$(\{1, 2\}, ?, \{2, 3\}, ?, ?, ?, ?)$$

But in this case any singleton can be the second entry, because in all cases we would find all numbers $1, 2, 3$ in the first three entries of the sequence. We would arrive at the same conclusion if we assume that the entry $\{2, 3\}$ is the sixth one, in that case, we would find that any singleton can be the seventh entry.
We assume therefore that the set $\{2, 3\}$ is the fourth entry, namely

$$(\{1, 2\}, ?, ?, \{2, 3\}, ?, ?, ?)$$

In this case we observe immediately that the second entry cannot be the singleton $\{3\}$. If the second entry is the singleton $\{2\}$ then the third entry must be either $\{1\}$ or $\{3\}$ but in both cases we find all numbers $1, 2, 3$ in the position from the second to the fourth one. So in this case we cannot fined a 3-admissible sequence. similarly if we assume that the second entry is $\{1\}$ then the third entry has to be either $\{2\}$ or $\{3\}$ but in both cases we find all numbers $1, 2, 3$ in the position from the second to the fourth one. We would arrive at the same conclusion if we assume that the entry $\{2, 3\}$ is the fifth one, in that case, we can repeat what we have done before looking at the sicth and seventh entry. We conclude that there are not 3-admissible sequences of length 7 and therefore it is also not possible to have 3-admissible sequences of length smaller than 7 with multiplicity $r = 3$, because from a 3-admissible sequence of length smaller than 7.
If $r = 2$, we claim that, up to a shifting and a permutation of the numbers $1, 2$ and 3, there is only one 2-admissible sequence of length 6 which is simple, namely

$$(\{1\}, \{1\}, \{2\}, \{2\}, \{3\}, \{3\}). \tag{3.8}$$

To prove the claim, we start by assuming, without loss of generality, that the first entry is $\{1\}$. If the second entry is $\{1\}$ again then the 3-rd and 4-th entry have to be equal in order to satisfy condition 2 of Definition 3.20, because otherwise we could find all numbers $1, 2$ and 3 in three consecutive entries of the sequence. Therefore, up to a permutation of numbers 2 and 3, we get the sequence (3.8). Assume now that the first and second entry are different, namley without loss of generality we can assume that the first entry is $\{1\}$ and the second entry is $\{2\}$. This implies that the number 3 cannot appear in the 3-rd and 6-th entry otherwise we could find the numbers $1, 2$ and 3 in three consecutive position of the sequence. We obtain

$$(\{1\}, \{2\}, ?, \{3\}, \{3\}, ?)$$

To satisfy condition 2 of Definition 3.20, the third entry has to be $\{2\}$ and the last one has to be $\{1\}$, and up to a shifting this is sequence (3.8).
From the simple 2-admissible sequence (3.8) we cannot get 2-admissible sequences of length shorter than 6: As soon as we substitute two different singletons in a row with a subset of two elements, WLOG, we get the sequence

$$(\{1\}, \{1, 2\}, \{2\}, \{3\}, \{3\}),$$

and therefore, by reordering, we get the sequence

$$(\{1\}, \{2\}, \{1\}, \{2\}, \{3\}, \{3\}),$$

that does not respect the second condition of Definition 3.20 because the numbers $1, 2$ and 3 appear in three consecutive entries of the sequence.
This concludes the proof of the claim that when $n = 3$ there are no admissible sequences of length $4, 5$ and 7.

- If $n = 4$ we have r-admissible sequences of the form (3.7) when r is at least 3:
  The sequence is

$$(\{1\}, \dots, \{1\}, \{1, 2\}, \{2\}, \dots, \{4\}, \{1, 4\})$$

  which has length $m = 4(r - 1)$ and is r-admissible:

  1. The first condition of Definition 3.20 is satisfied because all numbers $1, 2, 3, 4$ appear r times.

2. To prove the second condition of Definition 3.20, we consider any sequence $\underline{\sigma}$ obtained by choosing an ordering of the entries $\{1,2\},\{2,3\},\{3,4\}$ and $\{1,4\}$ and we prove that we cannot find $4r'$ consecuitve entries where all numbers $1,2,3,4$ appear with the same multiplicity $r' < r$. Since $r \geq 3$ we cannot find the numbers $1,2,3$ and 4 in 4 consecutive positions of some sequence $\underline{\sigma}$: If we take two consecutive entries where 1 and 2 appear there is at least one entry different from the number 3 both in the entry immediately before and in the entry immediately after the entries with the numbers 1 and 2 that we have chosen. Therefore to find $4r'$ consecutive entries where the numbers $1,2,3$ and 4 appear with the same multiplicity r', we must have $r' > 1$. The best attempt that we can do to find $4r'$ consecutive entries of the sequence where all numbers $1,2,3,4$ appear with multiplicity r' consist in taking the $(r-2)$-th, $(r-1)$-th, r-th and $(r+1)$-th entry, where both the number 1 and the number 2 appear two times. But now we have to choose other consecutive entries in order to have also entries where the numbers 3 and 4 appear. This implies in particular that we have to take all entries from the first to the $(2r-2)$-th entry as well and yet all the entries are different from 3 and the numbers 1 and 2 appear with multiplicity $r-1$, then we want to have also $r-1$ entries equal to 3 and $r-1$ entries equal to 4. This implies that we also have to consider the $(2r-1)$-th, $(2r)$-th and $(2r+1)$-th entry to have two entries with the number 3. But in this case either the $(2r-1)$-th or the $(2r)$-th entry is equal to 2. This implies that the number 2 appears in r entries and therefore we cannot find $3r'$ consecutive entries of the sequence where $1,2,3,4$ appear all with multiplicity $r' < r$.

Therefore we know that we have r-admissible sequences of length $1 \leq m \leq 4$ and $m \geq 8$. We can consider the sequence

$$(\{1\},\{1\},\{2\},\{2,3\},\{3,4\},\{4\})$$

which is 2-admissible and has length 6. This implies that we can also have 2-admissible sequences of length 7. Now we want to show that we can not have admissible sequences of length 5.
If $r \geq 4$, from Lemma 3.26, we have $m \geq (n \cdot r)/3 \geq (4 \cdot 4)/3 > 5$. So we can focus on r-admissible sequences with $r \leq 3$. If $r = 3$, to see whether there are 3-admissible sequences of length 5, we have to understand how we can write the number 12 as a sum of five natural numbers smaller than or equal to 3. There are only two possibilities:

- $12 = 3+3+3+2+1$,
- $12 = 3+3+2+2+2$.

In both cases, since we have some entries which are set of ardinality equal to 3, we can assume without loss of generality that $\Sigma_1 = \{1,2,3\}$. This implies that 4 can not be in Σ_2 and Σ_5. But then the number 4 could be an element only of Σ_3 and Σ_4 and then it cannot appear in three different entries of $\underline{\Sigma}$. We conclude that it is not possible to have 3-admissible sequences of length 5.
If $r = 2$, to see whether there are 2-admissible sequences of length 5, we have to understand how we can write the number 8 as a sum of five natural numbers smaller than or equal to 3. There is only one possibility: $8 = 2+2+2+1+1$. We distinguish between two cases:

1. The two singletons that appear in the sequence are equal, without loss of generality they are both the singleton $\{1\}$,
2. The two singletons that appear in the sequence are different, without loss of generality we assume that they are the singleton $\{1\}$ and $\{2\}$.

In the first case the other three entries must be subsets with two different elements, namely they must be $\{2,3\},\{2,4\},\{3,4\}$. In this situation, for sure one singleton and

one subset of two elements have to appear in two consecutive entries of the sequence, hence we can assume that the first entry is the singleton $\{1\}$ and the second entry is the subset $\{2,3\}$. This implies that the subsets $\{2,4\}$ and $\{3,4\}$ cannot be the 3-rd and 5-th entry of the sequence, otherwise we could find all the numbers $1,2,3$ and 4 in three consecutive entries of the sequence. This implies that the subsets $\{2,4\}$ and $\{3,4\}$ must both be the 4-th entry of the sequence and this is not possible.
In the second case we should have three subsets of two different elements were the numbers 1 and 2 appear once and the numbers 3 and 4 appear twice. This implies that either we have the subsets $\{1,2\},\{3,4\}$ and $\{3,4\}$ or the subsets $\{1,3\},\{2,4\},\{3,4\}$.
If we have the subsets $\{1,2\},\{3,4\}$ and $\{3,4\}$ and if we assume that the first entry is $\{1,2\}$ then the subsets $\{3,4\}$ cannot be the 2-nd and 5-th entry and therefore they have to be the 3-rd and 4-th, but then the singletons $\{1\}$ and $\{2\}$ should be in the 2-nd and 4-th entry and this would imply that we can find all the numbers from 1 to 4 considering the 2-nd, 3-rd and 4-th entry, which means that we cannot find a 3-admissible sequence with these entries.
If we have the subsets $\{1,3\},\{2,4\},\{3,4\}$, without loss of generality we can assume that the first entry is $\{1,3\}$ which implies that the subset $\{2,4\}$ cannot be neither the 2-nd nor the 5-th entry otherwise we could find all the numbers from 1 to 4 in two consecutive entries of the sequence. If we assume that $\{2,4\}$ is in the 3-rd position, namely

$$(\{1,3\},?,\{2,4\},?,?)$$

we cannot find a 3-admissible sequence because the 2-nd entry has to be one set among $\{1\},\{2\},\{3,4\}$ and in all these cases we can find all the numbers from 1 to 4 in the first three position and this does not respect the second condition of Definition 3.20.
If we assume that $\{2,4\}$ is in the 4-th position, namely

$$(\{1,3\},?,?,\{2,4\},?)$$

then looking at the 5-th entry has to be one set among $\{1\},\{2\},\{3,4\}$ and again this is not possible because otherwise we could find all the numbers from 1 to 4 in the 5-th, 1-st and 2-nd entry and again and this does not respect the second condition of Definition 3.20.

- If $n \geq 5$ the sequences of the form 3.7, namely

$$(\{1\},\dots,\{1,2\},\dots,\{n-1,n\},\{n,1\})$$

are r-admissible for every $r \geq 2$:

1. The first condition of Definition 3.20 is satisfied because all numbers from 1 to n appear r times.
2. To prove that the second condition of Definition 3.20 is satisfied we consider any sequence $\underline{\sigma}$ obtained by choosing an ordering of each entry and we prove that we cannot find $r'n$ consecutive entries of $\underline{\sigma}$ where all numbers from 1 to n appear $r' < r$ times. If we want to find $r'n$ consecutive entries where the numbers $1,2,\dots,n$ appear all with the same multiplicity r' then for sure we have to consider the entries $r-1$ and r where the numbers 1 and 2 appear, but then to have that also the number 3 appears we also have to consider the entries from $r+1$ to $2r-1$ and $2r$. This implies that the number 2 appears r times, namely that we cannot find $r'n$ consecutive positions of some $\underline{\sigma}$ where all the numbers from 1 to n appear r' rimes with $r' < r$.

These sequences have length $(r-1)n$, for every $r \geq 2$. Therefore there are r-admissible sequences of any length between $(r-1)n$ and rn and hence there exist r-admissible sequences of any length bigger or equal than n. Moreover we know that for every n we

can always find r-admissible sequences of any length between 1 and n. We conclude that when $n \geq 5$ we can find r-admissible sequences of length m for every natural number m.

□

We introduced the Definition of r-admissible sequences because we were interested in finding conditions that a sequence $(\underline{\Sigma}) = (\Sigma_1, \dots, \Sigma_m)$ has to satisfy such that, given a multidegree $\underline{d}$, the open $\mathcal{U}(\underline{d}, \underline{\Sigma})$ admits a proper good moduli space.

Theorem 3.28. *Let X be a necklace of length n and genus $g = 1$. The map*

$$f\colon \left\{ \begin{array}{c} \textit{Open and connected substacks} \\ \mathcal{U} \subseteq \underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X) \textit{ that admit} \\ \textit{a proper good moduli space} \end{array} \right\} \longrightarrow \bigcup_{r \in \mathbb{N}_{\geq 1}} \left(\left\{ \begin{array}{c} r\textit{-admissible sequences} \\ (\underline{d}, \Sigma_1, \dots, \Sigma_m) \end{array} \right\}_{/\sim} \right)$$

defined by sending $\mathcal{U} = \mathcal{U}(N_{\Sigma_1}^{\underline{e}_1}) \cup \dots \cup \mathcal{U}(N_{\Sigma_m}^{\underline{e}_m})$ to

$$(\underline{e_1} + \mathcal{D}(\Sigma_1, O^-), \Sigma_1, \dots, \Sigma_m)$$

is a bijection.
The inverse map is defined by sending an r-admissible sequence $(\underline{d}, \underline{\Sigma}) = (\underline{d}, \Sigma_1, \dots, \Sigma_m)$ to the open $\mathcal{U}$ in $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ defined as:

$$\mathcal{U}(\underline{d}, \underline{\Sigma}) \coloneqq \mathcal{U}(N_{\Sigma_1}^{\underline{e}_1}) \cup \dots \cup \mathcal{U}(N_{\Sigma_m}^{\underline{e}_m}),$$

where

- $\underline{e}_1 = \underline{d} - \mathcal{D}(\Sigma_1, O^-)$,
- $\underline{e}_{i+1} = \underline{e}_i + \mathcal{D}(\Sigma_i, O^+) - \mathcal{D}(\Sigma_{i+1}, O^-)$.

The moduli space of the open $\mathcal{U}(\underline{d}, \underline{\Sigma})$ is a necklace of length m.

Remark 3.29. If we consider only fine compactified Jacobians, the statement of Theorem 3.28 becomes [2, Lemma 3.7]. In particular the r-admissible sequences are *simple*.

Before proving Theorem 3.28, we write explicitly the multidegrees of line bundles in the open $\mathcal{U}(\underline{d}, \underline{\Sigma})$ as defined in Theorem 3.28.

Lemma 3.30. *Let $(\underline{d}, \Sigma_1, \dots, \Sigma_m)$ be an r-admissible sequence and consider the open $\mathcal{U}(\underline{d}, \underline{\Sigma})$ as defined in Theorem 3.28. The multidegrees $\underline{f}$ corresponding to line bundles in $\mathcal{U}(\underline{d}, \underline{\Sigma})$ are*

$$\underline{f} = \underline{d} + \mathcal{D}'(\Sigma_1, O^+) + \dots + \mathcal{D}'(\Sigma_{i-1}, O^+) + \mathcal{D}'(S_i, O^+),$$

where $S_i \subseteq \Sigma_i$, for some $i = 1, \dots, m$.

Proof. By definition, we have

$$\mathcal{U}(\underline{d}, \underline{\Sigma}) = \mathcal{U}(N_{\Sigma_1}^{\underline{e}_1}) \cup \dots \cup \mathcal{U}(N_{\Sigma_n}^{\underline{e}_m}),$$

where

- $\underline{e}_1 = \underline{d} - \mathcal{D}(\Sigma_1, O^-)$,
- $\underline{e}_{i+1} = \underline{e}_i + \mathcal{D}(\Sigma_i, O^+) - \mathcal{D}(\Sigma_{i+1}, O^-)$.

This implies that, using the notation $\mathcal{D}'(\Sigma_1, O^+) = \mathcal{D}(\Sigma_i, O^+) - \mathcal{D}(\Sigma_i, O^-)$,

$$\underline{e}_i = \underline{d} + \mathcal{D}'(\Sigma_1, O^+) \cdots + \mathcal{D}'(\Sigma_{i-1}, O^+) - \mathcal{D}(\Sigma_i, O^-). \tag{3.9}$$

From Corollary 3.6, we know that the multidegrees of line bundles in $\mathcal{U}(N_{\Sigma_i}^{\underline{e}_i})$ are of the form

$$\underline{f} = \underline{e}_i + \mathcal{D}(\Sigma_i, O)$$

for some orientation $O = (s, t)$. We define

$$\Sigma'_i \coloneqq \{P_j \in \Sigma_i |\ s(P_j) = s_+(P_j)\}$$

and we can write

$$\mathcal{D}(\Sigma_i, O) = \mathcal{D}(\Sigma'_i, O^+) + \mathcal{D}(\Sigma_i \setminus \Sigma'_i, O^-).$$

We obtain

$$\underline{f} = \underline{e}_i + \mathcal{D}(\Sigma'_i, O^+) + \mathcal{D}(\Sigma_i \setminus \Sigma'_i, O^-),$$

which gives, by substituing $\underline{e}_i$ with Equation (3.9) and by using that $\mathcal{D}(\Sigma_i, O^-) = \mathcal{D}(\Sigma'_i, O^-) + \mathcal{D}(\Sigma_i \setminus \Sigma'_i, O^-)$, that $\underline{f}$ is

$$\underline{d} + \mathcal{D}'(\Sigma_1, O^+) \cdots + \mathcal{D}'(\Sigma_{i-1}, O^+) - [\mathcal{D}(\Sigma'_i, O^-) + \mathcal{D}(\Sigma_i \setminus \Sigma'_i, O^-)] + \mathcal{D}(\Sigma'_i, O^+) + \mathcal{D}(\Sigma_i \setminus \Sigma'_i, O^-),$$

Namely,

$$\underline{f} = \underline{d} + \mathcal{D}'(\Sigma_1, O^+) + \cdots + \mathcal{D}'(\Sigma_{i-1}, O^+) + \mathcal{D}'(\Sigma'_i, O^+).$$

□

Proof of Theorem 3.28. Let $\mathcal{U}$ be an open connected substack of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ that admits a proper good moduli space. From Proposition 3.17, we know that we can write

$$\mathcal{U} = \mathcal{U}(N_{\Sigma_1}^{\underline{e}_1}) \cup \cdots \cup \mathcal{U}(N_{\Sigma_m}^{\underline{e}_m}),$$

where

- $\{N_{\Sigma_1}^{\underline{e}_1}, \ldots, N_{\Sigma_m}^{\underline{e}_m}\}$ is the set of the most degenerate sheaves of $\mathcal{U}$,
- $\mathcal{U}(N_{\Sigma_i}^{\underline{e}_i})$ and $\mathcal{U}(N_{\Sigma_{i+1}}^{\underline{e}_{i+1}})$ are consecutive for every $i \in \mathbb{Z}/m\mathbb{Z}$.

The map f sends $\mathcal{U}$ to the sequence $(\underline{d}, \underline{\Sigma}) \coloneqq (\underline{e_1} + \mathcal{D}(\Sigma_1, O^-), \Sigma_1, \ldots, \Sigma_m)$.
Observe that we could have written

$$\mathcal{U} = \mathcal{U}(N_{\Sigma_i}^{\underline{e}_i}) \cup \mathcal{U}(N_{\Sigma_{i+1}}^{\underline{e}_{i+1}}) \cup \cdots \cup \mathcal{U}(N_{\Sigma_{i-1}}^{\underline{e}_{i-1}})$$

for every $i = 1, \ldots, m$. We show that the map f is well defined, namely we show that the sequences

$$\tau_i \coloneqq (\underline{e_i} + \mathcal{D}(\Sigma_i, O^-), \Sigma_i, \ldots, \Sigma_{i-1})$$

are equivalent in the sense of Definition 3.23. For every $i = 1, \ldots, m$, the sequence $\tau_i = (\underline{e_i} + \mathcal{D}(\Sigma_i, O^-), \Sigma_i, \ldots, \Sigma_{i-1})$ is equivalent to the sequence $(\underline{e_1} + \mathcal{D}(\Sigma_1, O^-), \Sigma_1, \ldots, \Sigma_m)$:
Indeed we have

- $\underline{e}_i + \mathcal{D}(\Sigma_i, O^-) = \underline{e}_1 + \mathcal{D}'(\Sigma_1, O^+) + \mathcal{D}'(\Sigma_2, O^+) \cdots + \mathcal{D}'(\Sigma_{i-1}, O^+)$, as it follows from Equation 3.6,
- $(\Sigma_i, \ldots, \Sigma_{i-1}) = \tau^{i-1}(\Sigma_1, \ldots, \Sigma_m)$,

where τ^i is the cyclic permutation that sends j to $j + i$ for every $j = 1, \ldots, m$.
Now we show that the sequence $f(\mathcal{U})$ is r-admissible:

1. We start by showing that the first condition of Definition 3.20 of r-admissible sequences is satisfied, namely we prove that there is a natural number r such that

   $$\{l \colon t \in \Sigma_l\} = r, \text{ for every } t = 1, \ldots, n.$$

   From the fact that the $\mathcal{U}(N_{\Sigma_i}^{\underline{e}_i})$'s are consecutive we have that

   $$\underline{e}_i + \mathcal{D}(\Sigma_i, O^+) = \underline{e}_{i+1} + \mathcal{D}(\Sigma_{i+1}, O^-).$$

   Hence, using the notation $\mathcal{D}'(\Sigma_i, O^+) \coloneqq \mathcal{D}(\Sigma_i, O^+) - \mathcal{D}(\Sigma_i, O^-)$, we have

$$\underline{e}_i = \underline{e}_1 + \mathcal{D}(\Sigma_1, O^+) + \mathcal{D}'(\Sigma_2, O^+) + \cdots + \mathcal{D}'(\Sigma_{i-1}, O^+) - \mathcal{D}(\Sigma_i, O^-).$$

Using that the $\mathcal{U}(N_{\Sigma_i}^{\underline{e}_i})$'s are cyclic, we have $\underline{e}_m + \mathcal{D}(\Sigma_m, O^+) = \underline{e}_1 + \mathcal{D}(\Sigma_1, O^-)$, which implies

$$\underline{e}_1 + \mathcal{D}(\Sigma_1, O^+) + \mathcal{D}'(\Sigma_2, O^+) + \cdots + \mathcal{D}'(\Sigma_{m-1}, O^+) - \mathcal{D}(\Sigma_m, O^-) + \mathcal{D}(\Sigma_m, O^+) = \underline{e}_1 + \mathcal{D}(\Sigma_1, O^-) \ .$$

We get the telescoping sum

$$\mathcal{D}'(\Sigma_1, O^+) + \cdots + \mathcal{D}'(\Sigma_m, O^+) = \underline{0} \ .$$

Therefore we have that the j-th entry of the vector $\mathcal{D}'(\Sigma_1, O^+) + \cdots + \mathcal{D}'(\Sigma_m, O^+)$ has to be zero for every $j = 1, \ldots, n$. Since we are working with the convention that $s_+(P_j) = s(P_j, O^+) = X_j$ and for every irreducible component X_i of X there is a unique node, namely P_i such that $s(P_i, O^+) = X_i$, we get that for every j the j-th entry d_j of $\mathcal{D}'(\Sigma_1, O^+) + \cdots + \mathcal{D}'(\Sigma_m, O^+)$ is

$$d_j = \#\{\Sigma_k | P_{j-1} \in \Sigma_k\} - \#\{\Sigma_k | P_j \in \Sigma_k\}.$$

This implies that, for every $j = 1, \ldots, n$, the nodes P_{j-1} and P_j have to appear in the same number of entries of $\underline{\Sigma}$. Hence, there exists a natural number r such that every node appears in r of the Σ_j's, namely

$$\{l \colon t \in \Sigma_l\} = r, \text{ for every } t = 1, \ldots, n \ .$$

2. We prove that the second condition of Definition 3.20 holds. We can assume that $m > 1$ because if $m = 1$ than $\Sigma_1 = \{1, \ldots, n\}$, namely $r = 1$ and the second condition of Definition 3.20 is obviously satisfied.
   We prove that the second condition of Definition 3.20 holds also when $m > 1$ by contradiction: Assume that there is a subsequence of $r'n$ consecutive entries of a sequence $\underline{\sigma}$ obtained from $\underline{\Sigma}$ by choosing an ordering of the elements in each Σ_i where all numbers from 1 to n appear with multiplicity $r' < r$. Namely we can write

   $$\underline{\sigma} = (\mathrm{ord}(\Sigma_1), \ldots, \mathrm{ord}(\Sigma_j \setminus \Sigma'_j), \mathrm{ord}(\Sigma'_j), \ldots, \mathrm{ord}(\Sigma'_{j+l}), \mathrm{ord}(\Sigma_{j+l} \setminus \Sigma'_{j+l}), \ldots \mathrm{ord}(\Sigma_m))$$

   where we assume that in the subsequence

   $$(\mathrm{ord}(\Sigma'_j), \ldots, \mathrm{ord}(\Sigma'_{j+l}))$$

   all numbers from 1 to n appear with the same multiplicity r'. Since we are assuming $r' < r$, either we have that $1 < l < m$ namely the opens $\mathcal{U}(N_{\Sigma'_j}^{\underline{e}_j})$ and $\mathcal{U}(N_{\Sigma'_{j+l}}^{\underline{e}_{j+l}})$ are not consecutive or we have that $\mathcal{U}(N_{\Sigma'_j}^{\underline{e}_j})$ and $\mathcal{U}(N_{\Sigma'_{j+l}}^{\underline{e}_{j+l}})$ are consecutive and in this case we must have $\Sigma'_j \subsetneq \Sigma_j$ and $\Sigma'_{j+l} \subsetneq \Sigma_{j+l}$ because if one of these subset is empty than we should have that all numbers from 1 to n appear in one entry of $\underline{\Sigma}$ which cannot be the case unless $\mathcal{U} = \mathcal{U}(N_{\Sigma_1}^{\underline{e}_1})$ and we are assuming $m > 1$.
   We can write

   $$\mathcal{D}'(\Sigma'_j, O^+) + \mathcal{D}'(\Sigma_{j+1}, O^+) \cdots + \mathcal{D}'(\Sigma_{j+l-1}, O^+) + \mathcal{D}'(\Sigma'_{j+l}, O^+) = \underline{0}.$$

   Therefore the line bundles of multidegree

   - $\underline{f}_j \coloneqq \underline{e}_1 + \mathcal{D}(\Sigma_1, O^+) + \mathcal{D}'(\Sigma_2, O^+) + \cdots + \mathcal{D}'(\Sigma_{j-1}, O^+) + \mathcal{D}'(\Sigma_j \setminus \Sigma'_j, O^+)$,
   - $\underline{f}_{j+l} \coloneqq \underline{f}_j + \mathcal{D}'(\Sigma'_j, O^+) + \mathcal{D}'(\Sigma_{j+1}, O^+) + \cdots + \mathcal{D}'(\Sigma_{j+l-1}, O^+) + \mathcal{D}'(\Sigma'_{j+l}, O^+)$,

are equal.
From Lemma 3.30, we get that the line bundles of multidegree $\underline{f}_j$ are in $\mathcal{U}(N^{\underline{e}_j}_{\Sigma_j})$ and the line bundles of multidegree $\underline{f}_{j+l}$ are in $\mathcal{U}(N^{\underline{e}_{j+l}}_{\Sigma_{j+l}})$. But by assumption the opens $\mathcal{U}(N^{\underline{e}_1}_{\Sigma_1}), \dots, \mathcal{U}(N^{\underline{e}_1}_{\Sigma_1})$ are cyclic and two non consecutive opens $\mathcal{U}(N^{\underline{e}_j}_{\Sigma_j})$ and $\mathcal{U}(N^{\underline{e}_{j+l}}_{\Sigma_{j+l}})$ are disjoint. This implies that the only possibility is that $l = 1$, but in this case the opens are consecutive and therefore the intersection is given by line bundles of multidegree

$$\underline{d}_j = \underline{e}_1 + \mathcal{D}(\Sigma_1, O^+) + \mathcal{D}'(\Sigma_2, O^+) + \cdots + \mathcal{D}'(\Sigma_j, O^+),$$

namely we must have $\Sigma'_j = \emptyset$ and we already explained that this cannot be the case because in this case we should have $m = 1$.

This concludes the proof that f is well-defined and that it associates to $\mathcal{U}$ an r-admissible sequence.

Conversely, we consider an r-admissible pair $(\underline{d}, \underline{\Sigma})$ and define

$$\mathcal{U} := \mathcal{U}(N^{\underline{e}_1}_{\Sigma_1}) \cup \cdots \cup \mathcal{U}(N^{\underline{e}_m}_{\Sigma_m}),$$

where

- $\underline{e}_1 = \underline{d} - \mathcal{D}(\Sigma_1, O^-)$,
- $\underline{e}_{i+1} = \underline{e}_i + \mathcal{D}(\Sigma_i, O^+) - \mathcal{D}(\Sigma_{i+1}, O^-)$.

We want to prove that $\mathcal{U} \subseteq \underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X)$ is open connected and has a proper good moduli space. From Proposition 3.17 it is enough to prove that the opens $\mathcal{U}(N^{\underline{e}_1}_{\Sigma_1}), \dots, \mathcal{U}(N^{\underline{e}_m}_{\Sigma_m})$ are cyclic.
We observe that $\mathcal{U}$ is open since it is given by the union of opens and it is connected since the opens $\mathcal{U}(N^{\underline{e}_i}_{\Sigma_i})$ and $\mathcal{U}(N^{\underline{e}_{i+1}}_{\Sigma_{i+1}})$, for every $i = 1, \dots, m-1$ are consecutive by construction. We need to prove that the opens $\mathcal{U}(N^{\underline{e}_1}_{\Sigma_1})$ and $\mathcal{U}(N^{\underline{e}_m}_{\Sigma_m})$ are consecutive and that two opens which are not consecutive have empty intersection. We show now that these properties follow from the first and second condition of Definition 3.20:

1. The first condition of Definition 3.20, namely the fact that the sequence $(\Sigma_1, \dots, \Sigma_m)$ has multiplicity r, tells us that the opens $\mathcal{U}(N^{\underline{e}_1}_{\Sigma_1})$ and $\mathcal{U}(N^{\underline{e}_m}_{\Sigma_m})$ are consecutive: We have that:

   - Condition 1 of Definition 3.20 can be rewritten as
   $$\mathcal{D}'(\Sigma_1, O^+) + \cdots + \mathcal{D}'(\Sigma_m, O^+) = \underline{0}$$
   because for a necklace X there is precisely one node in the intersection of two irreducible components X_i and X_{i+1},
   - By construction
   $$\underline{e}_m = \underline{e}_1 + \mathcal{D}(\Sigma_1, O^-) + \mathcal{D}'(\Sigma_2, O^+) + \cdots + \mathcal{D}'(\Sigma_{m-1}, O^+) - \mathcal{D}(\Sigma_m, O^-).$$
   Putting these two facts together (recall that $\mathcal{D}'(\Sigma_i, O^+) = \mathcal{D}(\Sigma_i, O^+) - \mathcal{D}(\Sigma_i, O^-)$), we obtain
   $$\underline{e}_m = \underline{e}_1 + \mathcal{D}(\Sigma_1, O^+) - \mathcal{D}(\Sigma_m O^+),$$
   which means that the opens $\mathcal{U}(N^{\underline{e}_1}_{\Sigma_1})$ and $\mathcal{U}(N^{\underline{e}_m,}_{\Sigma_m})$ are consecutive.

2. The second condition of Definition 3.20 tells us that the open $\mathcal{U}$ admits a good and separated quotient because two opens that are not consecutive have empty intersection: If by contradiction we assume that two non consecutive opens $\mathcal{U}(N^{\underline{e}_j}_{\Sigma_j})$ and $\mathcal{U}(N^{\underline{e}_{j+l}}_{\Sigma_{j+l}})$, for some $1 < l < m$ have non empty intersection, there exists a multidegree $\underline{d}$ such that both $N^{\underline{e}_j}_{\Sigma_j}$ and $N^{\underline{e}_{j+l}}_{\Sigma_{j+l}}$ can be obtained as degeneration of line bundles of multidegree $\underline{d}$.
   From Corollary 3.6 we know that the line bundles in $\mathcal{U}(N^{\underline{e}_j}_{\Sigma_j})$ have multidegree

$$\underline{e}_j + \mathcal{D}(\Sigma_j, O_j),$$

for some orientation O_j of Σ_j, and the line bundles in $\mathcal{U}(N_{\Sigma_{j+l}}^{\underline{e}_{j+l}})$ have multidegree

$$\underline{e}_{j+l} + \mathcal{D}(\Sigma_{j+l}, O_{j+l}),$$

for some orientation O_{j+l} of Σ_{j+l}.
Therefore there exist two orientations O_j of Σ_j and O_{j+l} of Σ_{j+l} such that

$$\underline{e}_j + \mathcal{D}(\Sigma_j, O_j) = \underline{e}_{j+l} + \mathcal{D}(\Sigma_{j+l}, O_{j+l}). \tag{3.10}$$

By construction, we have

- $\underline{e}_j = \underline{e}_1 + \mathcal{D}(\Sigma_1, O^+) + \mathcal{D}'(\Sigma_2, O^+) + \cdots + \mathcal{D}'(\Sigma_{j-1}, O^+) - \mathcal{D}(\Sigma_j, O^-)$,
- $\underline{e}_{j+l} = \underline{e}_1 + \mathcal{D}(\Sigma_1, O^-) + \mathcal{D}'(\Sigma_2, O^+) + \cdots + \mathcal{D}'(\Sigma_{j+l-1}, O^+) - \mathcal{D}(\Sigma_{j+l}, O^-)$,

which implies that

$$\underline{e}_{j+l} = \underline{e}_j + \mathcal{D}(\Sigma_j, O^+) + \mathcal{D}'(\Sigma_{j+1}, O^+) + \cdots + \mathcal{D}'(\Sigma_{j+l-1}, O^+) - \mathcal{D}(\Sigma_{j+l}, O^-).$$

Therefore, from Equation (3.10), we get

$$-\mathcal{D}(\Sigma_j, O_j)+\mathcal{D}(\Sigma_j, O^+)+\mathcal{D}'(\Sigma_{j+1}, O^+)+\cdots+\mathcal{D}'(\Sigma_{j+l-1}, O^+)-\mathcal{D}(\Sigma_{j+l}, O^-)+\mathcal{D}(\Sigma_{j+l}, O_{j+l}) = \underline{0} \tag{3.11}$$

If we denote Σ'_j the subset of Σ_j and Σ'_{j+l} the subset of Σ_j such that

- $O_j|_{\Sigma'_j} = O^-|_{\Sigma'_j}$,
- $O_j|_{\Sigma_j \setminus \Sigma'_j} = O^+|_{\Sigma_j \setminus \Sigma'_j}$,
- $O_{j+l}|_{\Sigma'_{j+l}} = O^-|_{\Sigma'_{j+l}}$,
- $O_{j+l}|_{\Sigma_{j+l} \setminus \Sigma'_{j+l}} = O^+|_{\Sigma_{j+l} \setminus \Sigma'_{j+l}}$.

we get that

- $\mathcal{D}(\Sigma_j, O_j) = \mathcal{D}(\Sigma'_j, O^-) + \mathcal{D}(\Sigma_j \setminus \Sigma'_j, O^+)$,
- $\mathcal{D}(\Sigma_{j+l}, O_{j+l}) = \mathcal{D}(\Sigma'_{j+l}, O^-) + \mathcal{D}(\Sigma_{j+l} \setminus \Sigma'_{j+l}, O^+)$.

Substituting these expressions in Equation (3.11) and using that

- $\mathcal{D}(\Sigma_j, O^+) = \mathcal{D}(\Sigma'_j, O^+) + \mathcal{D}(\Sigma_j \setminus \Sigma'_j, O^+)$,
- $\mathcal{D}(\Sigma_{j+l}, O^-) = \mathcal{D}(\Sigma'_{j+l}, O^-) + \mathcal{D}(\Sigma_{j+l} \setminus \Sigma'_{j+l}, O^-)$,

we obtain

$$\begin{gathered}-\mathcal{D}(\Sigma'_j, O^-) + \mathcal{D}(\Sigma'_j, O^+) + \mathcal{D}'(\Sigma_{j+1}, O^+) + \cdots + \mathcal{D}'(\Sigma_{j+l-1}, O^+)+ \\ -\mathcal{D}(\Sigma'_{j+l} \setminus \Sigma'_{j+l}, O^-) + \mathcal{D}(\Sigma_{j+l} \setminus \Sigma'_{j+l}, O^+) = \underline{0}\end{gathered}$$

namely, using the notation $\mathcal{D}'(\Sigma_j, O^+) = \mathcal{D}(\Sigma'_j, O^+) - \mathcal{D}(\Sigma'_j, O^-)$ and $\mathcal{D}'(\Sigma_{j+l}, O^+) = \mathcal{D}(\Sigma'_{j+l}, O^+) - \mathcal{D}(\Sigma'_{j+l}, O^-)$, we can write

$$\mathcal{D}'(\Sigma'_j, O^+) + \mathcal{D}'(\Sigma_{j+1}, O^+) \cdots + \mathcal{D}'(\Sigma_{j+l-1}, O^+) + \mathcal{D}'(\Sigma_{j+l} \setminus \Sigma'_{j+l}, O^+) = \underline{0}.$$

Therefore all numbers from 1 to n appear the same number of times r' in the subsets $\Sigma'_j, \Sigma_{j+1}, \ldots, \Sigma_{j+l-1}, \Sigma_{j+l} \setminus \Sigma'_{j+l}$. This means that if we consider the sequence

$$\underline{\sigma} = (\mathrm{ord}(\Sigma_1), \ldots, \mathrm{ord}(\Sigma_j \setminus \Sigma'_j), \mathrm{ord}(\Sigma'_j), \ldots, \mathrm{ord}(\Sigma_{j+l} \setminus \Sigma'_{j+l}), \mathrm{ord}(\Sigma'_{j+l}), \ldots \mathrm{ord}(\Sigma_m)),$$

for any choice of ordering $\mathrm{ord}(\Sigma_i)$ for every $i = 1, \ldots, m$, we get that in the subsequence

$$(\mathrm{ord}(\Sigma'_j), \dots, \mathrm{ord}(\Sigma_{j+l} \setminus \Sigma'_{j+l}))$$

the numbers $\{1, \dots, n\}$ appear all and with the same multiplicity r' and this leads to a contradiction with the second condition of Definition 3.20 of r-admissible sequences.

We conclude that the opens $\mathcal{U}(N^{\underline{e}_i}_{\Sigma_i})$'s are cyclic and that, as we have seen in Proposition 3.17, $\mathcal{U}$ admits a proper good moduli space which is a necklace of length m.

It follows directly from the definitions that the map f is a bijection which inverse given as explained in the statement. This concludes our proof. □

Definition 3.31.
We say that an open connected substack $\mathcal{U}$ of $\underline{\mathrm{Coh}}^{\mathrm{tf}}_1(X)$ that admits a proper good moduli space has length m if it corresponds to an r-admissible sequence $(\underline{d}, \Sigma_1, \dots, \Sigma_m)$, namely it has as good moduli space a necklace of length m.

Corollary 3.32. *Let X be a necklace of length $n \geq 2$ and genus $g = 1$.*
If $n = 2$ the good compactified Jacobians of X have length either 1 or 2.
If $n = 3$ there exist good compactified Jacobians of X of length m for every natural number but $4, 5$ and 7.
If $n = 4$ there exist good compactified Jacobians of X of length m for every natural number but 5.
If $n \geq 5$ there exist good compactified Jacobians of X of length m for every natural number.

Proof. A good compactified Jacobian of X, as described in Theorem 3.28, corresponds to an r-admissible pair $(\underline{d}, \Sigma_1, \dots, \Sigma_m)$. The statement of this Corollary is essentially the same as the statement of Lemma 3.27, where $(\underline{d}, \Sigma_1, \dots, \Sigma_m)$ is interpreted as a good compactified Jacobian of length m, via the bijection of Theorem 3.28. □

Remark 3.33. In the case of fine compactified Jacobians for a necklace of length $n \geq 3$ only the lengths $m = n \cdot k, k \in \mathbb{N}$ are possible. This recovers [2, Lemma 3.7].

4. NUMERICAL POLARIZATIONS

A classical approach to find compactifications of Jacobians consists in considering numerical polarizations. In this Section we recall how one can obtain compactifications of Jacobians starting from the notion of semistability for rank 1 torsion-free sheaves with respect to some numerical polarization. We refer to compactified Jacobians obtained starting from a numerical polarization as *classical* compactified Jacobians. After recalling the main definitions we prove, in Proposition 4.5, that all good compactified Jacobians for a necklace X of genus $g = 1$ corresponding to a 1-admissible sequence are classical, namely they can be obtained from a suitable numerical polarization.
We recall the definitions of numerical polarization and semistability, following [6, Section 3].

Definition 4.1.
Given a nodal curve X, a *numerical polarization* on X is a m-tuple $\phi = \{\phi_i = \phi_{X_i}\}_{i=1,\dots,m} \in \mathbb{Q}^m$ of rational numbers, one for every irreducible component X_i of X, such that $|\phi| := \sum_i \phi_i \in \mathbb{Z}$.
For every subcurve $Y \subseteq X$ and rank 1 torsion-free aheaf N on X, we denote by N_Y the biggest torsion-free quotient of the restriction of N to Y and by Y^C the closure of $X \setminus Y$ in X.
A rank 1 torsion-free sheaf N that is not locally free at the set of nodes Σ' is *ϕ-semistable* if $\deg(I) = |\phi|$ and

$$\deg(N_Y) \leq \sum_{X_i \subseteq Y} \phi_i - |Y \cap Y^C \cap \Sigma'| + \frac{|Y \cap Y^C|}{2}. \tag{4.1}$$

Starting from numerical polarizations Oda and Seshadri in [7] constructed compactifications of Jacobians: Given a nodal curve X and a numerical polarization ϕ one can obtain a compactified Jacobian, that we denote by $\bar{J}_X(\phi)$, as the union of all the rank 1 torsion-free sheaves on X which are ϕ-semistable.

Remark 4.2. Given an open connected substack of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ which admits a proper good moduli space good compactified Jacobian $\overline{J}_X$ corresponding to an r-admissible sequence $(\underline{d}, \Sigma_1, \dots, \Sigma_m)$ we have seen that the good moduli space of $\overline{J}_X$ is a necklace of length m.
We can distinguish between multidegrees of line bundles which are not contracted in the moduli space and multidegrees of line bundles which are contracted in the moduli space.
We spend a few lines to clarify the computation of these multidegrees starting from $(\underline{d}, \Sigma_1, \dots, \Sigma_m)$ and to point out that $\mathcal{U}(\underline{d}, \Sigma_1, \dots, \Sigma_m)$ is completely determined by the line bundles which are in $\mathcal{U}(\underline{d}, \Sigma_1, \dots, \Sigma_m)$..

Lemma 4.3. *Let $(\underline{d}, \Sigma_1, \dots, \Sigma_m)$ be an r-admissible pair and $\mathcal{U}(\underline{d}, \underline{\Sigma})$ the corresponding good compactified Jacobian. The multidegrees $\underline{d}_0, \dots, \underline{d}_{m-1}$ corresponding to line bundles in $\mathcal{U}(\underline{d}, \underline{\Sigma})$ which are not contracted to some node of the moduli space $U(\underline{d}, \underline{\Sigma})$ of $\mathcal{U}(\underline{d}, \underline{\Sigma})$ are*

- $\underline{d}_0 = \underline{d}$,
- $\underline{d}_i = \underline{d} - \mathcal{D}(\Sigma_1, O^-) + \mathcal{D}(\Sigma_1, O^+) + \cdots - \mathcal{D}(\Sigma_i, O^-) + \mathcal{D}(\Sigma_i, O^+)$.

The multidegrees $\underline{f}$ of line bundles which are in $\mathcal{U}(\underline{d}, \underline{\Sigma})$ can be computed as

$$\underline{f} = \underline{d}_{i-1} + \mathcal{D}(S_i, O^+) - \mathcal{D}(S_i, O^-),$$

where $S_i \subseteq \Sigma_i$, for some $i = 1, \dots, m$.

Proof. From Theorem 3.28, we can write

$$\mathcal{U}(\underline{d}, \underline{\Sigma}) = \mathcal{U}(N^{\underline{e}_1}_{\underline{\Sigma}_1}) \cup \cdots \cup \mathcal{U}(N^{\underline{e}_m}_{\underline{\Sigma}_n}),$$

where

- $\underline{e}_1 = \underline{d} - \mathcal{D}(\Sigma_1, O^-)$,
- $\underline{e}_{i+1} = \underline{e}_i - \mathcal{D}(\Sigma_i, O^-) + \mathcal{D}(\Sigma_{i+1}, O^+)$.

We call $\underline{d}_0, \underline{d}_1, \ldots, \underline{d}_{m-1}$ the multidegrees of the line bundles of $\mathcal{U}(\underline{d}, \underline{\Sigma})$ which are not contracted in the moduli space $U(\underline{d}, \underline{\Sigma})$. From Corollary 3.6, these line bundles have multidegree

$$\underline{e}_i + \mathcal{D}(\Sigma_i, O),$$

where O is either the clockwise O^+ or the anticlockwise orientation O^- of the dual graph Γ_X of X.
We start by observing that by definition the multidegree

$$\underline{d} = \underline{e}_1 + \mathcal{D}(\Sigma_1, O^-)$$

is one of the two multidegrees of line bundles of $\mathcal{U}(N^{\underline{e}_1}_{\underline{\Sigma}_1})$ which are not contracted in the moduli space $U(N^{\underline{e}_1}_{\underline{\Sigma}_1})$.
The other line bundles which are not contracted to the node of the moduli space $U(N^{\underline{e}_1}_{\underline{\Sigma}_1})$ of $\mathcal{U}(N^{\underline{e}_1}_{\underline{\Sigma}_1})$ have multidegree

$$\underline{d}_1 = \underline{e}_1 + \mathcal{D}(\Sigma_1, O^+) = \underline{d} - \mathcal{D}(\Sigma_1, O^-) + \mathcal{D}(\Sigma_1, O^+).$$

Now we consider the open $\mathcal{U}(N^{\underline{e}_2}_{\underline{\Sigma}_2})$ and we know, by construction, that the line bundles of multidegree $\underline{d}_1 = \underline{e}_2 + \mathcal{D}(\Sigma_2, O^-)$ are in $\mathcal{U}(N^{\underline{e}_2}_{\underline{\Sigma}_2})$ and they are not contracted in the moduli space. We denote by $\underline{d}_2$ the other multidegree of line bundles in $\mathcal{U}(N^{\underline{e}_2}_{\underline{\Sigma}_2})$ which are not contracted in the moduli space, namely

$$\underline{d}_2 = \underline{e}_2 + \mathcal{D}(\Sigma_2, O^+) = \underline{e}_1 - \mathcal{D}(\Sigma_1, O^-) + \mathcal{D}(\Sigma_1, O^+) + \mathcal{D}(\Sigma_2, O^+) =$$
$$= \underline{d} - \mathcal{D}(\Sigma_1, O^-) + \mathcal{D}(\Sigma_1, O^+) - \mathcal{D}(\Sigma_1, O^-) + \mathcal{D}(\Sigma_2, O^+).$$

Repeating the same argument we find that

$$\underline{d}_i = \underline{d} - \mathcal{D}(\Sigma_1, O^-) + \mathcal{D}(\Sigma_1, O^+) + \cdots - \mathcal{D}(\Sigma_i, O^-) + \mathcal{D}(\Sigma_i, O^+)$$

Consider now a multidegree $\underline{f}$ of line bundles in $\mathcal{U}(\underline{d}, \underline{\Sigma})$ which are contracted to the node of the moduli space $U(\underline{d}, \underline{\Sigma})$. From Lemma 3.30, we know that

$$\underline{f} = \underline{d} + \mathcal{D}'(\Sigma_1, O^+) + \cdots + \mathcal{D}'(\Sigma_{i-1}, O^+) + \mathcal{D}'(\Sigma'_i, O^+),$$

where $S_i \subseteq \Sigma_i$, for some $i = 1, \ldots, m$. Therefore we can write

$$\underline{f} = \underline{d}_{i-1} + \mathcal{D}(\Sigma'_i, O^+) - \mathcal{D}(\Sigma'_i, O^-).$$

□

Lemma 4.4. *Every good compactified Jacobian $\mathcal{U} := \mathcal{U}(\underline{d}, \underline{\Sigma})$ is completely determined by the set of its line bundles.*

Proof. Suppose that we know which are the multidegrees of the line bundles in $\mathcal{U}$. Consider two different multidegrees $\underline{d}_1$ and $\underline{d}_2$ of line bundles which are in $\mathcal{U}$. We have two possibilities:

- We can not find a sheaf that $N^{\underline{e}}_{\underline{\Sigma}'}$ such that

  $$\underline{e} = \underline{d}_1 - \mathcal{D}(\Sigma', O^+) = \underline{d}_2 - \mathcal{D}(\Sigma', O^-)$$

  this implies that the open $\mathcal{U}(N^{\underline{e}}_{\underline{\Sigma}'})$ is not in $\mathcal{U}$.

- There is a sheaf $N^{\underline{e}}_{\underline{\Sigma}'}$ such that

$$\underline{e} = \underline{d}_1 - \mathcal{D}(\Sigma', O^+) = \underline{d}_2 - \mathcal{D}(\Sigma', O^-). \tag{4.2}$$

  In this case we can consider the open $\mathcal{U}(N^{\underline{e}}_{\underline{\Sigma}'})$ and compare the line bundles of this open with the line bundles of $\mathcal{U}$. Either in $\mathcal{U}(N^{\underline{e}}_{\underline{\Sigma}'})$ there are line bundles which are not in $\mathcal{U}$ or $\mathcal{U}(N^{\underline{e}}_{\underline{\Sigma}'}) \subseteq \mathcal{U}$.

This process is finite since there are only finitely many different multidegrees of line bundles in $\mathcal{U}$. In this way we have that $\mathcal{U}$ is the union of all the opens $\mathcal{U}(N)$ that we have found, observe that indeed in this process we find for sure the most degenerate sheaves of $\mathcal{U}$ becuase for every most degenerate sheaf $N^{\underline{e}}_{\underline{\Sigma}'}$ of $\mathcal{U}$ there are two multidegrees $\underline{d}_1$ and $\underline{d}_2$ such that Equation (4.2) is satisfied. □

Proposition 4.5. *Let X be a necklace of genus $g = 1$ and $(\underline{d}, \Sigma_1, \dots, \Sigma_m)$ a 1-admissible sequence. The open $\mathcal{U}(\underline{d}, \underline{\Sigma})$ is equal to the classical compactified Jacobian $\overline{J}_X(\phi)$, where is the average of all the multidegrees of the line bundles of $\mathcal{U}(\underline{d}, \underline{\Sigma})$ that are not contracted in the moduli space $U(\underline{d}, \underline{\Sigma})$, namely*

$$\phi := \underline{d} + \frac{m-1}{m}[\mathcal{D}(\Sigma_1, O^+) - \mathcal{D}(\Sigma_1, O^-)] + \cdots + \frac{1}{m}[\mathcal{D}(\Sigma_{m-1}, O^+) - \mathcal{D}(\Sigma_{m-1}, O^-)]. \tag{4.3}$$

Proof. Let $\mathcal{U}(\underline{d}, \underline{\Sigma})$ be a smoothable good compactified Jacobian defined by the 1-admissible sequence $(\underline{d}, \Sigma_1, \dots, \Sigma_m)$, namely

$$\mathcal{U}(\underline{d}, \underline{\Sigma}) = \mathcal{U}(N^{\underline{e}_1}_{\Sigma_1}) \cup \cdots \cup \mathcal{U}(N^{\underline{e}_m}_{\Sigma_n}),$$

where

- $\underline{e}_1 = \underline{d} - \mathcal{D}(\Sigma_1, O^-)$,
- $\underline{e}_{i+1} = \underline{e}_i + \mathcal{D}(\Sigma_i, O^+) - \mathcal{D}(\Sigma_{i+1}, O^-)$.

We first prove that ϕ is indeed the average of all the multidegrees of the line bundles of $\mathcal{U}(\underline{d}, \underline{\Sigma})$ that are not contracted in the moduli space $U(\underline{d}, \underline{\Sigma})$. We call $\underline{d}_0, \underline{d}_1, \dots, \underline{d}_{m-1}$ the multidegrees of the line bundles of $\mathcal{U}(\underline{d}, \underline{\Sigma})$ which are not contracted in the moduli space.
As explained in Lemma 4.3, we have

- $\underline{d}_0 = \underline{d}$,
- $\underline{d}_i = \underline{d} + \mathcal{D}(\Sigma_1, O^+) - \mathcal{D}(\Sigma_1, O^-) + \cdots + \mathcal{D}(\Sigma_i, O^+) - \mathcal{D}(\Sigma_i, O^-)$.

therefore we get that the average of the multidegrees of the line bundles which are not contracted in the moduli space is:

$$\underline{d} + \tfrac{m-1}{m}[\mathcal{D}(\Sigma_1, O^+) - \mathcal{D}(\Sigma_1, O^-)] + \cdots + \tfrac{1}{m}[\mathcal{D}(\Sigma_{m-1}, O^+) - \mathcal{D}(\Sigma_{m-1}, O^-)],$$

namely ϕ as defined in Equation (4.3).

We prove now that $\mathcal{U}(\underline{d}, \underline{\Sigma}) = \overline{J}_X(\phi)$. From Lemma 4.4 it is enough to prove that all and only the line bundles in $\mathcal{U}(\underline{d}, \underline{\Sigma})$ are ϕ-semistable.
We first prove that all line bundles of $\mathcal{U}(\underline{d}, \underline{\Sigma})$ are ϕ-semistable and then we will prove that these are the only line bundles that are ϕ-semistable, namely we start by proving that $\overline{J}_X(\phi) = \mathcal{U}(\underline{d}, \underline{\Sigma})$.
To prove that all the line bundles of $\mathcal{U}(\underline{d}, \underline{\Sigma})$ are ϕ-semistable, given a subcurve $Y \subseteq X$ and a line bundle L of $\mathcal{U}(\underline{d}, \underline{\Sigma})$, we want to prove that

$$\deg(L_Y) \leq \sum_{X_i \subseteq Y} \phi_i + \frac{|Y \cap Y^C|}{2}. \tag{4.4}$$

Observe that $|Y \cap Y^C|$ is given by an even number of nodes $2k$. Assume for the moment that $k = 1$ and that $Y = X_1 \cup \cdots \cup X_k$, then $Y \cap Y^C = \{P_n, P_k\}$.
We first prove the inequality (4.4) for the line bundles of $\mathcal{U}(\underline{d}, \underline{\Sigma})$ which are not contracted in the moduli space and for the subcurve $Y = X_1 \cup \cdots \cup X_k$. Given a vector $\underline{v}$ we define

$$||\underline{v}||_Y := \sum_{X_i \subseteq Y} v_i.$$

For every $i = 0, \dots, m-1$ we want to prove that

$$||\underline{d}_i||_Y \leq$$
$$||\underline{d}||_Y + \tfrac{m-1}{m}||\mathcal{D}(\Sigma_1, O^+) - \mathcal{D}(\Sigma_1, O^-)||_Y + \cdots + \tfrac{1}{m}||\mathcal{D}(\Sigma_{m-1}, O^+) - \mathcal{D}(\Sigma_{m-1}, O^-)||_Y + 1.$$

Since

$$||\underline{d}_i||_Y = ||\underline{d}||_Y + ||\mathcal{D}(\Sigma_1, O^+) - \mathcal{D}(\Sigma_1, O^-)||_Y + \cdots + ||\mathcal{D}(\Sigma_i, O^+) - \mathcal{D}(\Sigma_i, O^-)||_Y,$$

we have to prove that

$$0 \leq -\frac{1}{m}||\mathcal{D}(\Sigma_1, O^+) - \mathcal{D}(\Sigma_1, O^-)||_Y - \cdots - \frac{i}{m}||\mathcal{D}(\Sigma_i, O^+) - \mathcal{D}(\Sigma_i, O^-)||_Y + \tag{4.5}$$
$$+\tfrac{m-i-1}{m}||\mathcal{D}(\Sigma_{i+1}, O^+) - \mathcal{D}(\Sigma_{i+1}, O^-)||_Y + \cdots + \tfrac{1}{m}||\mathcal{D}(\Sigma_{m-1}, O^+) - \mathcal{D}(\Sigma_{m-1}, O^-)||_Y + 1.$$

Observe that the quantity $||\mathcal{D}(\Sigma_i, O^+) - \mathcal{D}(\Sigma_i, O^-)||_Y$ can assume only the values $-1, 0$ and 1, because $\mathcal{D}(\Sigma_i, O^+) - \mathcal{D}(\Sigma_i, O^-)$ is a vector whose j-th entry is the difference between the number of nodes P in Σ_i such that $s_+(P) = j$ and the number of nodes Q in Σ_i such that $t_+(Q) = j$. Since we are working with a necklace curve and the clockwise orientation O^+, for every irreducible component X_r of X, there is precisely one node whose source is X_r and one node whose target is X_r.
Moreover only two situations can occour:

- $\exists!\ l$ such that $P_k, P_n \in \Sigma_l$, therefore for all the indeces $i = 1, \dots, m-1$, we have

$$||\mathcal{D}(\Sigma_i, O^+) - \mathcal{D}(\Sigma_i, O^-)||_Y = 0,$$

- $\exists!\ l, l'$ such that $l \neq l'$, $P_k \in \Sigma_l$ and $P_n \in \Sigma_{l'}$, therefore

$$||\mathcal{D}(\Sigma_l, O^+) - \mathcal{D}(\Sigma_l, O^-)||_Y = -1 \text{ and } ||\mathcal{D}(\Sigma_{l'}, O^+) - \mathcal{D}(\Sigma_{l'}, O^-)||_Y = 1,$$

  and for all the other indeces $k \neq l, l'$ we have

$$||\mathcal{D}(\Sigma_k, O^+) - \mathcal{D}(\Sigma_k, O^-)||_Y = 0.$$

The uniqueness of the existence of the indeces l, l' follows from the fact that the sequence $(\underline{d}, \underline{\Sigma})$ is 1-admissible and therefore every node belongs to precisely one of the Σ_l's.
In the first case the inequality 4.5 is satisfied because we get $0 \leq 1$.
In the second case we get that the RHS of 4.5 is bigger or equal than

$$-\tfrac{i}{m} - \tfrac{m-i-1}{m} + 1 = \tfrac{1}{m}$$

which is non-negative.
Now we prove that also the line bundles which are contracted in the moduli space satisfy the inequality 4.4 when we consider the subcurve $Y = X_1 \cup \cdots \cup X_k$.
Consider a line bundle of $\mathcal{U}(\underline{d}, \underline{\Sigma})$ of multidegree $\underline{f}$ which is contracted in the moduli space. Without loss of generality we can assume that there exists $S \subset \Sigma_1$ such that

$$\underline{f}' = \underline{d} + \mathcal{D}(S, O^+) - \mathcal{D}(S, O^-).$$

We have to prove that

$$||\mathcal{D}(S, O^+) - \mathcal{D}(S, O^-)||_Y \leq$$
$$\tfrac{m-1}{m}||\mathcal{D}(\Sigma_1, O^+) - \mathcal{D}(\Sigma_1, O^-)||_Y + \cdots + \tfrac{1}{m}||\mathcal{D}(\Sigma_{m-1}, O^+) - \mathcal{D}(\Sigma_{m-1}, O^-)||_Y + 1.$$

As observed above we have $||\mathcal{D}(S, O^+) - \mathcal{D}(S, O^-)||_Y \in \{-1, 0, 1\}$. The right hand side of the inequality we want to prove is bigger or equal than

$$-\tfrac{m-1}{m} + \tfrac{1}{m} + 1 = \tfrac{2}{m} \geq 0.$$

This means that the only case we have to worry about is when $||\mathcal{D}(S,O^+)-\mathcal{D}(S,O^-)||_Y = 1$. This case occours only when $P_n \in S$ and $P_k \notin S$, we get $1 \leq \frac{m-1}{m} - \frac{i}{m} + 1$, for $i \leq m-1$ and therefore the inequality holds.
Now we prove that what we have done so far can be generalized to any subcurve $Y \subseteq X$: Consider $Y \subseteq X$ with h connected components $Y_1, \dots, Y_h$ and let $Y \cap Y^C = \{P_{i_1}, \dots, P_{i_{2h}}\}$. We can write

$$||\mathcal{D}(S,O^+) - \mathcal{D}(S,O^-)||_Y = ||\mathcal{D}(S,O^+) - \mathcal{D}(S,O^-)||_{Y_1} + \cdots + ||\mathcal{D}(S,O^+) - \mathcal{D}(S,O^-)||_{Y_h}$$

and taking each Y_i individually we know that the inequality 4.4 holds. Taking the sum over all the connected components Y_i of Y we get

$$||\mathcal{D}(S,O^+) - \mathcal{D}(S,O^-)||_Y \leq$$
$$\tfrac{m-1}{m}||\mathcal{D}(\Sigma_1,O^+) - \mathcal{D}(\Sigma_1,O^-)||_Y + \cdots + \tfrac{1}{m}||\mathcal{D}(\Sigma_{m-1},O^+) - \mathcal{D}(\Sigma_{m-1},O^-)||_Y + h$$

which is the inequality we want since $h = \frac{|Y \cap Y^C|}{2}$.
So far we have proved that $\bar{J}_X \subseteq J_X(\phi)$.

To prove the inclusion $J_X(\phi) \subseteq \bar{J}_X$ we have to show that if a line bundle does not belong to $\bar{J}_X$, it is not ϕ-semistable. We know that $\bar{J}_X(\phi)$ defines a good compactified Jacobian. Observe that if $(\underline{d}, \Sigma_1, \dots, \Sigma_m)$ contains $(\underline{d}', \Sigma'_1, \dots, \Sigma'_{m'})$ there exists $i \in \{1, \dots, m\}$ such that $\Sigma_i \supseteq \Sigma_j \cup \Sigma_{j+1}$ for some $j \in \{1, \dots, m'\}$. Without loss of generality we can assume $j = 1$ and hence $\bar{J}_X(\phi)$ must contain also the good compactified Jacobian $\mathcal{U}(\underline{d}, \Sigma_1 \cup \Sigma_2, \dots, \Sigma_m)$ defined by the 1-admissible sequence $(\underline{d}, \Sigma_1 \cup \Sigma_2, \dots, \Sigma_m)$. Assume that $P_h \in \Sigma_2$, $P_k \in \Sigma_1$ and consider the curve $Y = X_{h+1} \cup \cdots \cup X_k$. We claim that the line bundle of multidegree $\underline{d} + (\mathcal{D}(\{P_h\}, O^+) - \mathcal{D}(\{P_h\}, O^-))$ which is in $\mathcal{U}(\underline{d}, \Sigma_1 \cup \Sigma_2, \dots, \Sigma_m)$ is not ϕ-semistable: We should have

$$||\mathcal{D}(\{P_h\},O^+) - \mathcal{D}(\{P_h\},O^-)||_Y \leq$$
$$\tfrac{m-1}{m}||\mathcal{D}(\Sigma_1,O^+) - \mathcal{D}(\Sigma_1,O^-)||_Y + \cdots + \tfrac{1}{m}||\mathcal{D}(\Sigma_{m-1},O^+) - \mathcal{D}(\Sigma_{m-1},O^-)||_Y + 1$$

which implies

$$1 \leq -\tfrac{m-1}{m} + \tfrac{m-2}{m} + 1 = 1 - \tfrac{1}{m}$$

which is not true.
Therefore $\overline{J}_X(\phi) \subseteq \mathcal{U}(\underline{d}, \underline{\Sigma})$ and we have that

$$\mathcal{U}(\underline{d}, \underline{\Sigma}) = \overline{J}_X(\phi).$$

This means that every smoothable good compactified Jacobian is classical, namely it can be obtained starting from a numerical polarization. $\square$

5. FAMILIES OF GOOD COMPACTIFIED JACOBIANS

In this chapter we consider families of good compactified Jacobians. In particular we consider all the necklaces X in $\overline{\mathcal{M}}_{1,n}$, namely stable necklaces of length smaller than or equal to n with n marked points.
We want to assign a good compactified Jacobian to every necklace X such that this assignment is compatible with curves' degenerations or equivalently with morphisms between the corresponding dual graphs.

Let us denote by $L(X)$ and $L(X')$ respectively the set of multidegrees of line bundles which are in the good compactified Jacobian associated to X and in the good compactified Jacobian associated to X'. Assume that $\gamma\colon \Gamma \to \Gamma'$ is a morphism between the dual graphs of the curves X and X'. We denote by $\gamma(\underline{e})$ the vector whose w-th entry $\gamma(\underline{e})_w$ is

$$\gamma(\underline{e})_w = \textstyle\sum_{v\in(\gamma')^{-1}(w)} e_v.$$

If a collection of good compactified Jacobians is compatible with curve degenerations, we have that

$$L(X') = \{\gamma(\underline{e}) \mid \underline{e} \in L(X)\}. \tag{5.1}$$

Definition 5.1.
An *admissible* assignment L^d of degree d for $\overline{\mathcal{M}}_{1,n}$ is a collection of good compactified Jacobians of degree d, one for every necklace of $\overline{\mathcal{M}}_{1,n}$, such that this collection is compatible with degenerations of curves.

In this chapter, fixed the total degree d, we give a combinatorial classification of the admissible assignments L^d_n for $\overline{\mathcal{M}}_{1,n}$.

Lemma 5.2. *Let X and X' be two necklaces and $\gamma\colon \Gamma_X \to \Gamma_{X'}$ be a morphism between their dual graphs. Let L^d_n be an admissible assignment and $\overline{J}_X$ be the good compactified Jacobian of X corresponding to the r-admissible sequence $(\underline{d}, \Sigma_1, \dots, \Sigma_m)$. The good compactified Jacobian of the admissible assignment L^d_n for X' corresponds to the r-admissible sequence*

$$(\gamma(\underline{d}), \gamma(\Sigma_1) \cap E(\Gamma_{X'}), \dots, \gamma(\Sigma_m) \cap E(\Gamma_{X'})). \tag{5.2}$$

Proof. From Lemma 4.4, the line bundles in a good compactified Jacobian completely define the good compactified Jacobian and, from Corollary 3.6, we have that the line bundles of the good compactified Jacobian corresponding to the r-admissible sequence $(\underline{d}, \Sigma_1, \dots, \Sigma_m)$ have multidegrees:

$$\underline{f} = \underline{d} + \mathcal{D}'(\Sigma_1, O^+) + \dots + \mathcal{D}'(\Sigma_{i-1}, O^+) + \mathcal{D}'(\Sigma'_i, O^+),$$

where $\Sigma'_i \subseteq \Sigma_i$, for some $i = 1, \dots, m$.
Therefore, since $\gamma(\mathcal{D}'(\Sigma_1, O^+)) = \mathcal{D}'(\gamma(\Sigma_1)) \cap E(\Gamma_{X'}), O^+)$, the line bundles of multidegree

$$\gamma(\underline{f}) = \gamma(\underline{d}) + \mathcal{D}'(\gamma(\Sigma_1) \cap E(\Gamma_{X'}), O^+) + \dots + \mathcal{D}'(\gamma(\Sigma'_i) \cap E(\Gamma_{X'}), O^+),$$

where $\gamma(\Sigma'_i) \cap E(\Gamma_{X'}) \subseteq \gamma(\Sigma_i) \cap E(\Gamma_{X'})$, for some $i = 1, \dots, m$, are in the good compactified Jacobian of X' and these are precisely the multidegrees of the good compactified Jacobian corresponding to the sequence (5.2). $\square$

Remark 5.3. If two admissible sequences are γ-compatible for some morphism of graphs γ, they have the same multiplicity.
For every necklace X, there is a morphism of graphs $\gamma\colon \Gamma_X \to \Gamma_Y$, where Y is a necklace of length 2. Since good compactified Jacobians for necklaces of length 2 are determined by 1-admissible sequences, the choice of a good compactified Jacobian for every necklace X must correspond to a 1-admissible sequence.

Definition 5.4.
Let X be a necklace. An open connected substack of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ that admits a proper good moduli space is *smoothable* if it corresponds to a 1-admissible sequence $(\underline{d}, \underline{\Sigma})$.

Since in Lemma 4.4 we observed that every good compactified Jacobian is completely determined by the set of its line bundles, from Equation (5.1), we deduce that to have an admissible assignment L_n^d we need to assign a smoothable good compactified Jacobian of degree d to every necklace of length n and the good compactified Jacobians of necklaces of shorter length are determined by compatibility.
The assignment of a good compactified Jacobian for every necklace of length n must satisfy that for every two morphisms $\gamma\colon \Gamma \to \Gamma''$ and $\gamma'\colon \Gamma' \to \Gamma''$ with the same target Γ'', we have that for every $\underline{d} \in L(\Gamma)$ there exists $\underline{d}' \in L(\Gamma')$ such that

$$\sum_{v \in \gamma^{-1}(w)} d_v = \sum_{v \in (\gamma')^{-1}(w)} d'_v \text{ , for all w} \in V(\Gamma''). \tag{5.3}$$

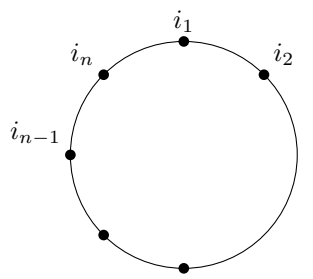
5.1 Smoothable good compactified Jacobians for necklaces of length n in $\overline{\mathcal{M}}_{1,n}$

In this section we focus on necklaces $X \in \overline{\mathcal{M}}_{1,n}$ of length n.
Starting from the numerical characterization given in [2, Proposition 3.3] to classify all *smoothable* fine compactified Jacobians of degree d for a necklace X of genus $g = 1$, we give a similiar numerical characterization to classify all *smoothable* good compactified Jacobians of degree d for a necklace X of genus 1.

The necklaces $X \in \overline{\mathcal{M}}_{1,n}$ of length n have precisely one marked point for each irreducible component and they correspond to cyclic orderings $\lhd$, up to the action of the dihedral group, of the labels $1, \dots, n$ of the marked points that identify the irreducible components:

$$i_1 \lhd \cdots \lhd i_n,$$

that we can represent as



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Given a curve $X \in \overline{\mathcal{M}}_{1,n}$, we fix a cyclic ordering $\lhd$ of the labels $1, \dots, n$ of its marked points such that $i_1 = 1$ and we refer to this curve as $X_\lhd$.
We denote by $\Sigma_\lhd := \{P_{1,\lhd}, \dots, P_{n,\lhd}\}$ the set of nodes of $X_\lhd$ and we assume that the node $P_{k,\lhd} \in X_{i_k} \cap X_{i_{k+1}}$.
If we consider the curve $X_<$ where the cyclic ordering is

$$1 < 2 < \cdots < n,$$

namely the standard ordering $<$ of the natural numbers $\{1, \dots, n\}$, we find the notation of Chapter 3: The node $P_i \in X_i \cap X_{i+1}$.

Definition 5.5.
Let $I_{n-1,\triangleleft}$ be the set of subsets of $\triangleleft$-consecutive integers of $\{2 \dots, n\}$. A function $f_\triangleleft \colon I_{n-1,\triangleleft} \to \mathbb{Z}$ is mildly superadditive if

$$0 \leq f_\triangleleft(I \cup J) - f_\triangleleft(I) - f_\triangleleft(J) \leq 1$$

whenever I, J are elements of $I_{n-1,\triangleleft}$ which are disjoint and consecutive.

Given an ordering $\triangleleft$, an integer d and a smoothable fine compactified Jacobian $\overline{J}^d_{X_\triangleleft}$ of degree d of the necklace $X_\triangleleft$, in [2, Proposition 3.3] the authors define a function $f_\triangleleft$ as

$$f_\triangleleft(I) = \min \left\{ \sum_{i \in I} d_i \middle| \text{ line bundles of multidegree } (d_1, \dots, d_n) \text{ are in } \overline{J}^d_{X_\triangleleft} \right\}, \tag{5.4}$$

for all $I \in I_{n-1,\triangleleft}$ and they prove that this function $f_\triangleleft$ is mildly superadditive.
Moreover they show that given any mildly superadditive function $f_\triangleleft$ there exists a unique smoothable fine compactified Jacobian $\overline{J}^d_{X_\triangleleft}$ of degree d such that

$$f_\triangleleft(I) = \min \left\{ \textstyle\sum_{i \in I} d_i \middle| \text{ line bundles of multidegree } (d_1, \dots, d_n) \text{ are in } \overline{J}^d_{X_\triangleleft} \right\},$$

for all $I \in I_{n-1,\triangleleft}$.

In this section we generalize [2, Proposition 3.3] to the case of smoothable *good* compactified Jacobians.
We show that we can associate a mildly superadditive function as in Equation (5.4) to every smoothable good compactified Jacobian of degree d and we explain the relation between the mildly superadditive function f associated to a fixed smoothable good compactified Jacobian of degree d and the only smoothable fine compactified Jacobian of degree d associated to this mildly superadditive function f.

Proposition 5.6. *Let $\overline{J}^d_{X_\triangleleft}$ be an open connected substack of the stack $\underline{\mathrm{Coh}}^{\mathrm{tf},d}_1(X)$ which admits a proper good moduli space defined by the 1-admissible sequence $(\underline{d}, \Sigma_1, \dots, \Sigma_m)$. The function $f \colon I_{n-1,\triangleleft} \to \mathbb{Z}$ defined by*

$$f(I) \coloneqq \min \left\{ \sum_{i \in I} d_i \middle| \text{ line bundles of multidegree } (d_1, \dots, d_n) \text{ are in } \overline{J}^d_{X_\triangleleft} \right\} \tag{5.5}$$

is mildly superadditive.
Moreover this function f corresponds to the smoothable fine compactified Jacobian of degree d defined by the simple 1-admissible sequence

$$(\underline{d},\ \mathrm{ord}_>(\Sigma_1), \dots, \mathrm{ord}_>(\Sigma_m)),$$

where $\mathrm{ord}_>(\Sigma_i)$ means that the elements of each Σ_i are arranged in decreasing order.

Proof. For simplicity of notation, in the proof we consider the ordering $<$ and we omit the pedix $<$.
Let $\overline{J}^d_X$ be an open connected substack of the stack $\underline{\mathrm{Coh}}^{\mathrm{tf},d}_1(X)$ which admits a proper good moduli space corresponding to the 1-admissible sequence $(\underline{d}, \Sigma_1, \dots, \Sigma_m)$. All fine compactified Jacobians defined by the 1-admissible sequence

$$(\underline{d}, \mathrm{ord}(\Sigma_1), \dots, \mathrm{ord}(\Sigma_m)),$$

for some ordering $\mathrm{ord}(\Sigma_i)$ for every $i = 1, \dots, m$, are contained in $\overline{J}^d_X$.
We claim that the values $f(I)$, for every $I \in I_{n-1}$, are reached by line bundles in the fine compactified Jacobian

$$\overline{J}^{d,\mathrm{simp}}_{>} := (\underline{d}, \mathrm{ord}_{>}(\Sigma_1), \dots, \mathrm{ord}_{>}(\Sigma_m))$$

corresponding to choosing the decreasing order for every Σ_i. Namely we want to prove that, for every $I \in I_{n-1}$,

$$f(I) = \min\left\{\sum_{i\in I} d_i \mid \text{ line bundles of multidegree } (d_1, \dots, d_n) \text{ are in } \overline{J}^{d,\ \mathrm{simp}}_{>}\right\}.$$

If we prove this claim than we have as a direct consequence of [2, Proposition 3.3] that the function f is mildly superadditive.
To prove the claim, let L_I be a line bundle of $\overline{J}^d_X$ of multidegree $(f_1, \dots, f_n)$ such that $f(I) = \sum_{i\in I} f_i$. Without loss of generality, we can assume that

$$\underline{f} = \underline{d} + \mathcal{D}(\Sigma', O^+) - \mathcal{D}(\Sigma', O^-)$$

where $\Sigma' \subseteq \Sigma_1$. We want to prove that in $\overline{J}^{d,\ \mathrm{simp}}_{>} = (\underline{d}, \mathrm{ord}_{>}(\Sigma_1), \dots, \mathrm{ord}_{>}(\Sigma_m))$ there is a line bundle that reaches the value $f(I) = \sum_{i\in I} f_i$.
We first show that for every $I \in I_{n-1}$ we have

$$\sum_{i\in I} d_i - 1 \leq f(I) \leq \sum_{i\in I} d_i.$$

If we call $\overline{d}$ the multidegree of some line bundles in $\overline{J}_X$ and we denote by

$$||\underline{v}||_{i,\dots,i+r} := \sum_{k=i,\dots i+r} v_k,$$

we have

$$\overline{d}_i + \dots \overline{d}_{i+r} = d_i + \dots + d_{i+r} + ||\mathcal{D}(\Sigma', O^+) - \mathcal{D}(\Sigma', O^-)||_{i,\dots,i+r} \tag{5.6}$$

for some $\Sigma' \subseteq \Sigma$: This follows from the explicit description of the multidegrees of line bundles in $\overline{J}_X$ that we gave in Lemma 3.30 and by observing that $\mathcal{D}(\Sigma_1, O) + \dots + \mathcal{D}(\Sigma_k, O) = \mathcal{D}(\Sigma_1 \cup \dots \cup \Sigma_k, O)$.
Moreover

$$||\mathcal{D}(\{P_k\}, O^+) - \mathcal{D}(\{P_k\}\, O^-)||_{i,\dots,i+r} \neq 0,$$

if and only if $k = i - 1$ or $k = i + r$. In particular we have:

- $||\mathcal{D}(\{P_{i-1}\}, O^+) - \mathcal{D}(\{P_{i-1}\}, O^-)||_{i,\dots,i+r} = 1$,
- $||\mathcal{D}(\{P_{i+r}\}, O^+) - \mathcal{D}(\{P_{i+r}\}, , O^-)||_{i,\dots,i+r} = -1$.

Therefore we have

$$\sum_{i\in I} d_i - 1 \leq f(I) \leq \sum_{i\in I} d_i.$$

If $f(I) = \sum_{i\in I} d_i$ then we conclude because the vector $\underline{d}$ is in every fine compactified Jacobian contained in $\overline{J}^d_X$, hence in particular it is in $\overline{J}^{d,\ \mathrm{simp}}_{>}$. We can focus on the case when

$$f(I) = \sum_{i\in I} d_i - 1.$$

In this case the node P_k, where $i_k = \max(I)$ is in Σ' and P_r, where $i_r = \min(I) - 1$ is not. If we order the elements of Σ_1 in a decreasing order, we have that the line bundles of multidegree

$$\underline{e} := \underline{d} + \mathcal{D}(\Sigma'', O^+) - \mathcal{D}(\Sigma'', O^-),$$

for Σ'' the subset of Σ' given by all the nodes P_j with $j \geq \max(I)$, are in $\overline{J}^{d,\ \mathrm{simp}}_{>}$. This implies that

$$\sum_{i\in I} e_i = \sum_{i\in I} d_i - 1 = f(I),$$

namely the function f that we defined in Equation (5.5) is the mildly superadditive function corresponding to the fine compactified Jacobian $\overline{J}^{d,\ \mathrm{simp}}_{>}$. □

Given a mildly superadditive function $f_\triangleleft$, we want to understand which are the smoothable good compactified Jacobians $\overline{J}^d_{X_\triangleleft}$ which have $f_\triangleleft$ as the mildly superadditive function defined in Equation (5.5).

Definition 5.7.
Let $\underline{\sigma} = (\sigma_i |\ i \in \mathbb{Z}/n\mathbb{Z}) = (P_{i_1}, \dots, P_{i_n})$ be a simple 1-admissible sequence.
The *decreasing* partition $P_{\underline{\sigma}}(\Sigma)$ of the set of nodes $\Sigma = \{P_1, \dots, P_n\}$ associated with $\underline{\sigma}$ is the partition formed by grouping consecutive entries of $\underline{\sigma} = (P_{i_1}, \dots, P_{i_n})$ if and only if their indices i_k's are decreasing.

Remark 5.8. If $\underline{\sigma}$ and $\underline{\sigma}'$ are two simple 1-admissible sequences such that

$$\underline{\sigma} = \tau^k \underline{\sigma}'$$

where τ^k is the cyclic permutation that sends i to $i+k$ for every $i = 1, \dots, n$, Then the partitions $P_{\underline{\sigma}}(\Sigma)$ and $P_{\underline{\sigma}'}(\Sigma)$ are equal.

Definition 5.9.
Let d be an integer and f a mildly superadditive function that corresponds to the degree d fine compactified Jacobian defined by the simple 1-admissible sequence $(\underline{d}, \underline{\sigma})$.
We define the partition of Σ associated to f as $P_f(\Sigma) \coloneqq P_{\underline{\sigma}}(\Sigma)$.

Remark 5.10. The definition of $P_f(\Sigma)$ is well posed because if two different admissible sequences $(\underline{d}, \underline{\sigma})$ and $(\underline{d}', \underline{\sigma}')$ define the same fine compactified Jacobian, then we know, from Theorem 3.28 that they are equivalent in the sense of Definition 3.23. Namely we have that

$$\underline{\sigma} = \tau^k \underline{\sigma}',$$

where τ^k is the cyclic permutation that sends i to $i+k$ for every $i = 1, \dots, n$. Therefore the partitions $P_{\underline{\sigma}}(\Sigma)$ and $P_{\underline{\sigma}'}(\Sigma)$ coincide.

Lemma 5.11. *Let $\overline{J}^d_X = (\underline{d}, \Sigma_1, \dots, \Sigma_m)$ be a good compactified Jacobian with associated mildly superadditive function f. The partition $\Sigma_1 \cup \dots \cup \Sigma_m$ is a refinement of the partition $P_f(\Sigma)$.*

Proof. From Proposition 5.6 we know that the mildly superadditive function f via the bijection of [2, Proposition 3.3] corresponds to the smoothable fine compactified Jacobian defined by the simple 1-admissible sequence $(\underline{d}, \mathrm{ord}_>(\Sigma_1), \dots, \mathrm{ord}_>(\Sigma_m))$. Therefore, if we denote by $\underline{\sigma} = (\mathrm{ord}_>(\Sigma_1), \dots, \mathrm{ord}_>(\Sigma_m))$, we have that every element of $P_{\underline{\sigma}}(\Sigma)$ is an union of consecutive Σ_i's, because the elements of each Σ_i's are in decreasing order in consecutive positions of $\underline{\sigma}$ and therefore they are grouped together, moreover if $\min(\Sigma_i) > \max(\Sigma_{i+1})$ then also the elements of Σ_{i+1} are grouped together with the element of Σ_i in $P_{\underline{\sigma}}(\Sigma)$. This means that $\Sigma_1 \cup \dots \cup \Sigma_m$ is a refinement of the partition $P_f(\Sigma)$. □

The idea now is that to generalize the bijection given in [2, Proposition 3.3] between mildly superadditive functions and smoothable fine compactified Jacobians we have to consider not just mildly superadditive functions but also partitions of the set of nodes Σ.

Definition 5.12.
Let $X_\triangleleft$ be a necklace and $f_\triangleleft$ a mildly superadditive function. Consider a partition $P(\Sigma_\triangleleft)$ of the set of nodes $\Sigma_\triangleleft$ of $X_\triangleleft$.
The function $f_\triangleleft$ is *compatible* with the partition $P(\Sigma_\triangleleft)$ if

- $P(\Sigma_\triangleleft)$ is a refinement of the partition $P_f(\Sigma_\triangleleft)$,
- For every $I, J \in I_{n-1,\triangleleft}$ disjoint and consecutive such that $\{P_{k,\triangleleft}, P_{k+r,\triangleleft} |\ i_k = \max_\triangleleft(I), i_{k+r} = \max_\triangleleft(J)\}$ belongs to the same element of the partition $P(\Sigma_\triangleleft)$, we have

$$f_\triangleleft(I \cup J) - f_\triangleleft(I) - f_\triangleleft(J) = 1.$$

A pair $(f_\triangleleft, P(\Sigma_\triangleleft))$ is *compatible* if $f_\triangleleft$ is a mildly superadditive function compatible with the partition $P(\Sigma_\triangleleft)$ of the set of nodes $\Sigma_\triangleleft$.

Proposition 5.13. *Let d be an integer and $X_\triangleleft$ be a necklace in $\overline{\mathcal{M}}_{1,n}$ of length n. The map*

$$f\colon \left\{ \begin{array}{c} \textit{Smoothable good compactified} \\ \textit{Jacobians of } X_\triangleleft \textit{ of degree } d \end{array} \right\} \longrightarrow \left\{ \begin{array}{c} \textit{compatible pairs} \\ (f_\triangleleft, P(\Sigma_\triangleleft)) \end{array} \right\}$$

that sends the 1-admissible sequence $(\underline{d}, \Sigma_1, \dots, \Sigma_m)$ to the pair $(f_\triangleleft, P(\Sigma_\triangleleft))$ where

- *the function $f_\triangleleft\colon I_{n-1,\triangleleft} \to \mathbb{Z}$ is defined as*

$$f_\triangleleft(I) \coloneqq \min\left\{ \textstyle\sum_{i\in I} d_i \,|\, \text{ line bundles of multidegree } (d_1, \dots, d_n) \text{ are in } \overline{J}^d_{X_\triangleleft} \right\}$$

- $P(\Sigma_\triangleleft) = \Sigma_1 \cup \dots \cup \Sigma_m$

is a bijection.
Given a compatible pair $(f_\triangleleft, P(\Sigma_\triangleleft))$ there is a unique smoothable good compactified Jacobian $\overline{J}^d_{X_\triangleleft} = (\underline{d}, \Sigma_1, \dots, \Sigma_m)$ of degree d such that the mildly superadditive function associated to $\overline{J}^d_{X_\triangleleft}$ as in Proposition 5.6 is $f_\triangleleft$ and $\Sigma_1 \cup \dots \cup \Sigma_m = P(\Sigma_\triangleleft)$.

Proof. In the proof, for simplicity of notation, we consider the curve $X_<$ corresponding to the standard order of natural numbers and we omit the pedix $<$.
Consider $\overline{J}_X$ an open connected substack of $\underline{\mathrm{Coh}}_1^{\mathrm{tf},d}(X)$ which admits a proper good moduli space defined by the 1-admissible sequence $(\underline{d}, \Sigma_1, \dots, \Sigma_m)$.
We want to prove that the pair $(f, P(\Sigma))$ defined in the statement is compatible. First of all, we observe that the function f is mildly superadditive as we have explained in Proposition 5.6 ans $P(\Sigma) \coloneqq \{\Sigma_1, \dots, \Sigma_m\}$ is a partition of Σ since by assumption $(\underline{d}, \Sigma_1, \dots, \Sigma_m)$ is 1-admissible and $P(\Sigma)$ is a refinement of the partition $P_f(\Sigma)$ as we have observed in Lemma 5.11.
We prove now that f is compatible with the partition $P(\Sigma)$: Consider $I = \{i, \dots, i+r\}, J = \{i+r+1, \dots, i+r+k\} \in I_{n-1}$ disjoint and consecutive such that $\{P_{i+r}, P_{i+r+k}\}$ belong to the same element of $P(\Sigma)$. Without loss of generality we can assume that $\{P_{i+r}, P_{i+r+k}\} \subseteq \Sigma_1$. The line bundles of multidegree

- $\underline{d} + \mathcal{D}(\{P_{i+r}\}, O^+) - \mathcal{D}(\{P_{i+r}\}, O^-) = (d_1, \dots, d_{i+r} - 1, d_{i+r+1} + 1, \dots, d_n)$,
- $\underline{d} + \mathcal{D}(\{P_{i+r+k}\}, O^+) - \mathcal{D}(\{P_{i+r+k}\}, O^-) = (d_1, \dots, d_{i+r+k} - 1, d_{i+r+k+1} + 1, \dots, d_n)$

are in $\overline{J}_X$. Therefore we get

- $f(I) = \sum_{i\in I} d_i - 1$,
- $f(J) = \sum_{j\in J} d_j - 1$,
- $f(I \cup J) = \sum_{k\in I\cup J} d_k - 1$,

which implies that $f(I \cup J) - f(I) - f(J) = 1$. We can conclude that the pair $(f, P(\Sigma))$ that we have defined is compatible.
For the other direction assume that we have a mildly superadditive function f compatible with a partition $P(\Sigma) = \{\Sigma_1, \dots, \Sigma_m\}$ of the set of nodes Σ. From [2, Proposition 3.3], we know that we can find a unique fine compactified Jacobian associated to f, namely a 1-admissible sequence $(\underline{d}, \underline{\sigma})$. Since by assumption the pair $(f, P(\Sigma))$ is compatible we can define the good compactified Jacobian $\overline{J}_X$ associated to this pair to be $(\underline{d}', \Sigma_1, \dots, \Sigma_m)$ such that the vector $(\underline{d}', \tau^k\underline{\sigma}) \sim (\underline{d}, \underline{\sigma})$ is obtained from the 1-admissible pair $(\underline{d}', \Sigma_1, \dots, \Sigma_m)$ by choosing the decreasing ordering for every Σ_j. $\square$

In Proposition 5.13 we have shown that every smoothable good compactified Jacobian of $X_\triangleleft$ is defined by a compatible pair $(f_\triangleleft, P(\Sigma_\triangleleft))$.
When we consider all the necklaces of length n, for every disjoint subsets $I, J \subseteq \{2, \dots, n\}$ there is a cyclic ordering $\triangleleft$ such that $I, J, I \cup J \in I_{n-1,\triangleleft}$.
This leads to the idea of extending the Definition 5.5 of mildly superadditive function to some function defined on the set of all nonempty subsets of $\{2, \dots, n\}$ that carries all the information about all the mildly superadditive function $f_\triangleleft$ that we associate to all the necklaces $X_\triangleleft \in \overline{\mathcal{M}}_{1,n}$ of length n.

Notation 5.14.
We denote by $\mathcal{P}^+_{n-1}$ the set of all nonempty subsets of $\{2, \dots, n\}$.

Definition 5.15.
A function $f\colon \mathcal{P}^+_{n-1} \to \mathbb{Z}$ is *globally mildly superadditive* if

$$0 \leq f(I \cup J) - f(I) - f(J) \leq 1, \text{ for all } I, J \subseteq \{1, \dots, n-1\},\ I \cap J = \emptyset.$$

Remark 5.16. Starting from a compatible assignment L^d_n for $\overline{\mathcal{M}}_{1,n}$, we can define a globally mildly superadditive function f as follows: Let I be an element of $\mathcal{P}^+_{n-1}$ define

$$f(I) \coloneqq f_\triangleleft(I)$$

for any curve $X_\triangleleft$ of length n and $I \in \mathcal{P}^+_{n-1}$ such that $I \in I_{n-1,\triangleleft}$.
Observe that f is well-defined becuase, by compatibility of the assignment L^d_n, the value $f_X(I)$ is the same for all the curves X whose dual graph Γ_X admits a morphism to the dual graph of the necklace of length 2 whose one of the two irreducible components contains precisely the marked points which are in I.
We get therefore a function $f\colon \mathcal{P}^+_{n-1} \to \mathbb{Z}$, which is globally mildly superadditive since it is defined by glueing mildly superadditive functions.

Remark 5.17. The value $f(I)$ tells which is the smallest integer $-e$ such that the line bundles of multidegree $(e+d, -e)$ are in the good compactified Jacobian of the necklace of length 2 where one irreducible component contains precisely the marked points which are in I.

5.2 Necklaces of length 2 in $\overline{\mathcal{M}}_{1,n}$

In this section we focus on necklaces of length 2 in $\overline{\mathcal{M}}_{1,n}$.
There is a bijection between the set of all nonempty subsets of $\{2, \dots, n\}$ and necklaces in $\overline{\mathcal{M}}_{1,n}$ of length 2, given by associating to every subset $I \subseteq \{2, \dots, n\}$ the n-pointed necklace of length 2 where one irreducible component contains precisely the marked points which are in I.

Notation 5.18.
Given $I \in \mathcal{P}^+_{n-1}$ we denote by X_I the n-pointed necklace of length 2 where one irreducible component contains precisely the marked points which are in I.

To know which is the good compactified Jacobian of X_I, it is enough to remember the smallest integer $-e$ such that the multidegree $(e+d, -e)$ is a line bundle of the good compactified Jacobian of X_I and the information wheter all the sheaves are simple or not:

- If all the sheaves are simple we have the 1-admissible sequence $((e+d, -e), \{1\}, \{2\})$,
- If there is a sheaf not locally free at both nodes we have the 1-admissible sequence $((e+d, -e), \{1, 2\})$.

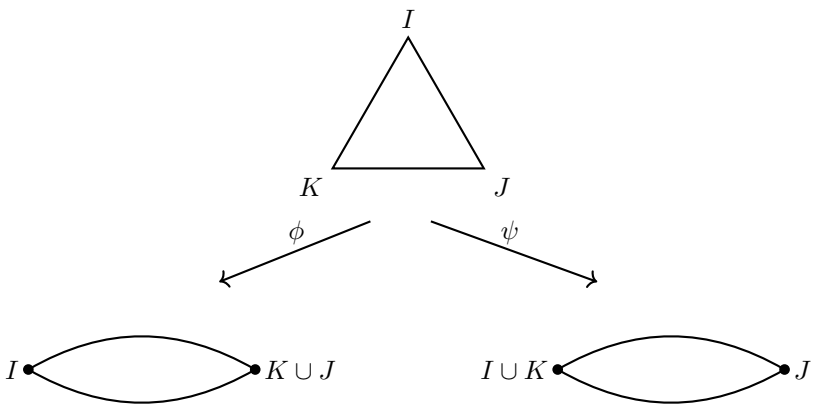
Proposition 5.19. *Let L^d_n be an admissible assignment for $\overline{\mathcal{M}}_{1,n}$, with $n \geq 3$. Let I, J be elements of $\mathcal{P}^+_{n-1}$ such that the compactified Jacobians of the necklaces X_I and X_J of length 2 have a sheaf which is not locally free at both nodes.*

1. *If $I \cap J = \emptyset$, the good compactified Jacobian of the assignment L_n^d for the necklace $X_{I \cup J}$ has a sheaf which is not locally free at both nodes.*

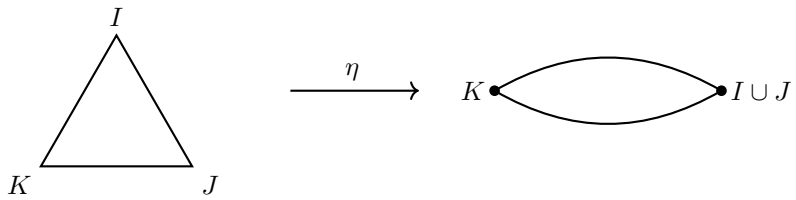
2. *If $J \subseteq I$ the good compactified Jacobian of the assignment L_n^d for the necklace $X_{I \setminus J}$ has a sheaf which is not locally free at both nodes.*

Proof. Assume that I and J are elements of $\mathcal{P}_{n-1}^+$ such that the compactified Jacobians of the necklaces X_I and X_J of length 2 have a sheaf which is not locally free at both nodes.

1. We show that if $I \cap J = \emptyset$, the good compactified Jacobians of the necklace $X_{I \cup J}$ has a sheaf which is not locally free at both nodes. Consider the necklace of length 3 where the elements of I and J define precisely the marked points of two of the three irreducible components of the curve. We can consider the morphisms of graphs

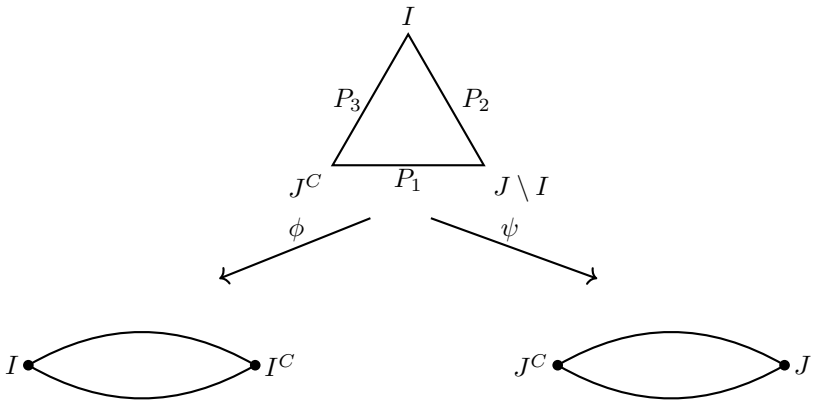


where the morphism ϕ is the contraction of the edge KJ and the morphism ψ is the contraction of the edge IK. Since we are assuming that the necklaces of length 2 corresponding to the subsets I and J of $\mathcal{P}_{n-1}^+$ have a sheaf which is not locally free at both nodes, by compatibility of the assignment L, we have that the good compactified Jacobian for the necklace of length 3 must have a sheaf which is not locally free at the nodes IJ and JK and a sheaf which is not locally free at the nodes IJ and IK. This implies that we must have a sheaf in the good compactified Jacobians corresponding to this necklace of length 3 which is not locally free at all the 3 nodes at the same time since the good compactified Jacobian of the necklace of length 3 has to be *smoothable*, namely it is defined by a 1-admissible pair $(\underline{d}, \underline{\Sigma})$, namely the vector $\underline{\Sigma}$ provides a partition of the set of nodes.
This implies, by compatibility of L, that a sheaf which is not locally free at both nodes has to appear in the good compactified Jacobian for the necklace $X_{I \cup J}$ of length 2, because of the morphism η

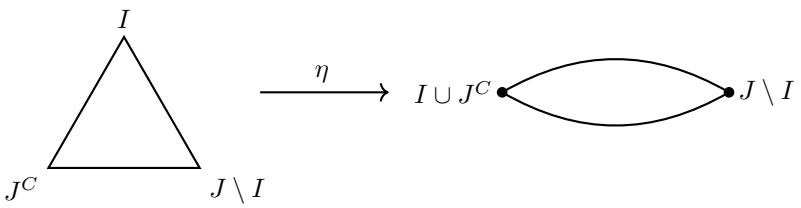


given by the contraction of the edge IJ.

2. We show that if $I \subsetneq J$ then the good compactified Jacobian of the necklace $X_{J \setminus I}$ has a sheaf which is not locally free at both nodes. Consider the necklace of length 3 where the elements of I and $J \setminus I$ define precisely the marked points of two of the three irreducible components of the curve. Consider the morphisms ϕ and ψ which are respectively contraction of the edge P_1 and P_2:



This implies that the good compactified Jacobian for the necklace of length 3 must have a sheaf which fails to be locally free at both the nodes P_2 and P_3 and a sheaf which fails to be locally free at both the nodes P_1 and P_3 which implies that this good compactified Jacobian must contain a sheaf which is not locally free at any of the nodes. Therefore if we consider the morphism η



given by the contraction of the node P_3, we find that also the good compactified Jacobian of the necklace $X_{J\setminus I}$ of length 2 must contain a sheaf which fails to be locally free at all the nodes.

□

Definition 5.20.
A subset $\mathcal{P}$ of $\mathcal{P}^+_{n-1}$ is *compatible* if the two following conditions are satisfied:

- If $I, J \in \mathcal{P}$ and are disjoint, $I \cup J \in \mathcal{P}$,
- If $I \subseteq J$, then $J \setminus I \in \mathcal{P}$.

Remark 5.21. Given a compatible assignment L^d_n, the subset $\mathcal{P}$ of $\mathcal{P}^+_{n-1}$ defined by $I \in \mathcal{P}$ if and only if the good compactified Jacobian of X_I has a sheaf that is not locally free at both nodes is compatible because of Proposition 5.19.

5.3 Admissible assignments for necklaces in $\overline{\mathcal{M}}_{1,n}$

To give an admissible assignment L^d_n in $\overline{\mathcal{M}}_{1,n}$ we need to assign a smoothable good compactified Jacobian to every necklace of length n.
If we consider the set of all the necklaces of length 2 in $\overline{\mathcal{M}}_{1,n}$ we can capture all the information about all the smoothable good compactified Jacobians of necklaces of length 2 by considering a pair $(f, \mathcal{P})$, where f carries the information for every necklace X of length 2 regarding the smallest integer $-e$ such that the line bundles of multidegree $(e+d, -e)$ are in the good compactified Jacobian of X, and $\mathcal{P}$ carries the information of which necklaces of length 2 have a good compactified Jacobian which contains a sheaf that is not locally free at both nodes.
We introduce the notion of *compatible* pair $(f, \mathcal{P})$ and we explain how one can obtain an admissible assignment L^d_n starting from a compatible pair $(f, \mathcal{P})$.

Definition 5.22.
We say that a pair $(f, \mathcal{P})$ is *compatible* if:

- f is a globally mildly superadditive function,
- $\mathcal{P}$ is a compatible subset of $\mathcal{P}^+_{n-1}$ in the sense of Definition 5.20,
- For every $I \in \mathcal{P}$ and J disjoint from I we have

$$f(I \cup J) - f(I) - f(J) = 1.$$

Theorem 5.23. *The map*

$$g\colon \left\{ \text{ Admissible assignments } L^d_n \text{ of degree } d \right\} \longrightarrow \left\{ \begin{array}{c} \text{compatible pairs} \\ (f, \mathcal{P}) \end{array} \right\}$$

that sends L^d_n to the pair $(f, \mathcal{P})$, where

- *f is defined as in Remark 5.16,*
- *$\mathcal{P}$ is the subset of $\mathcal{P}^+_{n-1}$, such that $I \in \mathcal{P}$ if and only if the good compactified Jacobian of X_I has a sheaf that is not locally free at both nodes,*

is a bijection.

Proof. Let L^d_n be an admissible assignment. We first show that given an admissible assignment L^d_n, the pair $(f, \mathcal{P})$ defined defined in the statement is a compatible pair. We have already explained in Remark 5.16 that the function $f\colon \mathcal{P}^+_{n-1} \to \mathbb{Z}$ is globally mildly superadditive since it is defined by glueing mildly superadditive functions. The subset $\mathcal{P}$ of $\mathcal{P}^+_{n-1}$ consisting of all the elements corresponding to necklaces of length 2, whose associated good compactified Jacobian has a sheaf that is not locally free at both nodes is compatible as observed in Remark 5.21.
We want to prove that the pair $(f, \mathcal{P})$ that we have defined is compatible, namely we want to show that for every $J \in \mathcal{P}$ and $I \in \mathcal{P}^+_{n-1}$ disjoint from J, we have

$$f(I \cup J) - f(I) - f(J) = 1.$$

Observe that there exists an ordering $\triangleleft$ such that $I, J, I \cup J \in I_{n-1,\triangleleft}$ and therefore

$$f(I \cup J) - f(I) - f(J) = f_\triangleleft(I \cup J) - f_\triangleleft(I) - f_\triangleleft(J) = 1$$

because the nodes $\{P_{k,\triangleleft}, P_{k+r,\triangleleft} |\ i_k = \max_\triangleleft(I), i_{k+r} = \max_\triangleleft(J)\}$ must be in the same element of the partition $P(\Sigma_\triangleleft)$ since $\Gamma_{X_\triangleleft}$ admits a morphism to Γ_{X_J} and, since J is in $\mathcal{P}$, there must be a sheaf in $\overline{J}_{X_\triangleleft}$ which is not locally free at $P_{k,\triangleleft}$ and $P_{k+r,,\triangleleft}$ at the same time. This concludes the proof that the pair $(f, \mathcal{P})$ is compatible.

We show now that the map g of the statement is a bijection by constructing an inverse:

$$h\colon \left\{ \begin{array}{c} \text{compatible pairs} \\ (f, \mathcal{P}) \end{array} \right\} \longrightarrow \left\{ \text{ Admissible assignments } L^d \text{ of degree } d \right\}.$$

Therefore we start from a compatible pair $(f, \mathcal{P})$ and we explain how we can obtain an admissible assignment L^d_n, namely a collection of good compactified Jacobians of degree d for every necklace of length smaller than or equal to n, such that the choice of these good compactified Jacobians is compatible with degenerations of curves.
For every necklace X_I of length 2 we consider the unique good compactified Jacobian $\overline{J}^d_{X_I}$ of degree d such that $f(I)$ is the smallest integer e such that the line bundles $(d-e, e)$ are in$\overline{J}^d_{X_I}$ and that has only simple sheaves if and only if $I \notin \mathcal{P}$.
We explain now how we assign a smoothable good compactified Jacobian, namley a compatible pair $(f_\triangleleft, P(\Sigma_\triangleleft))$ to every necklace $X_\triangleleft$ of length n such that this choice is compatible

with the choice of the good compactified Jacobians for necklaces of length 2.
From the globally mildly superadditive function f we can define $f_\triangleleft$ for every necklace $X_\triangleleft$ of length n by restricting f to $I_{n-1,\triangleleft}$.
Starting from the compatible subset $\mathcal{P}$ of $\mathcal{P}^+_{n-1}$, we can find for every curve $X_\triangleleft$ the partition $P(\Sigma_\triangleleft)$ of the set of nodes of $X_\triangleleft$: Consider $I \in \mathcal{P}$ and a necklace $X_\triangleleft$ of length n corresponding to an ordering $\triangleleft$ where the elements of I can be ordered in a consecutive way. We declare that the nodes of $X_\triangleleft$ which are in the intersection between the subcurve of $X_\triangleleft$ given by the irreducible components marked by I and the subcurve of $X_\triangleleft$ given by the irreducible components marked by I^C belong to the same subset of the partition of the set of nodes of $X_\triangleleft$. This gives indeed a partition of the set of nodes: Assume that:

- The nodes P_1 and P_2 are declared to be in the same element, namely there exists $I \in \mathcal{P}$ such that P_1 and P_2 are the intersection between the subcurve of X given by the irreducible components marked by I and the subcurve of X given by the irreducible components marked by I^C,
- The nodes P_2 and P_3 are declared to be in the same element of the partition, namely there exists $J \in \mathcal{P}$ such that P_2 and P_3 are the intersection between the subcurve of X given by the irreducible components marked by J and the subcurve of X given by the irreducible components marked by J^C.

This situation occours in two cases:

1. I and J are consecutive,
2. I is a proper subset of J such that $\min_\triangleleft(I) = \min_\triangleleft(J)$.

We want to show that P_1, P_2 and P_3 with our rule will be declared to be in the same element of the partition $P(\Sigma_\triangleleft)$. In the first (respectively second) case from the compatibility of $\mathcal{P}$ we have that $I \cup J \in \mathcal{P}$ (respectively $J \setminus I \in \mathcal{P}$). Therefore we declare that the nodes in the intersection between the subcurve of X given by the irreducible components marked by $I \cup J$ (respectively $J \setminus I$) and the subcurve of X given by the irreducible components marked by $(I \cup J)^C$ (respectively $(J \setminus I)^C$) must be in the same element of the partition, and this intersection is precisely given by the nodes P_1 and P_3.
Now we prove that the pair $(f_\triangleleft, P(\Sigma_\triangleleft))$ that we have just defined is compatible.
Assume that $I, J, I \cup J \in I_{n-1,\triangleleft}$ and $\{P_k, P_{k+r} |\ i_k = \max_\triangleleft(I), i_{k+r} = \max_\triangleleft(J)\}$ belong to the same element of the partition $P(\Sigma_\triangleleft)$. This is equivalent to say that J is in $\mathcal{P}$. This implies, since $\mathcal{P}$ is compatible, that

$$f_\triangleleft(I \cup J) - f_\triangleleft(I) - f_\triangleleft(J) = f(I \cup J) - f(I) - f(J) = 1.$$

Finally we want to prove that the partition $P(\Sigma_\triangleleft)$ is a refinement of the partition $P_{f_\triangleleft}(\Sigma_\triangleleft)$: By [2, Theorem 6.5], we know that there exists a unique fine compactified universal Jacobian $\overline{\mathcal{J}}^{\text{fine}}_{1,n}$ associated to a mildly superadditive function f. This implies that for every necklace $X_\triangleleft$ of length n, the good compactified Jacobian of $X_\triangleleft$ contains the fine compactified Jacobian $\overline{\mathcal{J}}^{\text{fine}}_{1,n}(X_\triangleleft)$ and as we have already explained since they have the same associated mildly super additive function f, the partition $P(\Sigma_\triangleleft)$ must be a refinement of the partition $P_f(\Sigma_\triangleleft)$.
In this way we have assigne a good compactified Jacobian to every necklace of length n and we have a good compactified Jacobian for every necklace of length 2 and the good compactified Jacobian induced from a necklace of length n to a necklace of length 2 is well defined.
We define the good compactified Jacobian $\overline{J}^d_Y$ of any necklace Y of length smaller than n by compatibility, namely by saying that the set of line bundles of $\overline{J}^d_Y$ is

$$L(Y) \coloneqq \{\gamma(\underline{e}) |\ \underline{e} \in L(X)\},$$

where X is any necklace of length n and $\gamma\colon \Gamma_X \to \Gamma_Y$ is a morphism between the dual graphs of X and Y.
We show that $L(Y)$ is well-defined: If there are two necklaces X_1 and X_2 of length n and two morphisms $\gamma_1\colon \Gamma_{X_1} \to \Gamma_Y$ and $\gamma_2\colon \Gamma_{X_2} \to \Gamma_Y$, then we claim that for every multidegree

$$\underline{d} := \gamma_1(\underline{d}_1)$$

for some multidegree $\underline{d}_1 \in L(X_1)$, there is a multidegree $\underline{d}_2 \in L(X_2)$ such that

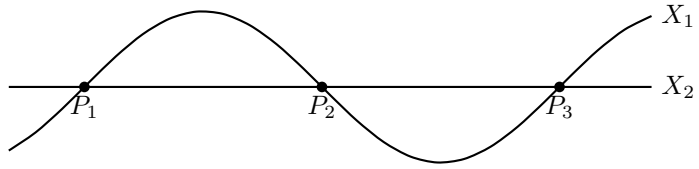
$$\underline{d} = \gamma_2(\underline{d}_1).$$

This must be the case because if not, we would find a necklace X_I of length 2 such that the morphisms of graphs $\Gamma_{X_1} \to \Gamma_{X_I}$ and $\Gamma_{X_2} \to \Gamma_{X_I}$ induce different good compactified Jacobians on X_I but this cannot be the case by the compatibility of good compactified Jacobians between necklaces of length n and necklaces of length 2 that we have by construction.
From the explicit construction of h it follows directly that $g \circ h$ and $h \circ g$ are the identity maps and therefore g is a bijection. $\square$

6. VINE CURVES WITH 3 NODES

In [4] it has been shown that to classify all possible choices of fine compactfied Jacobians for the universal family $\overline{\mathcal{C}}_{g,n}/\overline{\mathcal{M}}_{g,n}$, a crucial role is played by a class of curve, called vine curves, which are curves given by two copies of $\mathbb{P}^1$ which intersect transversally at t nodes.

In this chapter we apply the general theory developed in Chapter 2 to find all good compactified Jacobians for a vine curve with 3 nodes, namely a curve X given by two irreducible components X_1 and X_2 which are copies of $\mathbb{P}^1$ meeting transversally at three nodes that we call P_1, P_2 and P_3.



We want to find all the open connected substacks of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ which admit a proper good moduli space.
We denote by $\Sigma = \{P_1, P_2, P_3\}$ the set of nodes. For every node P_i we call S_i and T_i the preimages of P_i under the normalization map $\pi_\Sigma \colon X_\Sigma \to X$ such that $S_i \in X_1$ and $T_i \in X_2$. We start considering a sheaf $N := N_\Sigma^{\underline{e}}$ that is the push-forward of a line bundle of multidegree $\underline{e}$ on the total normalization X_Σ of X at Σ and therefore it is not locally free at all the nodes of X. We denote by $O_0 = (s_0, t_0)$ the orientation of the dual graph Γ_X of X such that $s_0(P_i) = X_2$ for every $i = 1, 2, 3$.

Proposition 6.1. *Let $N_\Sigma^{\underline{e}}$ be a sheaf that is the push-forward of a line bundle of multidegree $\underline{e}$ on the total normalization X_Σ of X at Σ and therefore it fails to be locally free at all the nodes of X. The étale morphism $\theta_{\underline{e}}$ of Proposition 2.28 for the sheaf $N_\Sigma^{\underline{e}}$ is*

$$\theta_{\underline{e}} \colon \left[\mathrm{Spec}(R_\Sigma)/\mathbb{G}_m^2\right] \to \underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$$

where $R_\Sigma = k[x_1, y_1, x_2, y_2, x_3, y_3]/(x_1y_1, x_2y_2, x_3y_3)$ and $\underline{\sigma} \in \mathbb{G}_m^2$ acts on $\mathrm{Spec}(R)$ as

- $\underline{\sigma} \cdot x_i := \sigma_1 x_i (\sigma_2)^{-1}$, *for every* $i = 1, 2, 3$,
- $\underline{\sigma} \cdot y_i := (\sigma_1)^{-1} y_i \sigma_2$, *for every* $i = 1, 2, 3$.

The image of this morphism is the open in $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ given by all the sheaves that specialize to $N_\Sigma^{\underline{e}}$.

Proof. This is Proposition 2.35 in the case of a vine curve X with 3 nodes where as fixed orientation of Γ_X we take the orientation O_0, which is the orientation such that $s_0(P_i) = X_2$, for every node P_i and $T_{\Sigma'}$ is $\mathbb{G}_m^2$. □

For every orientation $O = (s, t)$ of the dual graph Γ_X of X and multidegree $\underline{d} = \underline{e} + \mathcal{D}(\Sigma, O)$, we have

$$\textstyle\prod_{P \in \Sigma} \mathrm{Hom}^{\underline{d}}(L_{s(P_i)}, L_{t(P_i)}) = \mathrm{Spec}(k[t_1, t_2, t_3]),$$

where $t_i = x_i$ if $s(P_i) = X_1$ and $t_i = y_i$ if $s(P_i) = X_2$.
Taking all the possible orientations of the dual graph Γ_X of X we get 8 copies of $\mathbb{A}^3$ and the unramified morphisms of Proposition 2.23 are

$$\theta_O^N \colon [\mathbb{A}^3/\mathbb{G}_m^2] \to \underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X),$$

where the images of points of $\mathrm{Spec}(k[t_1, t_2, t_3])$ are given by the kernel of

$$\psi_O \colon \pi_{\Sigma,*}(\mathcal{O}_1(d_1 \cdot z_1) \oplus \mathcal{O}_2(d_2 \cdot z_2)) \to k_{P_1} \oplus k_{P_2} \oplus k_{P_3},$$

where $\underline{d} = (d_1, d_2) = \underline{e} + \mathcal{D}(\Sigma, O)$ and ψ_O sends (f_1, f_2) to

$$\begin{cases} t_i f_1(S_i) - f_2(T_i), & \text{if } t_i = x_i \\ f_1(S_i) - t_i f_2(T_i), & \text{if } t_i = y_i. \end{cases}$$

The images of points of $\mathrm{Spec}(k[t_1, t_2, t_3])$ with the coordinates $t_i = 0$ for every $i \in I$ are sheaves which fail to be locally free at the set of nodes $\Sigma' := \{P_i \mid i \in I\}$ and are the push-forward of a line bundle of multidegree $\underline{e} + \mathcal{D}(\Sigma \setminus \Sigma', O)$.
We can write explicitly the image of the morphism $\theta_{N_\Sigma^{\underline{e}}}$:

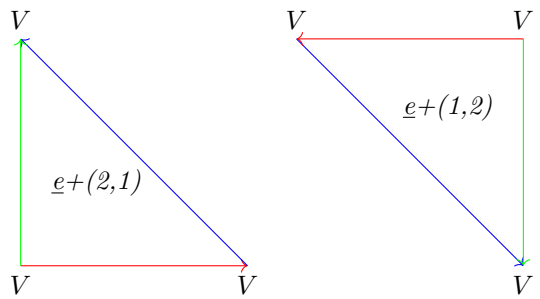
- The images of points of $\mathrm{Spec}(k[x_1, x_2, x_3])$ with all the coordinates different from zero are line bundles of multidegree $\underline{e} + \mathcal{D}(\Sigma, O_0) = \underline{e} + (0, 3)$, the points with some coordinates x_i, $i \in S \subset \{1, 2, 3\}$ equal to zero represent sheaves which are not locally free at the nodes P_i, for i in S, which are the push-forward of a line bundle of multidegree $\underline{e} + (0, 3 - |S|)$ on the total normalization X_Σ of X at Σ.

- In $\mathrm{Spec}(k[y_1, x_2, x_3]), \mathrm{Spec}(k[x_1, y_2, x_3])$ and $\mathrm{Spec}(k[x_1, x_2, y_3])$ the points with all the coordinates different from zero represent line bundles of multidegree $\underline{e} + (1, 2)$.

- In $\mathrm{Spec}(k[x_1, y_2, y_3]), \mathrm{Spec}(k[y_1, x_2, y_3]), \mathrm{Spec}(k[y_1, y_2, x_3])$ the images of points with all the coordinates different from zero are line bundles of multidegree $\underline{e} + (2, 1)$

- In $\mathrm{Spec}(k[y_1, y_2, y_3])$ the points with all the coordinates different from zero represent line bundles of multidegree $\underline{e} + (3, 0)$, the points with some coordinates y_i, $i \in S \subset \{1, 2, 3\}$ equal to zero represent sheaves which are not locally free at the nodes P_i which are the push-forward of a line bundle of multidegree $\underline{e} + (3 - |S|, 0)$.

Proposition 6.2. *Let $N_\Sigma^{\underline{e}}$ be the sheaf that is the push-forward of a line bundle of multidegree $\underline{e}$ on the total normalization X_Σ of X at Σ and therefore it fails to be locally free at all the nodes of X. The open $\mathcal{U}(N_\Sigma^{\underline{e}}) := \mathrm{Im}(\theta_{N_\Sigma^{\underline{e}}})$ in $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ admits a proper good moduli space that we can represent as*



where the two triangles represent two copies of $\mathbb{P}^2$. The edges of the triangles, which are copies of $\mathbb{P}^1$, get identified if they have the same colour and all the vertices get identified as well.

- *The points of one triangle, which do not belong to the edges, represent line bundles of multidegree $\underline{e} + (2, 1)$ and the points outside the edges of the other triangle represent line bundles of multidegree $\underline{e} + (1, 2)$*

- *The copy of $\mathbb{G}_m$ obtained from the blue copy of $\mathbb{P}^1$ removing the point V represents the sheaves $N_3^{\underline{e}+(1,1)}$*

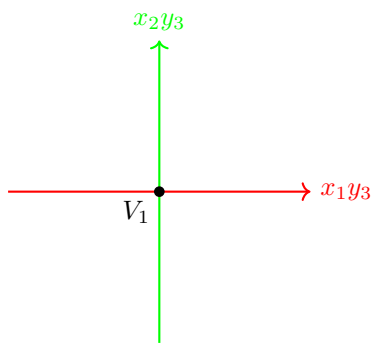
- *The copy of $\mathbb{G}_m$ obtained from the red copy of $\mathbb{P}^1$ removing the point V represents the sheaves $N_2^{\underline{e}+(1,1)}$*
- *The copy of $\mathbb{G}_m$ obtained from the green copy of $\mathbb{P}^1$ removing the point V represents the sheaves $N_1^{\underline{e}+(1,1)}$*
- *All the other sheaves in $\mathcal{U}(N_\Sigma^{\underline{e}})$ get identified in the moduli space an they are represented by the point V.*

Proof. We want to understand the image in our moduli space of the GIT quotient $\mathrm{Spec}(R)//\mathbb{G}_m^2$. The scheme $\mathrm{Spec}(R)$ is given by the union of 8 copies of $\mathbb{A}^3$. We write explicitly these copies of $\mathbb{A}^3$ and the GIT quotients $\mathbb{A}^3//\mathbb{G}_m^2$.
We first consider $\mathrm{Spec}(k[x_1, x_2, x_3])$ which has a point as GIT quotient

$$\bullet_V$$

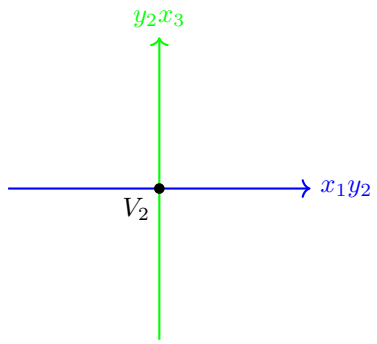
Therefore all the sheaves represented by points of $\mathrm{Spec}(k[x_1, x_2, x_3])$ get identified in the moduli space: We have the identification of the line bundles of multidegree $\underline{e} + (0, 3)$ and the sheaves $N_S^{\underline{e}'}$ with $S \subseteq \Sigma$ and $\underline{e}' = \underline{e} + (0, 3 - |S|)$.
We consider

- $\mathrm{Spec}(k[x_1, x_2, y_3])$ with GIT quotient $\mathrm{Spec}(k[x_1y_3, x_2y_3])$



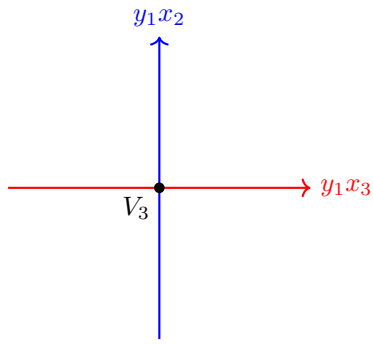
  In the moduli space the points outside the axes represent line bundles of multidegree $\underline{e} + (1, 2)$, the points on the red axis outside the origin represent $N_2^{\underline{e}+(1,1)}$, the points on the green axis represent $N_1^{\underline{e}+(1,1)}$ and in the moduli space the sheaves $N_\Sigma^{\underline{e}}, N_{12}^{\underline{e}+(1,0)}, N_{13}^{\underline{e}+(0,1)}, N_{23}^{\underline{e}+(0,1)}$ and $N_3^{\underline{e}+(0,2)}$ are identified at the origin V_1.

- $\mathrm{Spec}(k[x_1, y_2, x_3])$ has GIT quotient $\mathrm{Spec}(k[x_1y_2, y_2x_3])$



  In the moduli space the points outside the axes represent line bundles of multidegree $\underline{e} + (2, 1)$, the points on the green axis outside the origin represent $N_1^{\underline{e}+(1,1)}$, the points on the blue axis represent $N_3^{\underline{e}+(1,1)}$ and the sheaves $N_\Sigma^{\underline{e}}, N_{12}^{\underline{e}+(1,0)}, N_{13}^{\underline{e}+(1,0)}$ and $N_{23}^{\underline{e}+(0,1)}$ are identified at the origin V_2.

- $\mathrm{Spec}(k[y_1, x_2, x_3])$ has GIT quotient $\mathrm{Spec}(k[y_1x_2, y_1x_3])$



In the moduli space the points outside the axes represent line bundles of multidegree $\underline{e} + (1, 2)$, the points on the red axes outside the origin represent $N_2^{\underline{e}+(1,1)}$ and the points on the blue axes represent $N_3^{\underline{e}+(1,1)}$ V_2 the sheaves $N_{12}^{\underline{e}+(0,1)}, N_{13}^{\underline{e}+(1,0)}, N_{23}^{\underline{e}+(0,1)}$ and $N_\Sigma^{\underline{e}}$ are identified at the origin V_3. In the moduli space the points outside the axes represent line bundles of multidegree $\underline{e} + (2, 1)$, the points on the green axis outside the origin represent $N_1^{\underline{e}+(1,1)}$, the points on the blue axis represent $N_3^{\underline{e}+(1,1)}$ and the sheaves $N_\Sigma^{\underline{e}}, N_{12}^{\underline{e}+(1,0)}, N_{13}^{\underline{e}+(1,0)}$ and $N_{23}^{\underline{e}+(0,1)}$ are identified at the origin V_2.

We claim that the glueing of these three copies of $\mathbb{A}^2$ in the moduli space gives a copy of $\mathbb{P}^2$ where the coordinate points get identified. We define the following maps:

- $k[x_1y_3, x_2y_3] \to \mathbb{P}^2,\ (x_1y_3, x_2y_3) \mapsto [x_1y_3 : x_2y_3 : 1]$,
- $k[x_1y_2, y_2x_3] \to \mathbb{P}^2,\ (x_1y_2, y_2x_3) \mapsto [x_1y_2 : 1 : y_2x_3]$,
- $k[y_1yx_2, y_1x_3] \to \mathbb{P}^2,\ (y_1x_2, y_1x_3) \mapsto [1 : y_1x_2 : y_1x_3]$.

We have to prove that two points of two different copies of $\mathbb{A}^2$ that get identified in $\mathbb{P}^2$ represent the same sheaf in the moduli space and that two different points of $\mathbb{P}^2$ (which are not the coordinate points) do not get identified, namely they represent different sheaves:
Consider for example the points $[x_1y_3 : x_2y_3 : 1]$ and $[\bar{x}_1\bar{y}_2 : 1 : \bar{y}_2\bar{x}_3]$ which correspond to the sheaves defined by the short exact sequences

$$\begin{array}{ccccccc} 0 \longrightarrow & I_{(x_1,x_2,y_3)} & \longrightarrow & \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) & \xrightarrow{\psi_{x_1,x_2,y_3}} & k_{P_1} \oplus k_{P_2} \oplus k_{P_3} & \longrightarrow 0 \\ & & & (f_1, f_2) & \longmapsto & (x_1f_1(S_1) - f_2(T_1), x_2f_1(S_2) - f_2(T_2), f_1(S_3) - y_3f_2(T_3)). & \end{array}$$

and

$$\begin{array}{ccccccc} 0 \longrightarrow & I_{(\bar{x}_1,\bar{y}_2,\bar{x}_3)} & \longrightarrow & \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) & \xrightarrow{\psi_{\bar{x}_1,\bar{y}_2,\bar{x}_3}} & k_{P_1} \oplus k_{P_2} \oplus k_{P_3} & \longrightarrow 0 \\ & & & (f_1, f_2) & \longmapsto & (\bar{x}_1f_1(S_1) - f_2(T_1), f_1(S_2) - \bar{y}_2f_2(T_2), \bar{x}_3f_1(S_3) - f_2(T_3)). & \end{array}$$

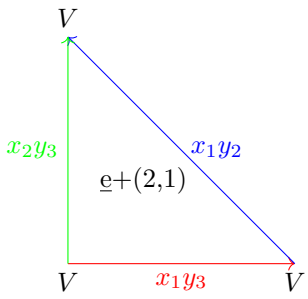
respectively.
The two sheaves coincide if and only if there exist $\alpha_1, \alpha_2, \beta_1, \beta_2, \beta_3 \in k^\times$ such that the following diagram commutes

$$\begin{array}{ccc} \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) & \xrightarrow{\psi_{x_1,x_2,y_3}} & k_{P_1} \oplus k_{P_2} \oplus k_{P_3} \\ \downarrow{\scriptstyle \cdot(\alpha_1\alpha_2)} & & \downarrow{\scriptstyle \cdot(\beta_1\beta_2\beta_3)} \\ \pi_{\Sigma,*}(\mathcal{O}_1 \oplus \mathcal{O}_2) & \xrightarrow{\psi_{\bar{x}_1,\bar{y}_2,\bar{x}_3}} & k_{P_1} \oplus k_{P_2} \oplus k_{P_3} \end{array}$$

which is equivalent to ask that there exists $t \in k^\times$ such that $x_1 = t\bar{x}_1, x_2 = t(\bar{y}_2)^{-1}$ and $(y_3)^{-1} = t\bar{x}_3$ namely if and only if

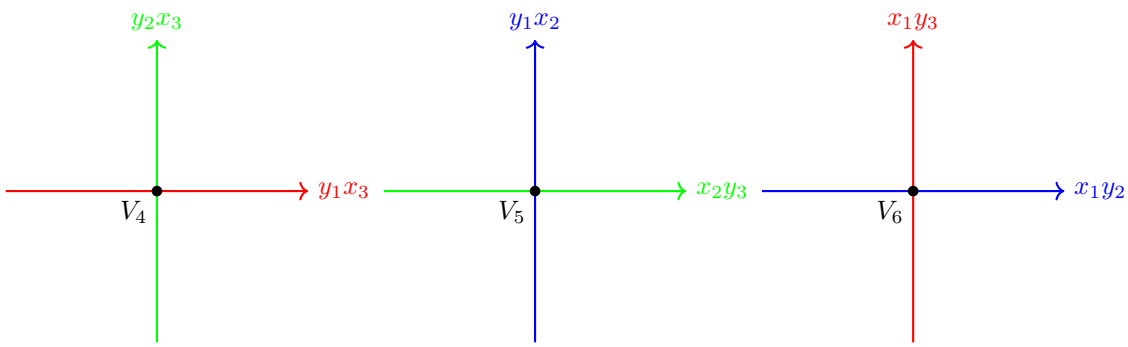
$$[x_1y_3 : x_2y_3 : 1] = [\bar{x}_1(\bar{x}_3)^{-1} : (\bar{y}_2\bar{x}_3)^{-1} : 1] = [\bar{x}_1\bar{y}_2 : 1 : \bar{y}_2\bar{x}_3].$$

The coordinate points get identified since the sheaf $N_\Sigma^{\underline{e}}$ is in the closure of the orbits of all the shaves identified in the origins V_1, V_2 and V_3 of the three copies of $\mathbb{A}^2$ that we are glueing. Therefore we get:

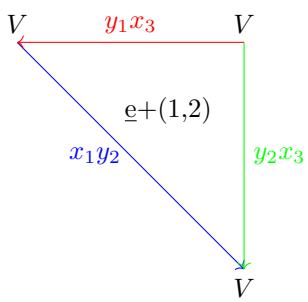


We can do something similiar for the copies $\mathrm{Spec}(k[x_1, y_2, y_3]), \mathrm{Spec}(k[y_1, x_2, y_3]), \mathrm{Spec}(k[y_1, y_2, x_3])$ and we get

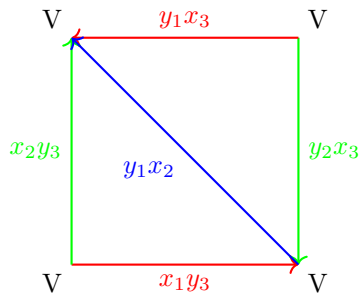
- $\mathrm{Spec}(k[y_1, y_2, x_3])$ with GIT quotient $\mathrm{Spec}(k[y_1x_3, y_2x_3])$ and in the moduli space the sheaves $N_\Sigma^{\underline{e}}, N_{13}^{\underline{e}+(1,0)}, N_{23}^{\underline{e}+(1,0)}, N_3^{\underline{e}+(2,0)}$ get identified at the origin.
- $\mathrm{Spec}(k[y_1, x_2, y_3])$ with GIT quotient $\mathrm{Spec}(k[y_1x_2, x_2y_3])$ and in the moduli space the sheaves $N_\Sigma^{\underline{e}}, N_{12}^{\underline{e}+(1,0)}, N_{23}^{\underline{e}+(1,0)}, N_2^{\underline{e}+(2,0)}$ get identified at the origin.
- $\mathrm{Spec}(k[x_1, y_2, y_3])$ with GIT quotient $\mathrm{Spec}(k[x_1y_2, x_1y_3])$ and in the moduli space the sheaves $N_\Sigma^{\underline{e}}, N_{12}^{\underline{e}+(1,0)}, N_{13}^{\underline{e}+(1,0)}, N_1^{\underline{e}+(2,0)}$ get identified at the origin.



and gluing we get a copy of $\mathbb{P}^2$ that we represent again as a triangle and whose vertices get identified and we write again as V because the points V_4, V_5 and V_6 represent sheaves whose orbits' closure contains the sheaf $N_\Sigma^{\underline{e}}$. Therefore in the moduli space we get:



Finally we consider $\mathrm{Spec}(k[y_1, y_2, y_3])$ with GIT quotient a point and in the modulil problem the line bundles of multidegree $\underline{e} + (3, 0)$ and the sheaves $N_{S'}^{\underline{e}'}$ with $S \subseteq \Sigma$ and $\underline{e}' = \underline{e} + (3 - |S|, 0)$ get identified, so this point is V in the moduli space.
Putting all these pieces together we get



which proves the statement. □

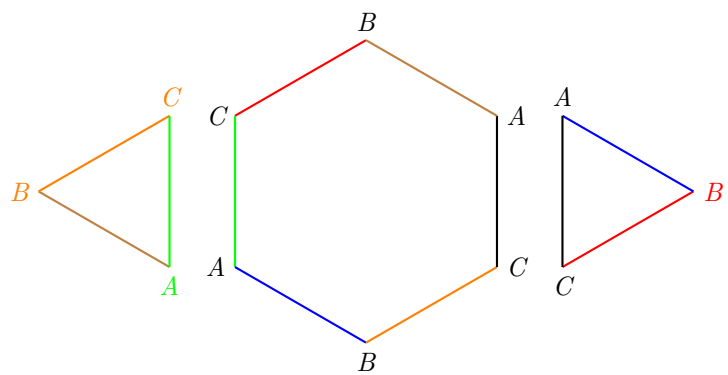
We have just found that if an open connected substacks of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ admits a proper good moduli space and it contains a sheaf $N_\Sigma^{\underline{e}}$ which is not locally free at all the nodes then this open is $U(N_\Sigma^{\underline{e}})$ and we have given an explicit description of the moduli space in this case.
Our aim is to classify *all* the open connected substacks of $\underline{\mathrm{Coh}}_1^{\mathrm{tf}}(X)$ which admits a proper good moduli space. If we assume that $\mathcal{U}$ is such an open without sheaves not locally free at any of the nodes we end up in the simple case, namely we are inside the stack $\mathrm{Simp}(X)$ of rank 1 torsion-free *simple* sheaves which are sheaves whose automorphism group is $\mathbb{G}_m$. A complete classification of the fine compactified Jacobians of X is given in [[3], Section 4, Example 4.6].
We can rewrite the classification of the fine compactified Jacobians for a vine curve with t nodes with our notation as follows.

Lemma 6.3. *If $\mathcal{U}$ is a fine compactified Jacobians for a vine curve with t nodes, there exists a sheaf N which is not locally free at all the nodes and such that the open $\mathcal{U}$ is contained in $\mathcal{U}(N)$ and it is the image under θ_N of the open defined by $x_i \neq 0$ for some $i = 1, 2, 3$.*

Proof. In the article [3] the fine compactified Jacobians are defined by a stability condition σ and in the case of vine curves every condition σ corresponds to giving an integer λ. We define $N := N_\Sigma^{(\lambda, d-\lambda-t)}$ and the image under θ_N of the open defined by $x_i \neq 0$ for some $i = 1, 2, 3$ is precisely the union of the rank 1 torsion-free sheaves listed in [3, Section 4, Example 4.6]. □

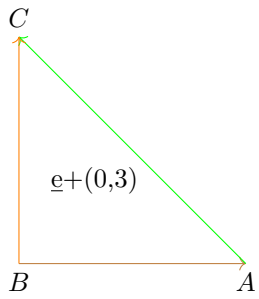
Remark 6.4. The open inside $\mathcal{U}(N_\Sigma^{\underline{e}})$ defined by $y_i \neq 0$ for some $i = 1, 2, 3$ is also a fine compactified Jacobian and it is equal to the open inside $\mathcal{U}(N_\Sigma^{\underline{e}'})$ defined by $x_i \neq 0$ for some $i = 1, 2, 3$ where $e' = \underline{e} + (1, -1)$.

Proposition 6.5. *The open inside $\mathcal{U}(N_\Sigma^{\underline{e}})$ defined by $x_i \neq 0$ for some $i \in \{1, 2, 3\}$ admits a proper good moduli space:*



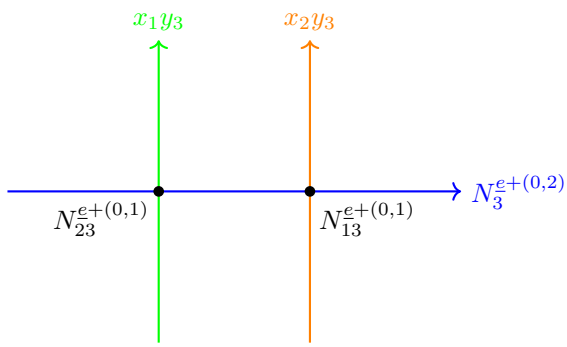
where the two triangles represent two copies of $\mathbb{P}^2$ and the hexagon represent a copy of $\mathbb{P}^2$ with blowups at the coordinate vertices. The edges with the some colours and the vertices with the same label get identified.

Proof. We first consider $\mathrm{Spec}(k[x_1, x_2, x_3])$ which has a copy of $\mathbb{P}^2$ as GIT quotient since the origin is removed in the open we are considering.



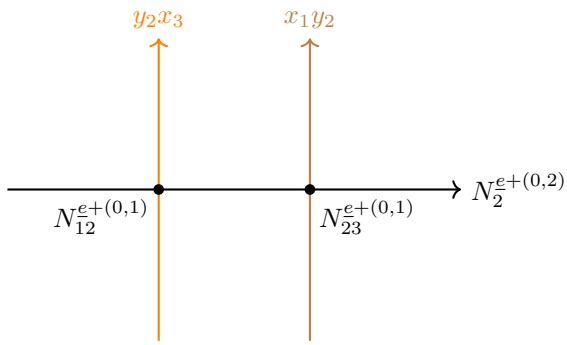
We represent $\mathbb{P}^2$ by a triangle. In the moduli space the points outside the edges represent line bundles of multidegree $\underline{e}+(0,3)$, the copies of $\mathbb{G}_m$ that we get removing the vertices represent sheaves that fail to be locally free at only one node, in particular we have $N_1^{\underline{e}+(0,2)}, N_2^{\underline{e}+(0,2)}$ and $N_3^{\underline{e}+(0,2)}$. The vertices A, B and C represent sheaves that fail to be locally free at two nodes: $N_{12}^{\underline{e}+(0,1)}, N_{13}^{\underline{e}+(0,1)}$ and $N_{23}^{\underline{e}+(0,1)}$.
We consider

- $\mathrm{Spec}(k[x_1, x_2, y_3])$ with GIT quotient a copy of $\mathbb{A}^2$ with a blowup at the origin: The GIT quotient $\mathrm{Spec}(k[x_1, x_2, y_3])//\mathbb{G}_m^2$ is a copy of $\mathbb{A}^2$ where at the origin the sheaves $N_{\Sigma}^{\underline{e}}, N_3^{\underline{e}+(0,2)}, N_{13}^{\underline{e}+(0,1)}$ and $N_{23}^{\underline{e}+(0,1)}$ are identified and the identification comes from the diagonal action on the plane $\mathrm{Spec}(k[x_1, x_2])$ defined by $y_3 = 0$. In the open we are considering we are removing the origin of this plane and therefore from the plane defined by $y_3 = 0$ we get as quotient a copy of $\mathbb{P}^1$ instead of the previous identification, where the points 0 and ∞ of $\mathbb{P}^1$ represent the sheaves $N_{13}^{\underline{e}+(0,1)}$ and $N_{23}^{\underline{e}+(0,1)}$ and the points of the copy of $\mathbb{G}_m$ obtained by removing the points 0 and ∞ represent the sheaves $N_3^{\underline{e}+(0,2)}$. Therefore we have as quotient a copy of $\mathbb{A}^2$ with a blowup at the origin:



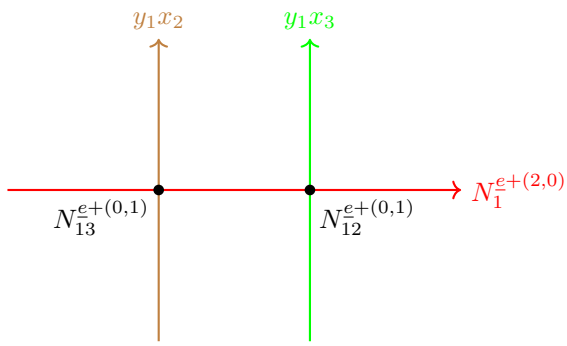
  The points of the orange axis different from $N_{12}^{\underline{e}+(0,1)}$ represent the sheaves $N_1^{\underline{e}+(1,1)}$ and the point of the brown axis different from $N_{23}^{\underline{e}+(0,1)}$ represent the sheaves $N_3^{\underline{e}+(1,1)}$.

- Something similiar holds for $\mathrm{Spec}(k[x_1, y_2, x_3])$ which has as GIT quotient a copy of $\mathbb{A}^2$ with a blowup at the origin



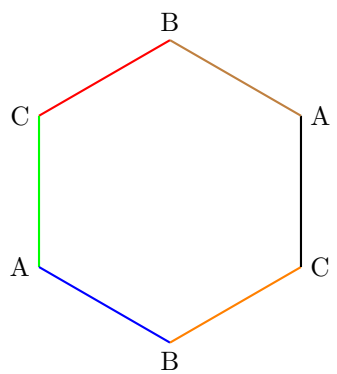
The points outside the axes represent line bundles of multidegree $\underline{e}+(1,2)$, the points on the orange axis different from the point $N_{12}^{\underline{e}+(0,1)}$ represent the sheaves $N_2^{\underline{e}+(1,1)}$, the points on the black axis different from $N_{23}^{\underline{e}+(0,1)}$ represent the sheaves $N_2^{\underline{e}+(0,2)}$ and the point on the brown axis different from $N_{23}^{\underline{e}+(0,1)}$ represent the sheaves $N_3^{\underline{e}+(1,1)}$

- $\mathrm{Spec}(k[y_1,x_2,x_3])$ has GIT quotient $\mathrm{Spec}(k[y_1x_2,y_1x_3])$



In the moduli space the points outside the axes represent line bundles of multidegree $\underline{e}+(1,2)$, the points on the brown axis outside the point $N_{13}^{\underline{e}+(0,1)}$ represent $N_3^{\underline{e}+(1,1)}$, the points on the red axis different from $N_{13}^{\underline{e}+(0,1)}$ and $N_{12}^{\underline{e}+(0,1)}$ represent the sheaves $N_1^{\underline{e}+(2,0)}$ and the points on the green axis different from $N_{12}^{\underline{e}+(0,1)}$ represent the sheaves $N_2^{\underline{e}+(1,1)}$.

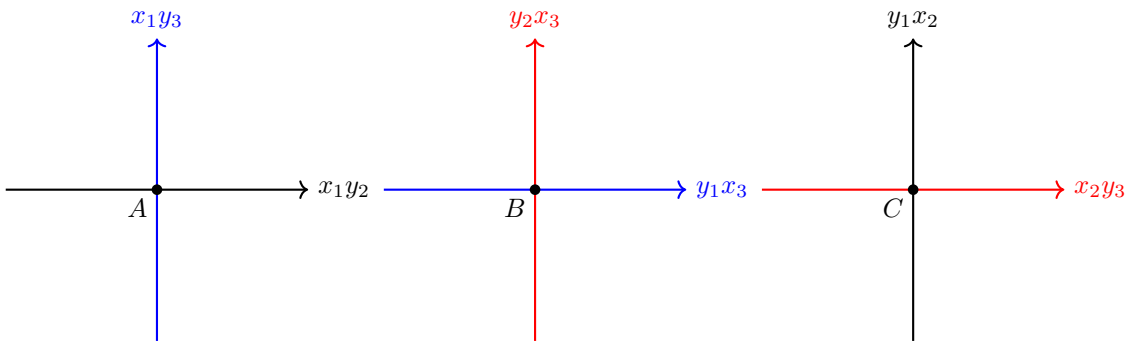
From the three copies of $\mathbb{A}^2$ with blowup at the origin we get a copy of $\mathbb{P}^2$ with blowup at the coordinate points, namley in the moduli space we get:



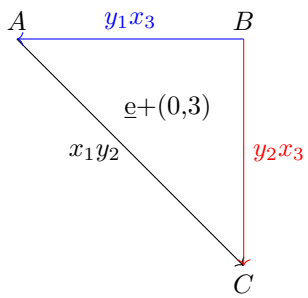
Now we consider the copies $\mathrm{Spec}(k[x_1,y_2,y_3]),\mathrm{Spec}(k[y_1,x_2,y_3]),\mathrm{Spec}(k[y_1,y_2,x_3])$ and we get

- $\mathrm{Spec}(k[y_1, y_2, x_3])$ has as GIT quotient $\mathrm{Spec}(k[y_1x_3, y_2x_3])$, now the origin in the moduli space represents only the sheaf $N_{12}^{\underline{e}+(0,1)}$ and the points outside the axes represent line bundles of multidegree $\underline{e} + (0, 3)$
- $\mathrm{Spec}(k[y_1, x_2, y_3])$ with GIT quotient a copy of $\mathbb{A}^2$ where the origin represents the sheaf $N_{13}^{\underline{e}+(0,1)}$
- $\mathrm{Spec}(k[x_1, y_2, y_3])$ with GIT quotient a copy of $\mathbb{A}^2$ where the origin represents $N_{23}^{\underline{e}+(0,1)}$.

We can draw the quotients as



and gluing in the moduli space we get a copy of $\mathbb{P}^2$ that we represent again by a triangle:



Putting all these pieces together we get the moduli space of the statement. $\square$

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